

INVARIANT TWO-JETS AND EFFECTIVE HYPERBOLICITY FOR COMPLEMENTS OF TWO PLANE CURVES

LEI HOU, PENGCHAO WANG, AND SONG-YAN XIE

ABSTRACT. Let $D = C_1 + C_2 \subset \mathbb{P}^2$ be a simple normal crossing union of smooth plane curves of degrees $1 \leq d_1 \leq d_2$. We prove an effective Second Main Theorem for a general ordered pair whenever

$$d_1, d_2 \geq 3, \quad \text{or} \quad d_1 = 2, d_2 \geq 5, \quad \text{or} \quad d_1 = 1, d_2 \geq 8.$$

For each admissible degree pair, there is a nonempty Zariski-open set of ordered pairs (C_1, C_2) for which every algebraically nondegenerate entire curve $f : \mathbb{C} \rightarrow \mathbb{P}^2$ whose image is not contained in $D = C_1 + C_2$ satisfies

$$T_f(r) \leq \mathcal{A}_{d_1, d_2} N_f^{[1]}(r, D) + o(T_f(r)) \parallel.$$

For two cubics one may take $\mathcal{A}_{3,3} = 57$; for a conic and a quintic, $\mathcal{A}_{2,5} = 45$; and for a line and an octic, $\mathcal{A}_{1,8} = 69$. Intersecting the resulting Zariski-open parameter locus with Xi Chen's very-general algebraic-hyperbolicity locus yields Kobayashi hyperbolicity and hyperbolic embedding of the complement.

The proof first constructs one negatively twisted invariant two-jet differential. It then obtains a second equation either from a Demailly–El Goul zero-locus argument or by differentiating with mixed $\mathcal{O}_{\mathbb{P}^2}(3)$ slanted vector fields. A finite calculation is needed only for a short list of low twists. In those cases, exact rank certificates over finite fields prove the required Key Vanishing Lemma.

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1. INTRODUCTION

Throughout, a property holds for a *general* member of a parameter space if it holds on a nonempty Zariski-open subset. It holds for a *very general* member if its failure locus is contained in a countable union of proper Zariski-closed subsets.

1.1. Motivation and previous results. The logarithmic Kobayashi conjecture predicts that the complement of a general plane curve of degree at least

$$2 \dim \mathbb{P}^2 + 1 = 5$$

is hyperbolic. It is the higher-dimensional analogue of the Little Picard theorem for the complement of three points in $\mathbb{P}^1$.

Jet differentials provide a way to study this conjecture. Negatively twisted sections of

$$E_{k,m}T_X^*(\log D)$$

give differential equations for entire curves in $X \setminus D$; see [3, 5]. The main difficulty is to obtain enough independent equations to control their common zero set.

A Second Main Theorem is a quantitative form of this problem. For a reduced divisor $D = C_1 + C_2$ and an entire curve $f : \mathbb{C} \rightarrow \mathbb{P}^2$, the desired estimate has the form

$$T_f(r) \leq \mathcal{A}N_f^{[1]}(r, D) + o(T_f(r)) \parallel, \quad N_f^{[1]}(r, D) = N_f^{[1]}(r, C_1) + N_f^{[1]}(r, C_2).$$

It measures how frequently an algebraically nondegenerate curve must meet the two components.

For one smooth plane curve, Siu and Yeung proved hyperbolicity in sufficiently high degree [22]. Rousseau later obtained the effective bound $d \geq 14$ by combining logarithmic vector fields with Siu's slanted-field strategy [19, 15, 16]. Recent work of Hou–Huynh–Merker–Xie lowered this bound to 12 by introducing finite key-vanishing calculations for invariant two-jets [8].

The same circle of ideas applies to divisors with several components. A particularly useful model is the union of three general conics, for which an effective Second Main Theorem was proved in [7]. Its exponent bounds convert an a priori infinite vanishing problem into finite exact algebra. The symmetry organization was later sharpened in [10].

The case of two components is more rigid. Let

$$D = C_1 + C_2 \subset \mathbb{P}^2, \quad 1 \leq d_1 := \deg C_1 \leq d_2 := \deg C_2,$$

where C_1 and C_2 are smooth and transverse. Rousseau's first result [17, Théorème 1] proves hyperbolicity in the ranges $d_1 \geq 5$, $d_1 = 4, d_2 \geq 7$, $(d_1, d_2) = (4, 4)$, $d_1 = 3, d_2 \geq 9$, and $d_1 = 2, d_2 \geq 12$.

Rousseau's later logarithmic vector-field construction [16, Theorem 3] improves these ranges to

$$d_1 \geq 4, \quad d_1 = 3, d_2 \geq 5, \quad d_1 = 2, d_2 \geq 8, \quad d_1 = 1, d_2 \geq 11.$$

The present paper reaches every pair with $d_1, d_2 \geq 3$, every pair with $d_1 = 2, d_2 \geq 5$, and every pair with $d_1 = 1, d_2 \geq 8$.

For $d \geq 1$, put

$$V_d = H^0(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(d)), \quad \mathcal{S}_{d_1, d_2} = \mathbb{P}(V_{d_1}) \times \mathbb{P}(V_{d_2}).$$

We denote by $\mathcal{S}_{d_1, d_2}^{\text{snc}}$ the nonempty open locus of ordered smooth transverse pairs. The order is kept even when $d_1 = d_2$.

Throughout the paper, an entire curve in $\mathbb{P}^2$ is called *algebraically nondegenerate* when its image is Zariski dense in $\mathbb{P}^2$. A logarithmic lift is algebraically degenerate when the Zariski closure of its image is a proper algebraic subset of the corresponding Sample level.

1.2. Two basic inputs. We begin with two basic analytic results. The first is the jet differential Second Main Theorem, recalled as Theorem 2.5. If

$$0 \neq \omega \in H^0(\mathbb{P}^2, E_{k,m}T_{\mathbb{P}^2}^*(\log D) \otimes \mathcal{O}_{\mathbb{P}^2}(-t))$$

and $f^*\omega \neq 0$, then

$$T_f(r) \leq \frac{m}{t} N_f^{[1]}(r, D) + o(T_f(r)) \parallel.$$

Thus the ratio m/t is the coefficient contributed by the first differential.

The second input is McQuillan's theorem, stated in Theorem 2.6. If the first logarithmic lift $f_{[1]}$ is algebraically degenerate and $d_{\text{tot}} = d_1 + d_2 \geq 4$, then

$$T_f(r) \leq \frac{1}{d_{\text{tot}} - 3} N_f^{[1]}(r, D) + o(T_f(r)) \parallel.$$

It remains to treat the case in which the first differential vanishes on the lifted curve but the first lift is not yet known to be degenerate.

The central task is therefore to construct a second equation. We use two complementary mechanisms. Siu's method differentiates the first equation by slanted vector fields, while the Demailly–El Goul method produces a section on an irreducible component of its zero divisor [4]. A finite Key Vanishing Lemma separates the numerical ranges in which these two mechanisms do not immediately apply.

1.3. Main statements.

Theorem 1.1 (Effective Second Main Theorem). *Let $1 \leq d_1 \leq d_2$ be integers. Assume that*

$$d_1, d_2 \geq 3, \quad \text{or} \quad d_1 = 2, d_2 \geq 5, \quad \text{or} \quad d_1 = 1, d_2 \geq 8.$$

There exist a nonempty Zariski-open subset $\mathcal{U}_{\text{SMT}}(d_1, d_2) \subset \mathcal{S}_{d_1, d_2}^{\text{Snc}}$ and an effectively computable constant $\mathcal{A}_{d_1, d_2} > 0$, both depending only on the ordered degree pair.

For every a in this subset, write $D_a = C_{1,a} + C_{2,a}$. Then every algebraically nondegenerate entire curve $f : \mathbb{C} \rightarrow \mathbb{P}^2$ whose image is not contained in D_a satisfies

$$(1.1) \quad T_f(r) \leq \mathcal{A}_{d_1, d_2} N_f^{[1]}(r, D_a) + o(T_f(r)).$$

For $(d_1, d_2) = (3, 3)$ one may take $\mathcal{A}_{3,3} = 57$. For $(d_1, d_2) = (2, 5)$ one may take $\mathcal{A}_{2,5} = 45$. For $(d_1, d_2) = (1, 8)$ one may take $\mathcal{A}_{1,8} = 69$.

The proof of Theorem 1.1 is given in Subsection 6.8.

For $a \in \mathcal{S}_{d_1, d_2}^{\text{Snc}}$, let $D_a = C_{1,a} + C_{2,a}$. If $\Gamma \not\subset D_a$ is an irreducible curve with normalization

$$\nu : \tilde{\Gamma} \longrightarrow \Gamma \hookrightarrow \mathbb{P}^2, \quad B_\nu := \text{Supp } \nu^* D_a,$$

then the logarithmic differential

$$d\nu : T_{\tilde{\Gamma}}(-\log B_\nu) \longrightarrow \nu^* T_{\mathbb{P}^2}(-\log D_a)$$

is nonzero on a dense Zariski-open subset. Its projectivization extends uniquely across the remaining finitely many points to a morphism

$$(1.2) \quad \nu_{[1]} : \tilde{\Gamma} \longrightarrow \mathbb{P}(T_{\mathbb{P}^2}(-\log D_a)), \quad x \longmapsto \left[d\nu_x(T_{\tilde{\Gamma}, x}(-\log B_\nu)) \right],$$

called the *logarithmic tangent lift* of Γ (or of ν). We call Γ *log-parabolic* with respect to D_a when

$$2g(\tilde{\Gamma}) - 2 + \#B_\nu \leq 0.$$

Theorem 1.2 (Uniform direction locus for log-parabolic curves). *Let $1 \leq d_1 \leq d_2$ satisfy one of the degree assumptions in Theorem 1.1. There exist a nonempty Zariski-open subset*

$$\mathcal{U}_{\text{dir}}(d_1, d_2) \subset \mathcal{U}_{\text{SMT}}(d_1, d_2) \subset \mathcal{S}_{d_1, d_2}^{\text{Snc}}$$

and a closed subscheme

$$\mathcal{R} \subset \mathcal{X}_{1, \mathcal{U}_{\text{dir}}} := \mathbb{P}(T_{\mathbb{P}^2 \times \mathcal{U}_{\text{dir}} / \mathcal{U}_{\text{dir}}}(-\log \mathcal{D}_{\mathcal{U}_{\text{dir}}})) ,$$

where $\mathcal{D}_{\mathcal{U}_{\text{dir}}}$ is the universal divisor, such that $\mathcal{R}$ is projective over $\mathcal{U}_{\text{dir}}$ and

$$\dim \mathcal{R}_a \leq 2 \quad (a \in \mathcal{U}_{\text{dir}}).$$

If $\Gamma \not\subset D_a$ is an irreducible curve with normalization ν and

$$2g(\tilde{\Gamma}) - 2 + \# \text{Supp } \nu^* D_a \leq 0,$$

then

$$\nu_{[1]}(\tilde{\Gamma}) \subset \mathcal{R}_a.$$

Thus the conclusion applies, in particular, to a rational curve meeting D_a in at most two points and to an elliptic curve disjoint from D_a .

The point is the uniformity: one algebraic set $\mathcal{R}$, defined over a single Zariski-open subset of the parameter space, contains the tangent lifts of all log-parabolic curves in all its fibers. Fiberwise proper subsets, allowed to vary with the curve, would not suffice. Moreover, each component of $\mathcal{R}_a$ that dominates $\mathbb{P}^2$ is generically finite over $\mathbb{P}^2$ and hence defines an algebraic multi-foliation; the remaining components project to proper algebraic subsets. This is the relative input used in forthcoming work of Xie and Zhao on the folklore general-pair form of the hyperbolicity conjecture; see Remark 2.14. The proof of Theorem 1.2 is given in Subsection 6.8.

Let $\mathcal{S}_{\text{Chen}}(d_1, d_2)$ denote the complement of the countable union of proper closed subsets excluded by Theorem 2.13.

Corollary 1.3 (Hyperbolicity and hyperbolic embedding). *Under the degree assumptions of Theorem 1.1, the complement $\mathbb{P}^2 \setminus D_a$ is Kobayashi hyperbolic and hyperbolically embedded in $\mathbb{P}^2$ for every*

$$a \in \mathcal{U}_{\text{SMT}}(d_1, d_2) \cap \mathcal{S}_{\text{Chen}}(d_1, d_2).$$

In particular, this conclusion holds for a very general ordered pair of smooth transverse curves in the stated degree range.

The proof of Corollary 1.3 is given in Subsection 6.8.

The two genericity conditions have different origins. The Second Main Theorem holds on the Zariski-open set $\mathcal{U}_{\text{SMT}}$. Xi Chen's theorem excludes a countable union of proper closed subsets. Since $\mathcal{S}_{d_1, d_2}^{\text{snc}}$ is irreducible, the two loci have a nonempty very-general intersection.

1.4. Significance and new ingredients. The main theorem has four useful features. First, the smallest total degree in the theorem comes from two cubics:

$$\deg(C_1 + C_2) = 6.$$

This is only one above the expected logarithmic Kobayashi bound $2 \dim \mathbb{P}^2 + 1 = 5$. The reducible two-cubic family is not the full parameter space of plane sextics, so this is not a direct case of the conjecture for a general irreducible curve.

Still, hyperbolicity with only two components and total degree six shows the power of invariant two-jets in low degree. The two-cubic divisor has the same total degree as three conics but one fewer component. Thus methods that require at least three logarithmic components do not apply directly.

Second, the conclusion is quantitative. We prove an effective Second Main Theorem with truncation at level one, rather than only the nonexistence of entire curves in the complement. It controls the intersections of every algebraically nondegenerate entire curve with the two components and gives explicit constants in the three boundary cases.

Third, we construct simultaneous mixed $\mathcal{O}_{\mathbb{P}^2}(3)$ slanted vector fields. The two logarithmic equations share the same base-jet variables, so their tangency conditions cannot be corrected independently. We solve the two Hessian-cancellation equations simultaneously and descend the resulting fields through both root gauges, the logarithmic double locus, and the Semple boundary.

The fact that twist three is enough is important for the constants. It strengthens the differentiated equation from twist $-t + 7$ to twist $-t + 3$. It also shortens the remaining low-twist range that must be treated by finite calculation. Separate generators for a line component and separate Taylor-remainder arguments for a conic make the same package available at the low-degree boundary.

Fourth, the finite Key Vanishing Lemma gives a reusable method. We turn the coupled extension conditions for two logarithmic residues into sparse integral matrices and organize them by the available symmetries. For equal degrees, character and component-exchange decompositions split the matrices into smaller independent blocks.

For unequal degrees, degree-sensitive normal forms keep the two homogeneities. Exact full-column-rank certificates over finite fields then give rigorous and reproducible vanishing

proofs. The same symmetry method can be used in similar finite jet-vanishing problems once their transition and divisibility conditions are written in this algebraic form.

1.5. Proof strategy. Riemann–Roch and logarithmic H^2 vanishing first produce a negatively twisted invariant two-jet differential. The Demailly–El Goul reduction then selects an irreducible horizontal factor

$$\omega_1 \in H^0(\mathbb{P}^2, E_{2,m} T_{\mathbb{P}^2}^*(\log D) \otimes \mathcal{O}_{\mathbb{P}^2}(-t)), \quad \frac{t}{m} \geq \rho_0.$$

After the fixed Semple-boundary factor is removed, its horizontal divisor has class

$$b_1 u_1 + b_2 u_2 - t h, \quad b_1 \geq 2b_2 \geq 0.$$

The proof has the following branches. The zero-locus threshold ρ_{DEG} is defined in Section 3.

branch	input used
$f^* \omega_1 \neq 0$	jet Second Main Theorem
$f^* \omega_1 \equiv 0, b_2 = 0$	algebraic degeneracy on the first Semple level
$f^* \omega_1 \equiv 0, b_2 > 0, t > 3$	mixed $\mathcal{O}_{\mathbb{P}^2}(3)$ differentiation
$f^* \omega_1 \equiv 0, b_2 > 0, t \leq 3, t/m < \rho_{\text{DEG}}$	restricted zero-locus theorem
remaining low-twist cases	finite Key Vanishing Lemma

The first four branches are geometric. They treat the full infinite range except for finitely many low-twist pairs. The Key Vanishing Lemma rules out exactly these remaining cases.

The constants 57, 45, and 69 use sharper rational splits than the basic division at $t = 3$. Section 6 gives the full calculation in a single comparison table and then proves each case.

The finite step concerns only the six boundary pairs

$$(3, 3), (3, 4), (2, 5), (2, 6), (1, 8), (1, 9).$$

For these pairs, Lemma 7.1 rules out a finite list of low-twist differentials. The exact lists and the statement of the lemma are given at the start of Section 7. This keeps the introduction focused on the proof rather than on the entries of the finite calculation.

Computational input. The finite-rank claims are exact statements about integral matrices. Full column rank modulo one good prime shows that one maximal integer minor is nonzero. The matrix therefore has full column rank over $\mathbb{Q}$ and $\mathbb{C}$.

The accompanying archive records the source, case list, matrix dimensions, ranks, nullities, certificates, hashes, and a separate checker. Runs at a second prime check the implementation; they are not extra mathematical assumptions. The archive will also be available on Song-Yan Xie’s homepage.¹

1.6. Structure of the article. The paper is organized so that readers may enter at the part most relevant to them. The following guide records the main dependencies.

¹The computational evidence and subsequent updates are maintained at <https://xiesongyan.github.io/>.

topic of interest	where to read
logarithmic jets and analytic inputs	Section 2
Chern numbers and the degree phase diagram	Section 3
existence and factorization of the first differential	Section 4
the simultaneous mixed $\mathcal{O}(3)$ construction	Section 5
the case split and the three explicit constants	Section 6, especially Subsection 6.8
the finite geometric reduction	Section 7
the exact rank certificates	Section 8
possible improvements of the constants	Section 9
SNC verification for the explicit test pairs	Appendix A
resource estimates for future computations	Appendix B

Section 2 separates standard inputs from the direct-image and zero-locus arguments proved here. Readers interested mainly in the new geometric construction may go from the main statements directly to Section 5. Readers interested mainly in the exact finite method may begin with Lemma 7.1 and then continue with Sections 7 and 8.

2. LOGARITHMIC JETS, DIRECT IMAGES, AND ANALYTIC INPUTS

This section has four parts. The first part fixes the jet notation and proves the direct-image facts used later. The second part states the two basic analytic theorems. The third part proves the cohomological and zero-locus results needed for the second equation. The final part treats algebraically degenerate curves. Thus not every result in this section is an external input.

For quick reference, the main notation is listed below. All bundles are logarithmic with respect to $D = C_1 + C_2$ unless another divisor is shown.

symbol	meaning
$\overline{T}$	$T_{\mathbb{P}^2}(-\log D)$
$\overline{K}$	$K_{\mathbb{P}^2} + D = \det(\overline{T}^*)$
$E_{2,m}\overline{T}^*$	invariant logarithmic two-jets of weighted degree m
X_1, X_2	the first two levels of the logarithmic Demailly–Semple tower
u_1, u_2, h	the two tautological classes and the pullback of the hyperplane class
Γ_2	the vertical divisor on X_2
$f_{[1]}, f_{[2]}$	the first and second logarithmic lifts of an entire curve f
$m, t, \rho = t/m$	jet weight, negative base twist, and their ratio
ρ_{RR}	upper ratio allowed by the Riemann–Roch existence argument
ρ_{DEG}	upper ratio allowed by the Demailly–El Goul zero-locus argument
$\tau_1(\rho)$	positive endpoint for the twist used on the zero divisor

2.1. Nevanlinna functions. Let ω_{FS} be the Fubini–Study form. We use the standard Nevanlinna functions and normalization; see [18]. For $f : \mathbb{C} \rightarrow \mathbb{P}^2$, set

$$T_f(r) = \int_1^r \frac{ds}{s} \int_{|z|<s} f^* \omega_{\text{FS}}.$$

If $E \subset \mathbb{P}^2$ is an irreducible divisor and $f(\mathbb{C}) \not\subset E$, let

$$N_f^{[1]}(r, E) = \int_1^r \sum_{|z| < s} \min\{1, \text{ord}_z f^*E\} \frac{ds}{s}.$$

For the reduced divisor $D = C_1 + C_2$, we use the componentwise convention

$$N_f^{[1]}(r, D) := N_f^{[1]}(r, C_1) + N_f^{[1]}(r, C_2).$$

The symbol $\parallel$ means that the inequality holds for all $r > 1$ outside a set of finite Lebesgue measure. For an algebraically nondegenerate curve in $\mathbb{P}^2$, the standard growth lemma gives

$$O(\log T_f(r) + \log r) = o(T_f(r)) \parallel,$$

so we shall replace the logarithmic error term by $o(T_f)$ after applying a jet-differential estimate.

2.2. Invariant logarithmic two-jets. We follow the standard invariant logarithmic jet notation of [3, 5]. Locally on a logarithmic surface, an invariant two-jet differential of weighted degree m has the form

$$(2.1) \quad \sum_{0 \leq j \leq \lfloor m/3 \rfloor} \sum_{\alpha_1 + \alpha_2 = m - 3j} R_{\alpha_1, \alpha_2, j} (f'_1)^{\alpha_1} (f'_2)^{\alpha_2} (f'_1 f'_2'' - f'_2 f_1'')^j.$$

Its associated graded bundle is

$$(2.2) \quad \text{Gr}^\bullet E_{2,m} \bar{T}^* = \bigoplus_{0 \leq j \leq \lfloor m/3 \rfloor} S^{m-3j} \bar{T}^* \otimes \bar{K}^j.$$

2.3. The logarithmic Demailly–Semple tower. Set $X_0 = \mathbb{P}^2$ and $V_0 = \bar{T}$. We use the convention that $\mathbb{P}(V)$ parametrizes lines in V , so that $\mathcal{O}_{\mathbb{P}(V)}(-1)$ is the tautological subbundle. Let

$$X_1 = \mathbb{P}(V_0), \quad X_2 = \mathbb{P}(V_1),$$

be the first two levels of the logarithmic Demailly–Semple tower, with projections $\pi_{2,1} : X_2 \rightarrow X_1$ and $\pi_{2,0} : X_2 \rightarrow X_0$. Write

$$u_1 = c_1(\pi_{2,1}^* \mathcal{O}_{X_1}(1)), \quad u_2 = c_1(\mathcal{O}_{X_2}(1)), \quad h = \pi_{2,0}^* c_1(\mathcal{O}_{\mathbb{P}^2}(1)),$$

and

$$\mathcal{O}_{X_2}(a, b) := \pi_{2,1}^* \mathcal{O}_{X_1}(a) \otimes \mathcal{O}_{X_2}(b).$$

Let $\Gamma_2 \subset X_2$ be the vertical divisor at the second stage. With the above convention,

$$\mathcal{O}_{X_2}(\Gamma_2) \simeq \mathcal{O}_{X_2}(-1, 1), \quad \mathcal{O}_{X_2}(3) \otimes \mathcal{O}_{X_2}(-2\Gamma_2) \simeq \mathcal{O}_{X_2}(2, 1).$$

Let $\mathcal{O}_{X_2}(m)$ denote the standard level-two Demailly–Semple tautological bundle of weighted degree m . The direct-image formula identifies invariant two-jets with its sections:

$$(2.3) \quad (\pi_{2,0})_* \mathcal{O}_{X_2}(m) \simeq E_{2,m} \bar{T}^*.$$

We shall also use the following more precise direct-image statement for the bundle $\mathcal{O}_{X_2}(2, 1)$. Its two-stage filtration proof is the one used in [7, Proposition 6.1]. Keeping every negative-degree graded piece gives the sharp criterion in terms of the single defect $a_1 - 2a_2$; we record the resulting statement and proof for later use.

Proposition 2.1 (Sharp higher direct images). *Let $a_1, a_2 \in \mathbb{Z}$. If $a_2 = -1$, then*

$$R^q(\pi_{2,0})_* \mathcal{O}_{X_2}(a_1, -1) = 0 \quad (q \geq 0).$$

Suppose that $a_2 \geq 0$ and put $d = a_1 - 2a_2$. Then

$$R^q(\pi_{2,0})_* \mathcal{O}_{X_2}(a_1, a_2) = 0 \quad (q \geq 2),$$

and

$$R^1(\pi_{2,0})_* \mathcal{O}_{X_2}(a_1, a_2) = 0 \iff d \geq -1.$$

If $d \leq -2$, the nonzero R^1 has a finite filtration with successive quotients

$$(2.4) \quad \overline{K}^{a_2-k-1} \otimes \text{Sym}^{-d-3k-2} \overline{T}, \quad 0 \leq k \leq \min \left\{ a_2, \left\lfloor \frac{-d-2}{3} \right\rfloor \right\}.$$

Proof. Write $q = \pi_{2,1}$ and $\pi = \pi_{1,0}$. If $a_2 = -1$, the restriction to each q -fiber is $\mathcal{O}_{\mathbb{P}^1}(-1)$, so every direct image under q , and thus under $\pi_{2,0}$, vanishes. Assume $a_2 \geq 0$. The projective-bundle formula gives

$$R^j q_* \mathcal{O}_{X_2}(a_1, a_2) = 0 \quad (j > 0), \quad q_* \mathcal{O}_{X_2}(a_1, a_2) = \mathcal{O}_{X_1}(a_1) \otimes \text{Sym}^{a_2} V_1^*.$$

The dual directed sequence

$$0 \longrightarrow \mathcal{O}_{X_1}(1) \longrightarrow V_1^* \longrightarrow \pi^* \overline{K} \otimes \mathcal{O}_{X_1}(-2) \longrightarrow 0$$

induces a filtration whose graded quotients are

$$(2.5) \quad \pi^* \overline{K}^{a_2-k} \otimes \mathcal{O}_{X_1}(d+3k), \quad 0 \leq k \leq a_2.$$

For the rank-two projective bundle of lines, $R^1 \pi_* \mathcal{O}_{X_1}(n) = 0$ for $n \geq -1$, while relative Serre duality gives, for $n \leq -2$,

$$R^1 \pi_* \mathcal{O}_{X_1}(n) \simeq \text{Sym}^{-n-2} \overline{T} \otimes \det \overline{T} = \text{Sym}^{-n-2} \overline{T} \otimes \overline{K}^{-1}.$$

Applying this to (2.5) gives exactly the quotients (2.4). In the bad range $d+3k \leq -2$, the corresponding ordinary direct image is zero; ordering the filtration from the good range to the bad range shows through the long exact sequences that these R^1 terms assemble by successive extensions. There are no direct images in degree at least two because both stages have relative dimension one and the first push has no positive higher direct images. The vanishing criterion is therefore exactly $d \geq -1$. $\square$

Remark 2.2 (Fiberwise interpretation). On a fiber $F \simeq \mathbb{F}_3$ of $\pi_{2,0}$,

$$\mathcal{O}_{X_2}(a_1, a_2)|_F \simeq \mathcal{O}_{\mathbb{F}_3}(a_2 E + (a_1 + a_2)f),$$

and for $a_2 \geq 0$ its cohomology is controlled by the line-bundle degrees $d+3k$, $0 \leq k \leq a_2$. Thus the defect $a_1 - 2a_2$ is sharp; conditions involving a_1 and a_2 separately do not suffice.

Lemma 2.3 (Direct images of $\mathcal{O}_{X_2}(2p, p)$). *For every integer $p \geq 1$ one has*

$$(2.6) \quad (\pi_{2,0})_* \mathcal{O}_{X_2}(2p, p) \simeq E_{2,3p} \overline{T}^*.$$

Also,

$$(2.7) \quad R^q(\pi_{2,0})_* \mathcal{O}_{X_2}(2p, p) = 0 \quad (q \geq 1).$$

Proof. The identification (2.6) is the standard Demailly–Semple direct-image formula for the weight- $3p$ subbundle; see [4, Lemma 3.3(c)]. The higher direct images vanish by Proposition 2.1, because $a_2 = p \geq 0$ and $a_1 - 2a_2 = 2p - 2p = 0$. $\square$

Thus the p -th power of $\mathcal{O}_{X_2}(2, 1)$ corresponds canonically to weighted degree $3p$.

We shall repeatedly use the following elementary fact about lifted curves.

Lemma 2.4 (The regular lift avoids the vertical divisor). *Let $f : \mathbb{C} \rightarrow X_0$ be nonconstant and satisfy $f(\mathbb{C}) \not\subset D$. Then its logarithmic lift $f_{[2]}$ is well defined after holomorphic extension across the isolated zeros of f' , and*

$$f_{[2]}(\mathbb{C}) \not\subset \Gamma_2.$$

Proof. On the nonempty open set on which $f' \neq 0$, the first lift is $f_{[1]}(z) = (f(z), [f'(z)])$. Its derivative is tangent to V_1 , and its projection by $d\pi_{1,0}$ is $f'(z) \neq 0$. Thus the line $[f'_{[1]}(z)] \subset V_{1, f_{[1]}(z)}$ is not vertical for $\pi_{1,0}$. By definition, Γ_2 consists exactly of the vertical lines at the second stage, so $f_{[2]}(z) \notin \Gamma_2$ on this open set. The standard removable-singularity construction for Semple lifts extends $f_{[1]}$ and $f_{[2]}$ through the isolated zeros of f' . Since the complement of Γ_2 has already met the lifted curve, the whole image cannot be contained in Γ_2 . $\square$

2.4. Jet Second Main Theorem and McQuillan's theorem. We use the following standard jet-differential estimate; compare [3] and [7, Theorem 1.2]. The invariant logarithmic jet bundle $E_{k,m}T_{\mathbb{P}^2}^*(\log D)$ is naturally a subbundle of the Green–Griffiths bundle appearing in the statement, so the estimate applies to the invariant differentials constructed below.

Theorem 2.5 (Jet differential Second Main Theorem). *Let $k, m, t \geq 1$ be integers. If*

$$0 \neq \omega \in H^0(\mathbb{P}^2, E_{k,m}^{\text{GG}}T_{\mathbb{P}^2}^*(\log D) \otimes \mathcal{O}(-t))$$

and $f : \mathbb{C} \rightarrow \mathbb{P}^2$ satisfies $f(\mathbb{C}) \not\subset D$ and $f^\omega \neq 0$, then*

$$(2.8) \quad T_f(r) \leq \frac{m}{t} N_f^{[1]}(r, D) + O(\log T_f(r) + \log r) \parallel.$$

We also use McQuillan's algebraic-degeneracy theorem in the following quantitative form; see [13] and [7, Theorem 1.1].

Theorem 2.6 (McQuillan). *Let $f : \mathbb{C} \rightarrow \mathbb{P}^2$ be algebraically nondegenerate. If its first logarithmic lift is algebraically degenerate, then for a normal-crossing divisor of total degree $d \geq 4$,*

$$(d-3)T_f(r) \leq N_f^{[1]}(r, D) + o(T_f(r)) \parallel.$$

For $d = d_{\text{tot}} = d_1 + d_2$ this gives

$$(2.9) \quad T_f(r) \leq \frac{1}{d_{\text{tot}} - 3} N_f^{[1]}(r, D) + o(T_f(r)) \parallel.$$

2.5. Cohomological and zero-locus inputs. We next record the vanishing and weighted-degree estimates that produce the negatively twisted differentials used in the two main cases.

Lemma 2.7 (Logarithmic Bogomolov vanishing). *Let $D = C_1 + C_2$ be a transverse pair of smooth plane curves, put $\kappa = d_1 + d_2 - 3 > 0$, $\bar{K} = K_{\mathbb{P}^2} + D$, and set $\bar{T} = T_{\mathbb{P}^2}(-\log D)$. For integers $p \geq 0$ and q ,*

$$H^0(\mathbb{P}^2, S^p \bar{T} \otimes \bar{K}^q) = 0 \quad \text{if} \quad p > 2q.$$

Proof. This is the logarithmic Bogomolov-type vanishing theorem [7, Theorem 2.1]. $\square$

Theorem 2.8 (The required logarithmic H^2 vanishing). *Let $D = C_1 + C_2$ be a transverse pair of smooth plane curves with $\kappa = d_1 + d_2 - 3 > 0$, and put $\bar{K} = K_{\mathbb{P}^2} + D \simeq \mathcal{O}_{\mathbb{P}^2}(\kappa)$. For every rational number $0 \leq \delta < 1/3$ and every sufficiently divisible $m \gg 1$,*

$$(2.10) \quad H^2(\mathbb{P}^2, E_{2,m}T_{\mathbb{P}^2}^*(\log D) \otimes \bar{K}^{-\delta m}) = 0.$$

Proof. By Serre duality, the dual of the group in (2.10) is

$$H^0(\mathbb{P}^2, K_{\mathbb{P}^2} \otimes (E_{2,m}\bar{T}^*)^\vee \otimes \bar{K}^{\delta m}).$$

The dual of the invariant-two-jet filtration has graded pieces

$$S^{m-3j}\bar{T} \otimes \bar{K}^{-j} \otimes K_{\mathbb{P}^2} \otimes \bar{K}^{\delta m}, \quad 0 \leq j \leq \lfloor m/3 \rfloor.$$

Because m is sufficiently divisible, δm is an integer. Multiplication by a nonzero section of $\mathcal{O}_{\mathbb{P}^2}(3)$ injects each graded piece into

$$(2.11) \quad S^{m-3j}\bar{T} \otimes \bar{K}^{\delta m-j}.$$

For (2.11), the inequality in Lemma 2.7 is equivalent to

$$(m-3j) - 2(\delta m - j) = (1-2\delta)m - j \geq (2/3-2\delta)m > 0.$$

Thus every graded piece has vanishing H^0 . Induction along the filtration gives the vanishing of the whole Serre-dual group, proving (2.10). $\square$

For the zero-locus argument, write

$$\bar{c}_i = c_i(T_{\mathbb{P}^2}^*(\log D)), \quad h = c_1(\mathcal{O}_{\mathbb{P}^2}(1)),$$

and identify these classes with their pullbacks to the Semple tower. For $\rho = t/m$, set

$$(2.12) \quad Q_{\text{DEG}}(\rho, \tau) = 3\tau^2 + (9\rho - 12(\bar{c}_1 \cdot h))\tau + (13\bar{c}_1^2 - 9\bar{c}_2) - 12(\bar{c}_1 \cdot h)\rho.$$

When its value at $\tau = 0$ is positive, denote its smaller positive root by

$$(2.13) \quad \tau_1(\rho) = \frac{12\kappa - 9\rho - \sqrt{(12\kappa - 9\rho)^2 - 12((13\bar{c}_1^2 - 9\bar{c}_2) - 12\kappa\rho)}}{6}.$$

Section 3 derives these expressions from the two-component Chern classes and the tower intersection relations.

Lemma 2.9 (Tautological evaluation of a restricted section). *Let $Z \subset X_2$ be an effective Cartier divisor and let*

$$\sigma \in H^0(Z, \mathcal{O}_{X_2}(2p, p) \otimes \pi_{2,0}^* \mathcal{O}_{\mathbb{P}^2}(-\tilde{t})|_Z), \quad p, \tilde{t} \in \mathbb{Z}_{>0}.$$

Let $f: \mathbb{C} \rightarrow \mathbb{P}^2$ be nonconstant, assume $f(\mathbb{C}) \not\subset D$, and suppose that its regular logarithmic second lift is contained in Z . On $\mathbb{C} \setminus f^{-1}(D)$, regard $df_{[1]}: T_{\mathbb{C}} \rightarrow f_{[1]}^ V_1$ as a differential in the directed structure. Wherever it is nonzero, its image is the tautological line $f_{[2]}^* \mathcal{O}_{X_2}(-1)$. It therefore defines, by meromorphic continuation across the remaining discrete set, the intrinsic tautological derivative*

$$\mathcal{D}_D f_{[1]} \in H_{\text{mer}}^0(\mathbb{C}, K_{\mathbb{C}} \otimes f_{[2]}^* \mathcal{O}_{X_2}(-1)).$$

Then the canonical inclusion

$$\iota_p: \mathcal{O}_{X_2}(2p, p) = \mathcal{O}_{X_2}(3p)(-2p\Gamma_2) \hookrightarrow \mathcal{O}_{X_2}(3p)$$

defines the intrinsic meromorphic evaluation

$$(2.14) \quad s_{\sigma, f} := \left\langle f_{[2]}^*(\iota_p \sigma), (\mathcal{D}_D f_{[1]})^{\otimes 3p} \right\rangle \in H_{\text{mer}}^0(\mathbb{C}, K_{\mathbb{C}}^{3p} \otimes f^* \mathcal{O}_{\mathbb{P}^2}(-\tilde{t})).$$

This construction does not require an ambient lift of σ . Also,

$$(2.15) \quad \text{div}_{\infty}(s_{\sigma, f}) \leq 3p \sum_{i=1}^2 (f^* C_i)_{\text{red}},$$

and

$$(2.16) \quad \int_0^{2\pi} \log^+ \|s_{\sigma, f}(re^{\sqrt{-1}\theta})\| \frac{d\theta}{2\pi} = O(\log^+ T_f(r) + \log r) \quad \parallel.$$

At an isolated zero of f' outside $f^{-1}(D)$, the evaluation has only a removable singularity or a zero. At a zero lying over D , no pole occurs beyond the logarithmic boundary poles already counted in (2.15).

Proof. Pairing its $3p$ -th tensor power with $f_{[2]}^*(\iota_p \sigma)$ gives (2.14). Since this is a pairing of dual tautological lines, the result is independent of all Semple coordinates and all local representatives of σ .

Equip the line bundles with smooth Hermitian metrics. Since Z is projective, the norm of the holomorphic section $\iota_p \sigma$ is bounded on Z . Thus

$$\|s_{\sigma, f}\| \leq C \|\mathcal{D}_D f_{[1]}\|^{3p}$$

for a uniform constant C .

In coordinates in which $D = \{x_1 \cdots x_q = 0\}$, the horizontal boundary components of the derivative of the regular first lift are $(x_i \circ f)' / (x_i \circ f)$, while the vertical projective-coordinate components are holomorphic. Thus the tautological derivative has at most a simple pole along each reduced pullback $(f^* C_i)_{\text{red}}$.

By Lemma 2.4, the Semple lift extends across the isolated zeros of f' . At a critical point outside $f^{-1}(D)$, the projective ratios extend and the evaluation has only a removable singularity or a zero. At a critical point over D , a logarithmic component may keep a simple pole; that pole is already counted by the reduced boundary pullback, and no additional Semple-type pole appears. This proves (2.15). Choose a finite logarithmic base cover and, above it, finitely many relatively compact Semple charts. A partition of unity is used only to estimate the norm of this single global section. On each chart, the components of $\mathcal{D}_D f_{[1]}$ are logarithmic derivatives of local coordinate functions. A projective Semple coordinate u is a rational function on X_1 and, in a logarithmic frame, a ratio of first logarithmic derivatives. Thus

$$T_{u \circ f_{[1]}}(r) = O(T_f(r) + \log r).$$

For a partition function χ supported in a relatively compact Semple chart, choose $a \notin \overline{u(\text{supp } \chi)}$. On that support, $u - a$ is bounded above and away from zero, and

$$(u \circ f_{[1]})' = (u \circ f_{[1]} - a) \frac{(u \circ f_{[1]})'}{u \circ f_{[1]} - a}.$$

The logarithmic-derivative lemma applied to $u \circ f_{[1]} - a$ controls the vertical component by

$$O(\log^+ T_f(r) + \log r).$$

Summing over the finite cover, together with the same estimates for the horizontal terms, gives (2.16). This is the argument of [9, Thm. 3.1, proof of (3.3)–(3.6)], with the Semple-coordinate adaptation made explicit. $\square$

Lemma 2.10 (Second Main Theorem for a section restricted to Z). *Let $Z \subset X_2$ be an effective Cartier divisor and let*

$$\sigma \in H^0(Z, \mathcal{O}_{X_2}(2p, p) \otimes \pi_{2,0}^* \mathcal{O}_{\mathbb{P}^2}(-\tilde{t})|_Z), \quad p, \tilde{t} \in \mathbb{Z}_{>0}.$$

Suppose that $f : \mathbb{C} \rightarrow \mathbb{P}^2$ is nonconstant, $f(\mathbb{C}) \not\subset D$, $f_{[2]}(\mathbb{C}) \subset Z$, and the canonical evaluation $s_{\sigma,f}$ of Lemma 2.9 is not identically zero. Then

$$(2.17) \quad \tilde{t} T_f(r) \leq 3p \sum_{i=1}^2 N_f^{[1]}(r, C_i) + O(\log T_f(r) + \log r) \quad \|.$$

In particular, if f is algebraically nondegenerate, then

$$T_f(r) \leq \frac{3p}{\tilde{t}} N_f^{[1]}(r, D) + o(T_f(r)) \quad \|.$$

Proof. By Lemma 2.9, the section $s_{\sigma,f}$ is globally and intrinsically defined, its pole divisor is bounded by (2.15), and its proximity term satisfies (2.16). Use the standard coordinate z on $\mathbb{C}$ to trivialize $K_{\mathbb{C}}^{3p}$ by the global frame $(dz)^{3p}$ and give it the flat metric. Equip $\mathcal{O}_{\mathbb{P}^2}(-1)$ with the metric dual to the Fubini–Study metric. Poincaré–Lelong and Jensen’s formula give

$$\begin{aligned} \tilde{t} T_f(r) &\leq 3p \sum_{i=1}^2 N_f^{[1]}(r, C_i) \\ &\quad + O(\log^+ T_f(r) + \log r) \quad \|. \end{aligned}$$

This is (2.17). It is the intrinsic restricted-section version of the argument in [7, Proposition 6.3 and Theorem 6.4]; the analytic norm estimate is the one in [9, Theorem 3.1, proof of (3.3)–(3.6)]. $\square$

Theorem 2.11 (Two-component zero-locus theorem and standard estimate). *Let ω_1 be an irreducible nonzero section of*

$$E_{2,m} \overline{T}^* \otimes \mathcal{O}(-t), \quad \rho = t/m,$$

whose zero divisor on X_2 is $Z + b\Gamma_2$, where Z is irreducible and reduced and

$$(2.18) \quad Z \sim b_1 u_1 + b_2 u_2 - th, \quad b_1 \geq 2b_2 > 0, \quad b_1 + b_2 = m.$$

Suppose

$$0 < \rho < \rho_{\text{DEG}} := \frac{13\bar{c}_1^2 - 9\bar{c}_2}{12(\bar{c}_1 \cdot h)}.$$

Assume $\kappa = d_1 + d_2 - 3 \geq 3$. Let $\tau_1(\rho)$ be the smaller positive root of $Q_{\text{DEG}}(\rho, \tau) = 0$, explicitly given in (2.13). For every rational

$$0 < \tau < \min\{3, \tau_1(\rho)\},$$

the $\mathbb{Q}$ -line bundle

$$(\mathcal{O}_{X_2}(2, 1) \otimes \pi_{2,0}^* \mathcal{O}_{\mathbb{P}^2}(-\tau))|_Z$$

is big. Also, if $f : \mathbb{C} \rightarrow \mathbb{P}^2$ is algebraically nondegenerate and $f^* \omega_1 \equiv 0$, then either

$$(2.19) \quad T_f(r) \leq \frac{3}{\tau} N_f^{[1]}(r, D) + o(T_f(r)) \quad \parallel,$$

or the first logarithmic lift $f_{[1]}$ is algebraically degenerate.

Proof. The restricted-bigness calculation follows the method of [7, Proposition 6.2], with the two-component intersection polynomial computed below. The local frame and analytic estimate are adapted from [7, Proposition 6.3 and Theorem 6.4] and are recorded in Lemma 2.10.

The proof has four steps. First, we show that the cubic intersection on Z is positive. Second, two cohomology vanishings turn this positive number into bigness. Third, we choose a section of a large power of the big bundle. Finally, we evaluate that section on the lifted curve. A nonzero evaluation gives the stated Second Main Theorem; a zero evaluation makes the first lift algebraically degenerate.

Put

$$L_\tau = \mathcal{O}_{X_2}(2, 1) \otimes \pi_{2,0}^* \mathcal{O}_{\mathbb{P}^2}(-\tau).$$

The intersection computation on the logarithmic Demailly–Semple tower gives

$$(2.20) \quad c_1(L_\tau)^3 \cdot Z = mQ_{\text{DEG}}(\rho, \tau).$$

By the definition of $\tau_1(\rho)$, the right-hand side is positive in the stated range.

We prove bigness without inferring it from (2.20) alone. Let $p \gg 1$ be sufficiently divisible, so that $p\tau/\kappa \in \mathbb{Z}$ (and thus $p\tau \in \mathbb{Z}$) and L_τ^p is a genuine line bundle. Since X_2 is smooth and Z is an effective Cartier divisor, the exact sequence

$$0 \rightarrow L_\tau^p(-Z) \rightarrow L_\tau^p \rightarrow L_\tau^p|_Z \rightarrow 0$$

and Hirzebruch–Riemann–Roch on the smooth fourfold X_2 give

$$(2.21) \quad \chi(Z, L_\tau^p|_Z) = \chi(X_2, L_\tau^p) - \chi(X_2, L_\tau^p(-Z))$$

$$(2.22) \quad = \int_{X_2} e^{pc_1(L_\tau)} (1 - e^{-[Z]}) \text{td}(T_{X_2})$$

$$(2.23) \quad = \frac{c_1(L_\tau)^3 \cdot Z}{3!} p^3 + O(p^2).$$

This ambient computation does not require Z to be smooth or normal.

We next show $H^2(Z, L_\tau^p|_Z) = 0$. Lemma 2.3, the projection formula, and Leray yield

$$H^2(X_2, L_\tau^p) \simeq H^2(\mathbb{P}^2, E_{2,3p} \bar{T}^* \otimes \mathcal{O}(-p\tau)) = 0,$$

where Theorem 2.8 is applied with $\delta = \tau/(3\kappa) < 1/3$.

Using (2.18),

$$L_\tau^p(-Z) \simeq \mathcal{O}_{X_2}(2p - b_1, p - b_2) \otimes \pi_{2,0}^* \mathcal{O}_{\mathbb{P}^2}(t - p\tau).$$

To prove the required H^3 vanishing, we use the filtration induced by the directed tangent sequence. The relevant relative defect is $2b_2 - b_1$, which may be negative, but only finitely many graded pieces contribute to the higher direct image.

Write $q = \pi_{2,1} : X_2 \rightarrow X_1$ and $\pi = \pi_{1,0} : X_1 \rightarrow \mathbb{P}^2$. For $p \gg 1$ one has $p - b_2 \geq 0$, so

$$R^r q_* L_\tau^p(-Z) = 0 \quad (r \geq 1)$$

and

$$q_* L_\tau^p(-Z) = \mathcal{O}_{X_1}(2p - b_1) \otimes \text{Sym}^{p-b_2} V_1^* \otimes \pi^* \mathcal{O}(t - p\tau).$$

The dual relative tangent sequence

$$(2.24) \quad 0 \longrightarrow \mathcal{O}_{X_1}(1) \longrightarrow V_1^* \longrightarrow \pi^* \overline{K} \otimes \mathcal{O}_{X_1}(-2) \longrightarrow 0$$

induces a filtration whose quotients, indexed by $0 \leq j \leq p - b_2$, are

$$\mathcal{O}_{X_1}(3p - m - 3j) \otimes \pi^* (\overline{K}^j \otimes \mathcal{O}(t - p\tau)).$$

Put $\ell = p - j$. The same quotients become

$$(2.25) \quad \mathcal{G}_{\ell,p} = \mathcal{O}_{X_1}(3\ell - m) \otimes \pi^* \mathcal{O}((\kappa - \tau)p + t - \kappa\ell), \quad b_2 \leq \ell \leq p.$$

If $3\ell - m \geq -1$, then $R^1 \pi_* \mathcal{O}_{X_1}(3\ell - m) = 0$, so $H^3(X_1, \mathcal{G}_{\ell,p}) = 0$ by Leray. The remaining indices satisfy

$$b_2 \leq \ell \leq \left\lfloor \frac{m-2}{3} \right\rfloor;$$

this is a fixed finite set, independent of p . For each such ℓ , the projection formula gives

$$R^1 \pi_* \mathcal{G}_{\ell,p} = R^1 \pi_* \mathcal{O}_{X_1}(3\ell - m) \otimes \mathcal{O}((\kappa - \tau)p + t - \kappa\ell).$$

The first factor is fixed, while $\kappa - \tau > 0$ because $\tau < 3 \leq \kappa$. Serre vanishing therefore kills its H^2 for $p \gg 1$. Leray, followed by induction through the filtration (2.24), yields

$$(2.26) \quad H^3(X_2, L_\tau^p(-Z)) = 0.$$

The divisor exact sequence now gives $H^2(Z, L_\tau^p|_Z) = 0$. Combining this with (2.21), we obtain

$$h^0(Z, L_\tau^p|_Z) = \chi(Z, L_\tau^p|_Z) + h^1(Z, L_\tau^p|_Z) + h^3(Z, L_\tau^p|_Z) \geq \chi(Z, L_\tau^p|_Z) \gg p^3,$$

which proves that the $\mathbb{Q}$ -line bundle $L_\tau|_Z$ is big.

Choose p in the same sufficiently divisible sequence and a nonzero section

$$\sigma \in H^0(Z, L_\tau^p|_Z), \quad \tilde{t} = p\tau \in \mathbb{Z}.$$

Since $f^* \omega_1 \equiv 0$ and Lemma 2.4 excludes containment in Γ_2 , one has $f_{[2]}(\mathbb{C}) \subset Z$. Suppose first that the canonical evaluation $s_{\sigma,f}$ is not identically zero. Since

$$L_\tau^p = \mathcal{O}_{X_2}(2p, p) \otimes \pi_{2,0}^* \mathcal{O}_{\mathbb{P}^2}(-\tilde{t}),$$

Lemma 2.10 yields

$$\tilde{t} T_f(r) \leq 3p N_f^{[1]}(r, D) + O(\log T_f(r) + \log r) \parallel.$$

The logarithmic derivative lemma makes the final term $o(T_f(r))$ for an algebraically nondegenerate curve. Since $\tilde{t} = p\tau$, this is (2.19).

If instead $s_{\sigma,f} \equiv 0$, the tautological derivative is nonzero away from a discrete set because f is nonconstant. The intrinsic pairing (2.14) therefore implies $f_{[2]}^* \sigma \equiv 0$. Thus $f_{[2]}(\mathbb{C})$ is contained in the proper divisor $\text{div}_Z(\sigma)$ of the irreducible threefold Z . This set has dimension at most two, so its image under $\pi_{2,1} : X_2 \rightarrow X_1$ is a proper algebraic subset of the threefold X_1 . Thus $f_{[1]}$ is algebraically degenerate. $\square$

Remark 2.12. The output used later is the following alternative. If $t/m < \rho_{\text{DEG}}$ and the first equation vanishes on the lifted curve, then either the coefficient is $3/\tau$ for every allowed τ , or $f_{[1]}$ is algebraically degenerate. The proof needs three separate facts: a positive cubic intersection, the H^2 and H^3 vanishings that give bigness, and the intrinsic evaluation estimate (2.19).

2.6. Algebraically degenerate curves. The preceding analytic estimates concern algebraically nondegenerate curves. For the hyperbolicity corollary, the algebraically degenerate case is supplied by the following specialization of Xi Chen’s theorem.

For an irreducible curve $\Gamma \not\subset D$, let $\nu : \tilde{\Gamma} \rightarrow \Gamma$ be the normalization and define

$$i(\Gamma, D) := \# \text{Supp } \nu^* D.$$

Theorem 2.13 (Xi Chen). *For a very general ordered pair of smooth transverse plane curves of total degree $d_{\text{tot}} = d_1 + d_2$ and every reduced irreducible curve $\Gamma \not\subset D$,*

$$(2.27) \quad 2g(\tilde{\Gamma}) - 2 + i(\Gamma, D) \geq (d_{\text{tot}} - 4) \deg \Gamma.$$

Source of the specialization. Xi Chen proves for a very general reducible plane divisor $D = \bigcup D_j$ of total degree d that

$$2g(\tilde{\Gamma}) - 2 + i(\Gamma, D) \geq (d - 4) \deg \Gamma.$$

Taking a very general ordered pair of degrees (d_1, d_2) gives $d = d_{\text{tot}}$ and thus (2.27); see [2, Corollary 1.19]. $\square$

Remark 2.14 (The general-pair problem). Theorem 2.13 is a very-general statement, whereas Theorem 1.1 holds on a single nonempty Zariski-open subset of the parameter space. This distinction is a genuine one. For plane boundaries with at most two irreducible components, Caporaso and Turchet observe that no analogous exceptional-set result is known without the “very general” hypothesis; in particular, their finite-exceptional-set conjecture remains open in this range [1, Introduction, Conjecture 1]. It is therefore natural to ask whether the specific consequence of Theorem 2.13 used in Corollary 1.3 can be made valid for a general ordered pair. This is the general-pair form of the folklore hyperbolicity expectation relevant here.

Theorem 1.2 supplies a uniform Zariski-open direction locus for this question. Indeed, after discarding the components whose images in $\mathbb{P}^2$ are proper, the components of $\mathcal{R}_a$ that dominate $\mathbb{P}^2$ define a finite algebraic multi-foliation. Every log-parabolic curve is either contained in the projected exceptional set or invariant under one of these directions. In forthcoming work, Song-Yan Xie and Shengyuan Zhao exploit precisely this additional structure to bound the degrees of invariant log-parabolic curves and exclude them for a general pair in the degree range of Theorem 1.1. Combined with the Second Main Theorem, this replaces the very-general qualification in Corollary 1.3 by general throughout that range.

Corollary 2.15. *If $d_{\text{tot}} \geq 5$, then for a very general pair there is no nonconstant algebraically degenerate entire curve in $\mathbb{P}^2 \setminus D$.*

Proof. If $f(\mathbb{C})$ is contained in $\Gamma \not\subset D$, it lifts to

$$\tilde{f} : \mathbb{C} \longrightarrow \tilde{\Gamma} \setminus \text{Supp } \nu^* D.$$

By (2.27),

$$2g(\tilde{\Gamma}) - 2 + i(\Gamma, D) > 0,$$

so the punctured Riemann surface on the right is hyperbolic. Thus $\tilde{f}$ and f are constant. $\square$

Output of this section. The later proof uses three alternatives. A nonzero jet evaluation gives the coefficient m/t by Theorem 2.5. An algebraically degenerate first lift gives the coefficient $1/(d_1 + d_2 - 3)$ by Theorem 2.6. On the zero divisor, Theorem 2.11 gives either the coefficient $3/\tau$ or algebraic degeneracy of the first lift. Corollary 2.15 is used only for the final hyperbolicity statement.

3. LOGARITHMIC CHERN NUMBERS AND THE DEGREE PHASE DIAGRAM

Two ratios organize the proof. The Riemann–Roch threshold ρ_{RR} allows the initial choice $t/m < \rho_{\text{RR}}$. The Demailly–El Goul threshold ρ_{DEG} tells us when an irreducible factor lies in the zero-locus range. Keeping both degrees as variables leaves only six pairs for finite verification.

The final degree split is shown here before the calculation.

degree family	finite boundary pairs	all larger pairs
$d_1, d_2 \geq 3$	$(3, 3), (3, 4)$	$\rho_{\text{DEG}} > 1$
$d_1 = 2, d_2 \geq 5$	$(2, 5), (2, 6)$	$d_2 \geq 7$
$d_1 = 1, d_2 \geq 8$	$(1, 8), (1, 9)$	$d_2 \geq 10$

For the six pairs in the middle column, Section 7 checks a finite list of low twists. For the pairs in the last column, the zero-locus threshold already covers every low twist.

3.1. The two-component Chern data. For $h = c_1(\mathcal{O}_{\mathbb{P}^2}(1))$ one has

$$\bar{K} = K_{\mathbb{P}^2} + D \simeq \mathcal{O}_{\mathbb{P}^2}(\kappa), \quad \bar{c}_1 = \kappa h, \quad \kappa = d_1 + d_2 - 3.$$

Using

$$c(\Omega_{\mathbb{P}^2}^1(\log D)) = c(\Omega_{\mathbb{P}^2}^1) \prod_{i=1}^2 (1 + [C_i] + [C_i]^2)$$

and $[C_i] = d_i h$, we obtain

$$(3.1) \quad \bar{c}_1^2 = \kappa^2, \quad \bar{c}_2 = d_1^2 + d_2^2 + d_1 d_2 - 3(d_1 + d_2) + 3, \quad \bar{c}_1 \cdot h = \kappa.$$

Two combinations will recur:

$$(3.2) \quad \bar{c}_1^2 - \bar{c}_2 = d_1 d_2 - 3(d_1 + d_2) + 6,$$

$$13\bar{c}_1^2 - 9\bar{c}_2 = A(d_1, d_2)$$

$$(3.3) \quad = 4(d_1^2 + d_2^2) + 17d_1 d_2 - 51(d_1 + d_2) + 90.$$

The first is positive in the classical one-jet range $d_1 \geq 5$, or $d_1 = 4, d_2 \geq 7$. The second is positive for every $d_1, d_2 \geq 3$; its minimum there is $A(3, 3) = 9$. On the degree-two boundary,

$$(3.4) \quad A(2, n) = 4n^2 - 17n + 4,$$

which is positive exactly for integers $n \geq 5$. On the degree-one boundary,

$$(3.5) \quad A(1, n) = 4n^2 - 34n + 43,$$

which, for integers $n \geq 2$, is positive exactly when $n \geq 7$.

3.2. Riemann–Roch threshold. This is the standard invariant-jet Riemann–Roch calculation introduced in Demailly’s foundational note [3]. We spell out only the two-component specialization of the leading coefficient because it determines the effective range used later. By (2.2), additivity of the Chern character gives

$$(3.6) \quad \chi(E_{2,m} \bar{T}^* \otimes \bar{K}^{-\delta m}) = \sum_{j=0}^{\lfloor m/3 \rfloor} \chi(S^{m-3j} \bar{T}^* \otimes \bar{K}^{j-\delta m}).$$

Let x_1, x_2 be the formal Chern roots of $\bar{T}^*$. For $q = m - 3j$,

$$\text{ch}(S^q \bar{T}^*) = \sum_{a=0}^q e^{ax_1 + (q-a)x_2}.$$

Keeping terms of cohomological degree two, multiplying by $e^{(j-\delta m)(x_1+x_2)} \text{td}(T_{\mathbb{P}^2})$, and integrating over $\mathbb{P}^2$ computes each summand in (3.6). The Todd terms of positive degree contribute only $O(m^3)$ after summation. For the degree-four term, the elementary identities

$$\sum_{a=0}^q 1 = q + 1, \quad \sum_{a=0}^q a = \frac{q(q+1)}{2}, \quad \sum_{a=0}^q a^2 = \frac{q(q+1)(2q+1)}{6}$$

followed by the sums in $0 \leq j \leq \lfloor m/3 \rfloor$ give

$$(3.7) \quad \sum_{j=0}^{\lfloor m/3 \rfloor} \int_{\mathbb{P}^2} \left[\text{ch}(S^{m-3j} \bar{T}^*) e^{(j-\delta m) \bar{c}_1} \right]_2 = \frac{m^4}{648} ((54\delta^2 - 48\delta + 13) \bar{c}_1^2 - 9\bar{c}_2) + O(m^3).$$

Define

$$Q_{\text{RR}}(\delta) = 54\bar{c}_1^2 \delta^2 - 48\bar{c}_1^2 \delta + (13\bar{c}_1^2 - 9\bar{c}_2).$$

Then, for $t = (\bar{c}_1 \cdot h) \delta m$,

$$(3.8) \quad \chi(\mathbb{P}^2, E_{2,m} \bar{T}^* \otimes \bar{K}^{-\delta m}) = \frac{Q_{\text{RR}}(\delta)}{648} m^4 + O(m^3).$$

Thus the leading coefficient is positive for sufficiently small positive δ . In the ratio $\rho = t/m = \kappa \delta$, its smaller positive root is

$$(3.9) \quad \rho_{\text{RR}}(d_1, d_2) = \frac{\kappa}{18} \left(8 - \sqrt{54 \frac{\bar{c}_2}{\bar{c}_1^2} - 14} \right) > 0.$$

The strict positivity is equivalent to $A(d_1, d_2) > 0$. For two cubics this specializes to $(8 - \sqrt{58})/6 > 1/16$.

3.3. Zero-locus threshold. Set $\rho = t/m$. Using the projective-bundle relations on the logarithmic Demailly–Semple tower, the cubic intersection occurring in Theorem 2.11 is

$$(3.10) \quad (2u_1 + u_2 - \tau h)^3 \cdot Z = m Q_{\text{DEG}}(\rho, \tau),$$

where, as in (2.12),

$$Q_{\text{DEG}}(\rho, \tau) = 3\tau^2 + (9\rho - 12(\bar{c}_1 \cdot h))\tau + (13\bar{c}_1^2 - 9\bar{c}_2) - 12(\bar{c}_1 \cdot h)\rho.$$

Here is the intersection calculation. Put $\Theta = 2u_1 + u_2$ and write $[Z] = b_1 u_1 + b_2 u_2 - th$, where $b_1 + b_2 = m$. Apply successively the rank-two projective-bundle formula

$$\pi_*(u) = 1, \quad \pi_*(u^2) = -c_1(V), \quad \pi_*(u^3) = c_1(V)^2 - c_2(V)$$

first to $\pi_{2,1}$ and then to $\pi_{1,0}$. With our convention that $\mathbb{P}(V)$ parametrizes lines, the intermediate identities are

$$u_1^2 - \bar{c}_1 u_1 + \bar{c}_2 = 0, \quad c_1(V_1) = u_1 - \bar{c}_1, \quad c_2(V_1) = 2\bar{c}_2 - \bar{c}_1 u_1.$$

Expanding with these relations gives the following three identities on the base:

$$(3.11) \quad \Theta h^2 \cdot Z = m,$$

$$(3.12) \quad \Theta^2 h \cdot Z = 4m(\bar{c}_1 \cdot h) - 3t,$$

$$(3.13) \quad \Theta^3 \cdot Z = m(13\bar{c}_1^2 - 9\bar{c}_2) - 12t(\bar{c}_1 \cdot h).$$

The terms involving b_1 and b_2 combine through $b_1 + b_2 = m$; this is why the answer is independent of the chosen irreducible factor. Since $h^3 = 0$ on the surface base, expansion of $(\Theta - \tau h)^3 \cdot Z$ and substitution of (3.11)–(3.13) yield

$$\begin{aligned} (\Theta - \tau h)^3 \cdot Z &= \Theta^3 \cdot Z - 3\tau \Theta^2 h \cdot Z + 3\tau^2 \Theta h^2 \cdot Z \\ &= m Q_{\text{DEG}}(\rho, \tau), \end{aligned}$$

which proves (3.10) directly. At $\tau = 0$ this is positive exactly when

$$(3.14) \quad \rho < \rho_{\text{DEG}} = \frac{13\bar{c}_1^2 - 9\bar{c}_2}{12(\bar{c}_1 \cdot h)}.$$

Using (3.3),

$$(3.15) \quad \rho_{\text{DEG}}(d_1, d_2) = \frac{A(d_1, d_2)}{12\kappa}.$$

Solving this quadratic recovers the smaller positive root $\tau_1(\rho)$ in (2.13). For (3, 3) this is

$$(3.16) \quad \tau_1(\rho) = 6 - \frac{3}{2}\rho - \frac{1}{2}\sqrt{132 - 24\rho + 9\rho^2}.$$

Equation (3.10) supplies the positive leading term; bigness and the second differential follow from the cohomological theorem, not from this number alone.

3.4. Why the proved range has six finite boundary pairs. For a genuine second-level component one has $b_2 > 0$ and $b_1 \geq 2b_2$, thus $m = b_1 + b_2 \geq 3$. If $t \leq 3$, then $t/m \leq 1$. Therefore no finite vanishing is needed whenever $\rho_{\text{DEG}} > 1$. We apply this observation only after the Riemann–Roch positivity condition $A(d_1, d_2) > 0$ has supplied the initial differential. Direct calculation gives

$$\rho_{\text{DEG}}(3, 3) = \frac{1}{4}, \quad \rho_{\text{DEG}}(3, 4) = \frac{37}{48}, \quad \rho_{\text{DEG}}(3, 5) = \frac{73}{60} > 1,$$

and $\rho_{\text{DEG}} > 1$ for every other pair with $3 \leq d_1 \leq d_2$, except (3, 3) and (3, 4). On the degree-two boundary,

$$\rho_{\text{DEG}}(2, 5) = \frac{19}{48}, \quad \rho_{\text{DEG}}(2, 6) = \frac{23}{30},$$

while $\rho_{\text{DEG}}(2, n) > 1$ for $n \geq 7$. This proves the phase diagram on the degree-two boundary. Finally,

$$\rho_{\text{DEG}}(1, 8) = \frac{3}{8}, \quad \rho_{\text{DEG}}(1, 9) = \frac{61}{84}, \quad \rho_{\text{DEG}}(1, 10) = \frac{103}{96} > 1,$$

and $\rho_{\text{DEG}}(1, n) > 1$ for every $n \geq 10$. Together these calculations reduce the stated line-component range $n \geq 8$ to (1, 8) and (1, 9). The omitted formal endpoint (1, 7) has $A(1, 7) = 1$ and $\rho_{\text{DEG}}(1, 7) = 1/60$; it is not a seventh certified boundary pair, because its required vanishing profile is neither proved nor claimed here. Remark 9.1 records that separate problem. Thus, in the range of the theorem, the pairs with $A(d_1, d_2) > 0$ but $\rho_{\text{DEG}} \leq 1$ are exactly (3, 3), (3, 4), (2, 5), (2, 6), (1, 8), (1, 9). These are exactly the six finite lists in Lemma 7.1.

Output of this section. The Riemann–Roch calculation gives the allowed initial ratio ρ_{RR} . The zero-locus calculation gives ρ_{DEG} and $\tau_1(\rho)$. The table above then separates the infinite geometric range from the six finite boundary pairs.

4. THE FIRST NEGATIVELY TWISTED DIFFERENTIAL

The construction has three steps. Riemann–Roch and H^2 vanishing first give one section with a degree pair independent of the parameter. The Demailly–El Goul reduction then selects a prime horizontal component with no loss in the twist-to-weight ratio. Finally, cohomology and base change spread its canonical reduced equation over one Zariski-open subset of the ordered parameter space. The output is therefore a fixed numerical type (m, t, b_1, b_2) and one algebraic family of prime horizontal divisors; these are exactly the uniform data used by the mixed-field construction. In short,

$$\begin{aligned} \text{Riemann–Roch existence} &\implies \text{prime horizontal factor} \\ &\implies \text{one algebraic family of equations.} \end{aligned}$$

4.1. Uniform existence.

Proposition 4.1 (Uniform existence). *Fix a rational number*

$$(4.1) \quad 0 < \rho_0 < \rho_{\text{RR}}(d_1, d_2).$$

There exist fixed positive integers M, T with $T/M = \rho_0$ such that for every $a \in \mathcal{S}_{d_1, d_2}^{\text{snc}}$ one has

$$(4.2) \quad H^0(\mathbb{P}^2, E_{2, M} T_{\mathbb{P}^2}^*(\log D_a) \otimes \mathcal{O}_{\mathbb{P}^2}(-T)) \neq 0.$$

Proof. Put $\delta_0 = \rho_0/(\bar{c}_1 \cdot h) = \rho_0/\kappa$. Write $\rho_0 = T_0/M_0$ with positive integers M_0, T_0 . Choose a sufficiently large N such that NT_0/κ is integral, and set

$$M = NM_0, \quad T = NT_0.$$

Then $\delta_0 M = T/\kappa \in \mathbb{Z}$, so the twist $\bar{K}^{-\delta_0 M}$ is a genuine line bundle. All smooth transverse pairs of degrees (d_1, d_2) have the same logarithmic Chern numbers (3.1). Thus the Riemann–Roch polynomial in (3.8), including its lower-order coefficients, is independent of the parameter $a \in \mathcal{S}_{d_1, d_2}^{\text{snc}}$. Since $Q_{\text{RR}}(\delta_0) > 0$, one may choose N once and for all so that this Euler characteristic is positive for every such a . Theorem 2.8, applied with the same δ_0 , also holds for every D_a . Therefore

$$h^0 = h^1 + \chi - h^2 \geq \chi > 0$$

for every $a \in \mathcal{S}_{d_1, d_2}^{\text{snc}}$, proving (4.2). $\square$

4.2. A prime horizontal factor. Here and below, a divisor on X_2 is called *horizontal* if its image under $\pi_{2,0} : X_2 \rightarrow \mathbb{P}^2$ is dense in $\mathbb{P}^2$. The following irreducible-component reduction is due to Demailly–El Goul [4, Lemma 3.3]; we use the implementation in [7, Section 4]. The reduction itself is unchanged in the two-component setting. We reproduce the short argument only to fix the notation and the ratio inequality needed below.

Lemma 4.2 (Irreducible-component reduction). *From a nonzero section of weighted degree M and twist T one may choose an irreducible horizontal factor*

$$0 \neq \omega_1 \in H^0(\mathbb{P}^2, E_{2,m} \bar{T}^* \otimes \mathcal{O}(-t))$$

whose zero divisor on X_2 has the form $Z + b\Gamma_2$, with Z irreducible, reduced, and horizontal, and such that

$$\frac{t}{m} \geq \frac{T}{M}.$$

Writing the horizontal component as

$$Z \sim b_1 u_1 + b_2 u_2 - th, \quad b_1 \geq 2b_2 \geq 0, \quad b_1 + b_2 = m,$$

one has the following alternative: if $b_2 = 0$, then Z is pulled back from a divisor on X_1 ; if $b_2 > 0$, it is a genuine second-level horizontal divisor. In particular, applying the lemma to the fixed pair (M, T) supplied by Proposition 4.1 yields an irreducible ω_1 satisfying

$$\frac{t}{m} \geq \rho_0, \quad \text{and thus} \quad \frac{m}{t} \leq \rho_0^{-1}.$$

Proof. Via (2.3), the original section is represented on X_2 by an effective divisor

$$Z_\omega \sim Mu_2 - Th.$$

Write its irreducible decomposition as

$$Z_\omega = \sum_j p_j Z_j, \quad Z_j \sim b_{1,j} u_1 + b_{2,j} u_2 - t_j h.$$

The standard classification of effective divisors on the second Demailly–Semple tower gives three possibilities:

- (i) $(b_{1,j}, b_{2,j}) = (0, 0)$ and Z_j is pulled back from the base; then $t_j \leq 0$;
- (ii) $(b_{1,j}, b_{2,j}) = (-1, 1)$, $t_j = 0$, and $Z_j = \Gamma_2$;
- (iii) Z_j is horizontal and $b_{1,j} \geq 2b_{2,j} \geq 0$, with $b_{1,j} + b_{2,j} > 0$.

For a horizontal component put

$$m_j = b_{1,j} + b_{2,j} > 0.$$

Since the class of Z_ω has $u_1 + u_2$ -weight M , while the first two types have weight zero, comparison of divisor classes gives

$$\sum_{j \text{ horizontal}} p_j m_j = M.$$

Comparison of the base twists gives

$$\sum_j p_j t_j = T.$$

The base-pullback components have $t_j \leq 0$ and Γ_2 has $t_j = 0$, so

$$\sum_{j \text{ horizontal}} p_j t_j \geq T.$$

Therefore at least one horizontal component satisfies

$$\frac{t_j}{m_j} \geq \frac{T}{M}.$$

Fix such a component and rename it Z . Because $\mathcal{O}_{X_2}(\Gamma_2) = \mathcal{O}_{X_2}(-1, 1)$, one has

$$Z + b_{1,j}\Gamma_2 \sim (b_{1,j} + b_{2,j})u_2 - t_j h = m_j u_2 - t_j h.$$

Thus, if s_j is the canonical reduced equation of the prime Cartier divisor Z and γ is the canonical section of Γ_2 , then $\gamma^{b_{1,j}} s_j$ is a section of $\mathcal{O}_{X_2}(m_j) \otimes \pi_{2,0}^* \mathcal{O}(-t_j)$, which corresponds by (2.3) to an invariant jet differential

$$\omega_j \in H^0(\mathbb{P}^2, E_{2,m_j} \bar{T}^* \otimes \mathcal{O}(-t_j)).$$

After saturation by the fixed vertical factor $\gamma^{b_{1,j}}$, its horizontal zero divisor is exactly the reduced prime Z ; no extraction of a root of a global section is involved. When $b_{2,j} = 0$, the classification identifies Z_j with the inverse image under $\pi_{2,1}$ of a divisor on X_1 . The ratio inequality proves the final assertion. $\square$

4.3. Extension over the parameter space. The following family-extension step follows the algebraic spread-out construction in [7, Section 4], applied to the ordered two-component parameter space. The only change is the base: the generic freeness, cohomology-and-base-change, and geometric-integrality arguments are the same.

Proposition 4.3 (Algebraic extension of the chosen factor). *Let (M, T) be the fixed integers supplied by Proposition 4.1. Then there exist integers (m, t) with $t/m \geq T/M$, integers*

$$b_1 \geq 2b_2 \geq 0, \quad b_1 + b_2 = m,$$

and a nonempty Zariski-open subset $\mathcal{U} \subset \mathcal{S}_{d_1, d_2}^{\text{snc}}$ with the following property. Let

$$p_{\mathcal{U}} : \mathcal{X}_2 \longrightarrow \mathcal{U}, \quad \Pi : \mathcal{X}_2 \longrightarrow \mathbb{P}^2 \times \mathcal{U}$$

be the relative second logarithmic Semple tower and its projection to level zero. There is a regular algebraic section

$$f \in H^0(\mathcal{X}_2, \mathcal{L}), \quad \mathcal{L} = \mathcal{O}_{\mathcal{X}_2}(b_1, b_2) \otimes \Pi^* \text{pr}_1^* \mathcal{O}_{\mathbb{P}^2}(-t),$$

whose zero scheme $\mathcal{Z} = (f = 0)$ is flat over $\mathcal{U}$ and has geometrically integral fibers. If γ denotes the canonical section of the relative vertical divisor Γ_2 , then

$$\omega_{1,a} = \gamma_a^{b_1} s_a \in H^0(\mathbb{P}^2, E_{2,m} T_{\mathbb{P}^2}^*(\log D_a) \otimes \mathcal{O}_{\mathbb{P}^2}(-t)), \quad a \in \mathcal{U},$$

is an algebraic family of invariant jet differentials, and its horizontal zero divisor is the irreducible reduced divisor $Z_a = \mathcal{Z}|_{X_{2,a}}$.

Proof. Work first over $S = \mathcal{S}_{d_1, d_2}^{\text{snc}}$, and denote its relative second logarithmic Semple tower by $p_S : \mathcal{X}_{2,S} \rightarrow S$. For every triple (b_1, b_2, t) consider the algebraic line bundle

$$\mathcal{L}_{b_1, b_2, t} = \mathcal{O}_{\mathcal{X}_{2,S}}(b_1, b_2) \otimes \Pi_S^* \text{pr}_1^* \mathcal{O}_{\mathbb{P}^2}(-t).$$

The sheaf $(p_S)_* \mathcal{L}_{b_1, b_2, t}$ is coherent. Generic freeness and generic cohomology and base change give a dense Zariski-open set $S_{b_1, b_2, t} \subset S$ on which it is locally free and the natural map

$$(4.3) \quad ((p_S)_* \mathcal{L}_{b_1, b_2, t}) \otimes k(a) \longrightarrow H^0(X_{2,a}, \mathcal{L}_{b_1, b_2, t}|_{X_{2,a}})$$

is an isomorphism. There are only countably many triples. Since the base field is the uncountable field $\mathbb{C}$, an irreducible variety of finite type over $\mathbb{C}$ cannot be the union of countably many proper closed subsets. Equivalently, every countable intersection of dense Zariski-open subsets has a complex point. We may therefore choose

$$a_0 \in \bigcap_{b_1, b_2, t} S_{b_1, b_2, t}.$$

This countable intersection is used only to select one numerical type and one prime factor; after that choice, the construction is spread over a single Zariski-open neighborhood. By Proposition 4.1, this fiber carries a section of the fixed weighted degree M and fixed twist T . Apply Lemma 4.2 on the fiber over a_0 and choose its reduced horizontal equation

$$0 \neq s_{a_0} \in H^0(X_{2, a_0}, \mathcal{L}_{b_1, b_2, t}|_{X_{2, a_0}})$$

with $b_1 \geq 2b_2 \geq 0$, $b_1 + b_2 = m$, and $t/m \geq T/M$. For this particular triple, the base-change isomorphism above identifies s_{a_0} with a vector in the fiber of the locally free sheaf $(p_S)_* \mathcal{L}_{b_1, b_2, t}$. After shrinking to an affine trivializing neighbourhood, extend that vector to a regular local section of the pushforward. The adjunction evaluation map

$$p_S^*(p_S)_* \mathcal{L}_{b_1, b_2, t} \longrightarrow \mathcal{L}_{b_1, b_2, t}$$

then produces a regular section f on the relative tower after replacing S by an affine Zariski-open neighborhood of a_0 .

The section s_{a_0} is nonzero and its zero scheme is the selected irreducible reduced horizontal divisor. Thus it is an integral Cartier divisor; over the algebraically closed field $\mathbb{C}$ it is also geometrically integral. Nonvanishing on every irreducible component of a fiber is an open condition, so after shrinking, $\mathcal{Z} = (f = 0)$ is a relative effective Cartier divisor. It is then flat over S because both $\mathcal{X}_{2, S}$ and the invertible ideal $\mathcal{L}^{-1}$ are flat and

$$0 \longrightarrow \mathcal{L}^{-1} \xrightarrow{f} \mathcal{O}_{\mathcal{X}_{2, S}} \longrightarrow \mathcal{O}_{\mathcal{Z}} \longrightarrow 0$$

remains exact on the fibers. Finally, geometric integrality of fibers is open for a proper flat morphism of finite presentation. A second shrinking therefore makes every Z_a geometrically integral. The relative divisor Γ_2 and its canonical section γ are algebraic. Therefore $\gamma^{b_1} f$ is a regular relative section of

$$\mathcal{O}_{\mathcal{X}_2}(m) \otimes \Pi^* \mathrm{pr}_1^* \mathcal{O}_{\mathbb{P}^2}(-t),$$

and the relative direct-image formula gives the required algebraic family $\omega_{1, a}$. This proves all statements after naming the resulting open set $\mathcal{U}$. $\square$

From now on we fix the irreducible differential given by Lemma 4.2 and denote its weighted degree and twist by (m, t) .

Remark 4.4 (Output of the first-differential section). After shrinking one nonempty Zariski-open subset of the ordered parameter space, we have fixed integers (m, t, b_1, b_2) satisfying

$$b_1 \geq 2b_2 \geq 0, \quad b_1 + b_2 = m, \quad \frac{t}{m} \geq \rho_0,$$

together with a relative geometrically integral horizontal divisor $\mathcal{Z}$ and its reduced equation f . The associated invariant jet differential is

$$\omega_1 = \gamma^{b_1} f.$$

All numerical data are constant on this open. The next section uses only this factor package.

5. MIXED $\mathcal{O}_{\mathbb{P}^2}(3)$ SLANTED VECTOR FIELDS

Goal and output. Starting with the relative prime divisor selected in Proposition 4.3, we shall construct a regular global slanted field of base twist $\mathcal{O}_{\mathbb{P}^2}(3)$ whose homogeneous derivative of the first equation is not divisible by the chosen prime equation. Following Siu's homogeneous-cone differentiation method [19, 21, 6], this derivative descends directly to the second jet differential of Proposition 5.8. The construction is lengthy because the local algebra, the projective descent, and the homogeneous differentiation must all be compatible.

By a *slanted vector field* we mean a vector field on the universal relative jet space that is tangent to the jet-incidence equations and may have components in the parameter directions. A *regular global twisted field* is a global section of the relative tangent bundle tensored with the stated twisting line bundles. These fields perform the actual differentiation. By contrast, the residual derivations introduced below are defined only over the generic incidence field and are used only to test independence.

We keep the standing notation

$$\mathcal{Z} = (f = 0), \quad \omega_{1,a} = \gamma_a^{b_1} s_a$$

from Proposition 4.3. Thus f is the Γ_2 -saturated horizontal equation, with the fixed vertical factor removed. The section differentiated in Phase VI is the full relative invariant equation

$$\omega_1 = \gamma^{b_1} f.$$

The factor f is kept separately only to test, at the generic point of $\mathcal{Z}$, whether the differentiated equation is independent of the first one.

The compact template. The compact pole-three argument is easiest to remember on the invariant fiber ring

$$K[u, v, W], \quad \text{wt}(u) = \text{wt}(v) = 1, \quad \text{wt}(W) = 3.$$

If every available slanted derivative vanished normally along a prime divisor ($P = 0$), then the principal ideal (P) would be stable under the corresponding residual derivations. In the compact case an acceleration field first supplies a nonzero multiple of ∂_W , so that P cannot depend on W ; a matrix field then acts on the remaining binary form and contradicts irreducibility in positive weight. This two-step contradiction is the expository model for what follows.

The two-component logarithmic adaptation. The compact fields cannot simply be copied. Each logarithmic component contributes its own root equation and its own value, gradient, and Hessian conditions. The same matrix motion T and the same quadratic acceleration C must therefore solve two Hessian-cancellation equations at once. On the transverse open set this determines a common correction $C = C_T$ through the inverse of the two-row gradient matrix; the resulting residual derivations are the fields D_T in (5.21). Unlike the compact model, no independent acceleration shear is inserted at this point. Instead, the commutators of D_H, D_E, D_F produce a nonzero polynomial multiple of ∂_W in (5.43). That curvature direction removes the W -dependence, after which the full binary $\mathfrak{sl}_2$ -action forces P to be constant. This is the essential two-component adaptation.

Remark 5.1 (Sources and new point). Finite-difference coefficient fields go back to [14, Section 3]. The compact matrix lift appears in [15, Lemma 1.1], and its logarithmic form appears in [16, Lemma 5]. The quadratic-acceleration field and the symmetry organization are related to Cui–Hou–Liu–Xie [10]. The simultaneous acceleration and independence arguments used here are developed and proved in this section. The new point is to couple two component equations while keeping one pole-three twist.

For navigation, the six phases are listed below.

phase	main object	output
I	two logarithmic jet equations	the invariant ring $K_0[u, v, W]$
II	coefficient directions	third-difference fields
III	two tangency systems	the residual derivations D_T
IV	parameter gauges and Semple charts	regular global fields with base twist $\mathcal{O}_{\mathbb{P}^2}(3)$
V	an invariant prime	residual rigidity
VI	the homogeneous first equation	a second global jet differential

Two kinds of vector fields occur, and they have different roles.

object	where it is defined	how it is used
global twisted field	globally, after twisting	performs the actual differentiation
residual derivation	over the generic field K_0	tests whether the prime ideal is invariant

A residual derivation is only a generic-field linear combination of global twisted fields; it is not claimed to be global.

The number three also has three different meanings here. Third differences remove Taylor terms of order at most two. Projective overlaps then give pole order at most three. Homogenization records this pole bound as the twist $\mathcal{O}_{\mathbb{P}^2}(3)$. We still work with invariant two-jets, not three-jets.

5.1. Phase I: two roots and the invariant fiber ring.

Goal. Identify the invariant generic fiber ring and record exactly which denominators belong to the generic incidence calculation.

On the affine chart $Z_0 \neq 0$, write the two universal equations as

$$f_i(z, a_i) = \sum_{|\alpha| \leq d_i} a_{i,\alpha} z^\alpha, \quad z = (z_1, z_2), \quad i = 1, 2.$$

Near a crossing of the two components one must keep the two logarithmic normal directions separate. We therefore introduce two roots

$$(5.1) \quad w_i^{d_i} + f_i(z, a_i) = 0, \quad i = 1, 2.$$

This construction is made first on affine charts of the two coefficient cones. Equivalently, over such a chart it is the fiber product of the two cyclic covers associated with the local trivializations of $\mathcal{O}_{\mathbb{P}^2}(d_i)$. The chartwise covers are auxiliary: deck equivariance, the product coefficient–root gauge in Lemma 5.3, and reparametrization equivariance below are exactly the descent data that produce vector fields on the relative logarithmic Semple tower. Thus no choice of a global root of either defining equation enters the theorem. The cover is smooth over the simple normal crossing (SNC) locus, including above $C_1 \cap C_2$. Its deck group is $\mu_{d_1} \times \mu_{d_2}$; when $d_i = 1$ the corresponding deck factor is trivial and the equation $w_i + f_i = 0$ simply defines a graph. The coefficient–root radial gauge is still present. The two independent radial fields on the coefficient cones are

$$(5.2) \quad R_i = d_i \sum_{|\alpha| \leq d_i} a_{i,\alpha} \partial_{a_{i,\alpha}} + w_i \partial_{w_i}.$$

Write the ordinary first and second derivatives as

$$\xi = (u, v), \quad \eta = (U, V),$$

and write the logarithmic root derivatives as

$$w'_i = w_i \chi_i, \quad w''_i = w_i (\psi_i + \chi_i^2).$$

Differentiating (5.1) zero, one, and two times gives

$$(5.3) \quad E_{i0} = w_i^{d_i} + f_i,$$

$$(5.4) \quad E_{i1} = d_i w_i^{d_i} \chi_i + df_i(\xi),$$

$$(5.5) \quad E_{i2} = d_i w_i^{d_i} \psi_i + d_i^2 w_i^{d_i} \chi_i^2 + df_i(\eta) + \text{Hess}(f_i)(\xi, \xi).$$

Here is the tangency test used throughout the construction. If a field has only an i -th coefficient component, represented by the polynomial $Q_i(Y)$, then its action on the three equations is

$$\begin{aligned} \delta E_{i0} &= Q_i(z), \\ \delta E_{i1} &= dQ_i|_z(\xi), \\ \delta E_{i2} &= dQ_i|_z(\eta) + \text{Hess}(Q_i)|_z(\xi, \xi). \end{aligned}$$

Thus a coefficient correction with $j_z^2 Q_i = 0$ is tangent to the entire two-jet system, independently of the chosen jet. This single observation is the reason that third-order Taylor remainders can be used below without repeating the tangency calculation for every coefficient field. On the dense open set $f_1 f_2 \neq 0$, these equations eliminate χ_i, ψ_i . The unipotent part of the two-jet reparametrization group acts by

$$(u, v, U, V) \longmapsto (u, v, U + \lambda u, V + \lambda v).$$

Putting $W = uV - vU$, one obtains

$$(5.6) \quad \mathbb{C}[u, v, U, V]^{G_{2, \text{unip}}} = \mathbb{C}[u, v, W], \quad \text{wt}(u) = \text{wt}(v) = 1, \quad \text{wt}(W) = 3.$$

Indeed, after localizing at u , the infinitesimal generator becomes $u\partial_U$ in the coordinates (u, v, U, W) . Its kernel is $\mathbb{C}[u, v, W]_u$, and the absence of a negative power of u follows by reducing a putative numerator modulo u . This also proves (5.6) over every characteristic-zero extension field.

We now specify the generic field used below. On fixed affine coefficient charts, let

$$\mathcal{B}^\circ = \left\{ (z, a_1, a_2) : f_1 f_2 \det \begin{pmatrix} \partial_1 f_1 & \partial_2 f_1 \\ \partial_1 f_2 & \partial_2 f_2 \end{pmatrix} \neq 0 \right\}.$$

It is a nonempty open subset of an affine space: for instance $f_1 = 1 + z_1$ and $f_2 = 1 + z_2$ at $z = 0$ give a point of it. Thus it is irreducible. We set

$$(5.7) \quad K_0 := \mathbb{C}(\mathcal{B}^\circ).$$

The affine two-jet incidence over it is cut out by the six equations $E_{i\nu} = 0$ in the variables $(w_i, \chi_i, \psi_i, u, v, U, V)$. On $f_1 f_2 \neq 0$, equations E_{i0} give a finite root extension and E_{i1}, E_{i2} eliminate χ_i, ψ_i ; its function field is therefore a finite extension of $K_0(u, v, U, V)$. Taking invariants of the unipotent reparametrization leaves $K_0[u, v, W]$ after descent from that finite extension. Every denominator in the generic residual coefficients divides a product of

$$(5.8) \quad f_1 f_2, \quad \det(g_1, g_2), \quad \ell_1(a_1) \ell_2(a_2),$$

on the corresponding affine incidence chart. The projective and Semple transition coordinates are different: their zero sets are not removed by a parameter shrinking. They are covered by the overlap and boundary calculations in Phase IV.

Output used later. Phase I produces the arena for the contradiction: an explicitly defined generic field K_0 , the invariant ring $K_0[u, v, W]$, and a recorded list of the genuine generic-incidence denominators. The next two phases construct polynomial generators whose generic residual actions operate on this ring.

5.2. Phase II: third differences and one-stage Taylor reduction.

Goal. Construct polynomial coefficient corrections that preserve the two-jet tangency equations and have base degree at most three.

We now build, separately in the two coefficient blocks, the third-difference Taylor package. Its finite-difference antecedent is [14, Section 3]; its symmetry organization is related to [10]. All formulas needed here are proved below. The point that requires care is that the two corrections are later coupled by the common base and jet-variable motion.

For $d_i \geq 3$ and multi-indices $\alpha \geq \gamma$ with $|\alpha| \leq d_i$ and $|\gamma| = 3$, define the third-difference field in the i -th coefficient block by

$$(5.9) \quad \mathcal{D}_{i,\alpha,\gamma} = \sum_{\beta \leq \gamma} (-1)^{|\beta|} \binom{\gamma}{\beta} z^\beta \partial_{a_{i,\alpha-\beta}}.$$

Its coefficient polynomial is $Y^{\alpha-\gamma}(Y-z)^\gamma$. It belongs to the third power $\mathfrak{m}_z^3$ of the maximal ideal at z and thus annihilates all three equations (5.3)–(5.5). Over any characteristic-zero field, the values of these fields span

$$(5.10) \quad \mathfrak{m}_z^3 \cap K[Y_1, Y_2]_{\leq d_i}.$$

This is just the Taylor basis: choose $\gamma \leq \delta$ with $|\gamma| = 3$ and expand $(Y-z)^{\delta-\gamma}$ in powers of Y .

This package is a correction engine, not a complete tangent frame. It can absorb exactly the coefficient polynomials in $\mathfrak{m}_z^3 \cap K[Y_1, Y_2]_{\leq d_i}$, while it contributes no base or jet-variable motion. The missing motions are supplied in Phase III by T , C , and N_i ; their remaining coefficient components will then be returned to this Taylor span.

We shall also use the following elementary reduction for $d_i \geq 2$. Suppose that a polynomial $Q^0(Y; z)$ has Y -degree at most $d_i + 1$, its top homogeneous part is independent of z , and its remaining coefficients have z -degree at most two. For every degree- $(d_i + 1)$ monomial choose once and for all $\gamma \leq \alpha$ with $|\gamma| = 3$, and subtract the corresponding multiple of

$$(Y-z)^\gamma Y^{\alpha-\gamma}.$$

The resulting polynomial Q has Y -degree at most d_i , $Q - Q^0 \in \mathfrak{m}_z^3$, and coefficient functions of z -degree at most three. In particular, Q and Q^0 have the same value, gradient, and Hessian at $Y = z$. For $d_i = 2$, the same reduction lowers degree three to degree two. More generally,

$$\mathfrak{m}_z^3 \cap K[Y_1, Y_2]_{\leq d_i} = 0 \quad (d_i \leq 2),$$

so no third-difference field exists or is needed in degrees one and two. An arbitrary raw polynomial (5.12) need not have degree at most one when $d_i = 1$; the compatible degree-one generators are constructed in Lemma 5.2 below.

Output used later. This stage supplies the degree-compatible coefficient fields; the next stage couples their values through one common base and jet-variable motion.

5.3. Phase III: simultaneous Hessian cancellation and mixed fields.

Goal. Couple one base and jet-variable motion to both component equations and identify its residual action on $K_0[u, v, W]$.

The following construction couples the classical matrix-lift input [15, Lemma 1.1] and [16, Lemma 5] with the quadratic-acceleration construction developed below. In contrast with the one-equation construction, the acceleration tensor is determined by two simultaneous Hessian equations.

Let $T \in \mathfrak{sl}_2(\mathbb{C})$, let $C : \text{Sym}^2 \mathbb{C}^2 \rightarrow \mathbb{C}^2$ be symmetric, and let $N_i \in (\mathbb{C}^2)^*$. Prescribe

$$(5.11) \quad \delta \xi = T\xi, \quad \delta \eta = T\eta + C(\xi, \xi), \quad \delta \chi_i = N_i(\xi), \quad \delta \psi_i = N_i(\eta).$$

In the i -th coefficient block begin with

$$(5.12) \quad Q_i^0(Y) = -df_i(Y)T(Y-z) + d_i f_i(Y)N_i(Y-z) - \frac{1}{2}df_i(Y)C(Y-z, Y-z).$$

For $d_i \geq 2$, apply the preceding one-stage reduction and denote the degree-at-most- d_i polynomial by Q_i . When $d_i = 1$, use instead the compatible degree-one fields of Lemma 5.2. The

vector fields obtained in either way, consisting of (5.11) and the coefficient vector represented by Q_i , satisfy the exact ideal identities

$$(5.13) \quad V(E_{i0}) = 0,$$

$$(5.14) \quad V(E_{i1}) = d_i N_i(\xi) E_{i0},$$

$$(5.15) \quad V(E_{i2}) = d_i N_i(\eta) E_{i0} + 2d_i N_i(\xi) E_{i1}.$$

The identities follow by direct substitution. The acceleration term in (5.12) has zero value and gradient at z and Hessian $-df_i \circ C$; this cancels the variation of $df_i(\eta)$ caused by $C(\xi, \xi)$. Any reduction term lies in $\mathfrak{m}_z^3$ and changes none of these identities; the degree-one identities are checked directly in Lemma 5.2.

We may now verify reparametrization equivariance without a forward reference. For (λ, μ) with $\lambda \neq 0$, the two-jet group acts by

$$\tilde{\xi} = \lambda\xi, \quad \tilde{\eta} = \lambda^2\eta + \mu\xi, \quad \tilde{\chi}_i = \lambda\chi_i, \quad \tilde{\psi}_i = \lambda^2\psi_i + \mu\chi_i.$$

Substitution in (5.11) gives

$$\begin{aligned} \delta\tilde{\xi} &= T\tilde{\xi}, & \delta\tilde{\eta} &= T\tilde{\eta} + C(\tilde{\xi}, \tilde{\xi}), \\ \delta\tilde{\chi}_i &= N_i(\tilde{\xi}), & \delta\tilde{\psi}_i &= N_i(\tilde{\eta}). \end{aligned}$$

Thus the mixed fields are equivariant for both the scaling and unipotent parts of the group and descend to invariant two-jets.

Put

$$F_i = f_i(z), \quad p_i = df_i|_z, \quad g_i = \frac{p_i}{F_i}, \quad K_i = \text{Hess}(\log f_i)|_z = \frac{\text{Hess}(f_i)|_z}{F_i} - \frac{p_i^\top p_i}{F_i^2}.$$

The value of Q_i^0 at z is zero. Its gradient vanishes exactly for

$$(5.16) \quad N_i = \frac{p_i T}{d_i F_i},$$

and, after this choice, its Hessian vanishes exactly when

$$(5.17) \quad g_i \circ C + T^\top K_i + K_i T = 0.$$

This identity is the reason that the matrix and acceleration parts must be treated together.

Work over the field K_0 in (5.7); by construction $g_1 \wedge g_2 \neq 0$. The map

$$G : K_0^2 \longrightarrow K_0^2, \quad G(\tau) = (g_1(\tau), g_2(\tau))$$

is then an isomorphism. For every $T \in \mathfrak{sl}_2(K_0)$, the two equations (5.17) have the unique simultaneous solution

$$(5.18) \quad C_T(\zeta, \zeta) = -G^{-1} \left(\zeta^\top (T^\top K_1 + K_1 T) \zeta \right)_{\zeta^\top (T^\top K_2 + K_2 T) \zeta}.$$

Formula (5.18) is the precise replacement for the free compact acceleration. A single component imposes one scalar quadratic identity; the two independent logarithmic gradients assemble the two identities into the invertible map G and thus determine the vector-valued quadratic correction C_T uniquely.

Lemma 5.2 (Compatible generators for a linear component). *Assume $d_1 = 1$ and $d_2 \geq 2$. Put*

$$x = Y - z, \quad f_1(Y) = F + p(x), \quad p = (p_1, p_2) = df_1, \quad v_p = (-p_2, p_1)^\top.$$

The line block of (5.12) is represented by a polynomial of Y -degree at most one if and only if

$$(5.19) \quad p(C(x, x)) = 2p(x)N_1(x),$$

and then

$$(5.20) \quad Q_1(Y) = (-pT + FN_1)x.$$

There is a finite polynomial family of ten regular fields with this property, split as $3 + 2 + 3 + 2$: three T -fields; two line-normal fields, for which

$$C_N(\zeta, \eta) = N(\zeta)\eta + N(\eta)\zeta, \quad N_1 = N;$$

three kernel-acceleration fields, for which

$$C_{p,q}(\zeta, \zeta) = v_p q(\zeta, \zeta), \quad q \in \{\zeta_1^2, \zeta_1 \zeta_2, \zeta_2^2\};$$

and two second-normal fields, for which only N_2 is nonzero. After the degree- d_2 Taylor reduction in the second block, these ten fields satisfy (5.13)–(5.15). Their span over K_0 contains the residual field determined by (5.16) and (5.18); for that residual field one has $Q_1 = 0$.

Proof. Since f_1 is linear, substitution in (5.12) gives

$$Q_1^0(Y) = -pTx + (F + px)N_1x - \frac{1}{2}p(C(x, x)).$$

Its quadratic part vanishes exactly under (5.19), and the remaining linear polynomial is (5.20). For the line-normal fields, $C_N(x, x) = 2N(x)x$, so the compatibility is automatic and $Q_1 = FN(x)$. For a kernel-acceleration field one has $p(v_p) = 0$, thus $Q_1 = 0$. The T -fields give $Q_1 = -pTx$, while a second-normal field contributes nothing in the line block. Thus all ten fields are represented by legitimate line-coefficient variations, and direct substitution gives the three tangent-ideal identities. The second block is handled by the degree- d_2 reduction already proved above. Although v_p depends linearly on the coefficients of the line, it is independent of z ; thus the degree and Taylor-remainder bounds used in that reduction are unchanged.

It remains to check that these polynomial fields span the required generic one. For a line,

$$K_1 = -\frac{p^\top p}{F^2}, \quad N_1 = \frac{pT}{F}.$$

Evaluating the first equation in (5.17) on (ζ, ζ) therefore gives

$$p(C_T(\zeta, \zeta)) = 2p(\zeta)N_1(\zeta) = p(C_{N_1}(\zeta, \zeta)).$$

On the generic incidence open $p \neq 0$, and v_p spans $\ker p$. Thus there is a scalar quadratic form q_T over K_0 such that

$$(C_T - C_{N_1})(\zeta, \zeta) = v_p q_T(\zeta, \zeta).$$

These four families therefore span the data (T, C_T, N_1, N_2) . Finally $-pT + FN_1 = 0$, so (5.20) gives $Q_1 = 0$. $\square$

Degree ledger. At the residual choice, $Q_i \in \mathfrak{m}_z^3$ in both blocks. If $d_i \geq 3$, the Taylor span (5.10) cancels Q_i by third differences. If $d_i = 2$, the reduced polynomial lies in $\mathfrak{m}_z^3$ and is therefore already zero. If $d_1 = 1$, the compatible family of Lemma 5.2 gives $Q_1 = 0$ directly. These are the only degree-dependent branches of the construction. On the invariant ring $K_0[u, v, W]$ the residual derivation is

$$(5.21) \quad D_T = (T\xi) \cdot \partial_\xi + S_T(u, v)\partial_W, \quad S_T(u, v) = \det(\xi, C_T(\xi, \xi)).$$

The rational functions in (5.16)–(5.18) select a K_0 -linear combination of the polynomial generators constructed above. They are not claimed to be coefficients of one rational vector field regular on the whole parameter space. This distinction lets us use the generic residual derivations without concealing any pole.

Output used later. The output of Phase III is algebraic rather than yet global: a finite collection of regular affine polynomial fields, together with the fact that its generic K_0 -span contains D_T for every $T \in \mathfrak{sl}_2(K_0)$. Phase IV globalizes the polynomial generators; it does not attempt to globalize the rational coefficients selecting D_T .

5.4. Phase IV: common twist and global extension.

Goal. Descend the affine generators through the coefficient gauges, base-chart overlaps, logarithmic corners, and Semple boundary with one common twist.

The preceding affine fields are not yet global on the projectivized parameter space or the Semple compactification. The product gauge first supplies the common parameter twist $\mathcal{O}(1) \boxtimes \mathcal{O}(1)$; the projective overlap and boundary calculations then supply the base twist $\mathcal{O}_{\mathbb{P}^2}(3)$. Together they give the regular global package used in residual rigidity. Parameter and root gauges.

Lemma 5.3 (Product coefficient–root gauge). *Let a cone field have parameter multidegree $e = (e_1, e_2)$. Thus, under independent coefficient scalings $a_j \mapsto \lambda_j a_j$, its nonparameter coefficients scale by $\lambda_1^{e_1} \lambda_2^{e_2}$, while its coefficient vector q_i in the i -th block has one additional factor λ_i . Write its root component as $b_i w_i \partial_{w_i}$. On coefficient charts $\ell_i(a_i) = 1$, put*

$$(5.22) \quad c_i = \ell_i(q_i), \quad \tilde{q}_i = q_i - c_i a_i, \quad \tilde{b}_i = b_i - \frac{c_i}{d_i}.$$

If only the i -th coefficient chart changes, with transition ratio s_i and induced coordinate map Φ_i , then

$$(5.23) \quad \tilde{q}'_j = s_i^{e_i} (\Phi_i)_* \tilde{q}_j, \quad j = 1, 2,$$

$$(5.24) \quad \tilde{b}'_j = s_i^{e_i} \tilde{b}_j, \quad j \neq i,$$

$$(5.25) \quad \tilde{b}'_i = s_i^{e_i} \left(\tilde{b}_i + \frac{1}{d_i} \tilde{q}_i (\log s_i) \right).$$

Therefore the normalized representatives glue with parameter twist $\mathcal{O}_{\mathbb{P}(V_{d_1})}(e_1) \boxtimes \mathcal{O}_{\mathbb{P}(V_{d_2})}(e_2)$.

Proof. The radial field R_i in (5.2) has coefficient component $d_i a_i$ and root component $w_i \partial_{w_i}$. Subtracting $(c_i/d_i)R_i$ therefore gives exactly (5.22); its coefficient component is tangent to $\ell_i(a_i) = 1$. On an overlap, the normalized coefficient representative is multiplied by $s_i^{e_i}$. The root changes by $w'_i = s_i^{1/d_i} w_i$, so its logarithmic derivative acquires $d_i^{-1} \tilde{q}_i (\log s_i)$. This gives (5.23)–(5.25). Performing the same calculation in the two blocks independently proves the product-twist statement. $\square$

When $d_i = 1$, the factors $1/d_i$ in the normalization and overlap formulas are simply one. Thus the triviality of the deck factor μ_1 removes a covering symmetry but does not remove the coefficient–root gauge.

The common parameter twist. When $d_1, d_2 \geq 2$, the reduced mixed family parametrized by (T, C, N_1, N_2) has parameter multidegree $(0, 0)$; no generator from the separate line-compatible package occurs. We multiply all these fields by $\ell_1 \ell_2$.

When $d_1 = 1$, Lemma 5.2 supplies the separate line-compatible package. Its T -, line-normal, and second-normal fields have multidegree $(0, 0)$, while its three kernel-acceleration fields are linear in the coefficients of the line and have multidegree $(1, 0)$. We multiply the former by $\ell_1 \ell_2$ and the latter by ℓ_2 . An i -th third-difference field has multidegree -1 in the i -th block and degree zero in the other block, so multiplication by $\ell_i^2 \ell_j$ ($j \neq i$) puts every summand in the common twist

$$(5.26) \quad \mathcal{O}_{\mathbb{P}(V_{d_1})}(1) \boxtimes \mathcal{O}_{\mathbb{P}(V_{d_2})}(1).$$

At the generic incidence point the values of the ℓ_i are units. The Taylor-span identity (5.10) therefore selects, in each block of degree at least three, a linear combination of the twisted third differences whose cone coefficient is the negative of Q_i . A quadratic block contributes nothing because its reduced Q_i is zero, and at the residual combination the linear block contributes nothing by Lemma 5.2. Lemma 5.3 shows at the same time that the associated scalars c_i , the projective coefficient components, and the root-gauge corrections cancel. Thus the subtraction producing D_T is a linear combination of regular common-twist fields, rather than a rational field whose coefficients are simply claimed to be globally regular.

The base-chart pole bound. We next check the base twist. On a projective overlap let $\widehat{z} = \phi(z)$ and let the i -th root frame change by $\widehat{w}_i = g_i(z)w_i$; put $h_i = \log g_i$. The ordinary and root jets transform as

$$(5.27) \quad \widehat{\xi} = D\phi \xi, \quad \widehat{\eta} = D\phi \eta + D^2\phi(\xi, \xi),$$

$$(5.28) \quad \widehat{\chi}_i = \chi_i + dh_i(\xi), \quad \widehat{\psi}_i = \psi_i + dh_i(\eta) + d^2h_i(\xi, \xi).$$

Varying these identities at a fixed base point and using (5.11), with $A = C(\xi, \xi)$, gives

$$(5.29) \quad \delta\widehat{\xi} = D\phi(T\xi),$$

$$(5.30) \quad \delta\widehat{\eta} = D\phi(T\eta + A) + 2D^2\phi(\xi, T\xi),$$

$$(5.31) \quad \delta\widehat{\chi}_i = N_i(\xi) + dh_i(T\xi),$$

$$(5.32) \quad \delta\widehat{\psi}_i = N_i(\eta) + dh_i(T\eta + A) + 2d^2h_i(\xi, T\xi).$$

To read the denominators explicitly, take

$$s = x^{-1}, \quad r = y/x,$$

and write $(\alpha, \beta) = (s', r')$ and $(B_1, B_2) = (s'', r'')$. Differentiation of $x = s^{-1}$ and $y = rs^{-1}$ gives

$$(5.33) \quad \xi_x = -\frac{\alpha}{s^2}, \quad \xi_y = \frac{\beta}{s} - \frac{r\alpha}{s^2},$$

$$(5.34) \quad \eta_x = -\frac{B_1}{s^2} + \frac{2\alpha^2}{s^3}, \quad \eta_y = \frac{B_2}{s} - \frac{rB_1}{s^2} - \frac{2\alpha\beta}{s^2} + \frac{2r\alpha^2}{s^3}.$$

Thus a first jet has pole order at most two and a second jet has pole order at most three. For the acceleration part write $\widehat{A}_x = s^4 A_x$ and $\widehat{A}_y = s^4 A_y$; these are polynomial in the new first jets. Formula (5.30) gives the exact identities

$$(5.35) \quad \delta(s'') = -\frac{\widehat{A}_x}{s^2}, \quad \delta(r'') = \frac{\widehat{A}_y - r\widehat{A}_x}{s^3}.$$

For the standard root transition $h_i = -\log x = \log s$, the additional normal term is

$$(5.36) \quad dh_i(A) = -sA_x = -\frac{\widehat{A}_x}{s^3}.$$

For the matrix part, equations (5.33)–(5.34) give $N_i(\eta) = O(s^{-3})$. The entries of $D\phi$ have orders at least one in s , those of $D^2\phi$ have orders at least two, while $dh_i = O(s)$ and $d^2h_i = O(s^2)$. Thus

$$D\phi(T\eta), \quad D^2\phi(\xi, T\xi), \quad dh_i(T\eta), \quad d^2h_i(\xi, T\xi) = O(s^{-2}),$$

and the only possible third-order matrix/root term is $N_i(\eta)$. Equations (5.29)–(5.32) therefore prove, term by term, that the complete mixed field has base pole order at most three. Since every reduced Q_i has coefficient degree at most three, all components homogenize with $\mathcal{O}_{\mathbb{P}^2}(3)$. In the linear block the coefficients in (5.20) have z -degree at most two; the parameter-linear kernel fields have no additional z -dependence, and their second-block reductions again have degree at most three. Thus the degree-one generators introduce no further base twist.

The logarithmic corner. It remains to extend across logarithmic corners and the Semple fiber boundary. In a regular logarithmic frame write

$$\zeta = (x, y), \quad \theta = (X, Y), \quad \delta\zeta = T\zeta, \quad \delta\theta = T\theta + A(\zeta, \zeta),$$

where A has regular coefficients. This form is stable under a change of logarithmic frame: the second jet changes by a term linear in θ plus a term quadratic in ζ .

At a point of $C_i \cap C_j$, put

$$P_{ij}(z) = \begin{pmatrix} df_i|_z \\ df_j|_z \end{pmatrix}.$$

Transversality makes P_{ij} invertible in a neighborhood. The first and second root-jet equations solve there as

$$(5.37) \quad \xi = -P_{ij}^{-1} \begin{pmatrix} d_i w_i^{d_i} \chi_i \\ d_j w_j^{d_j} \chi_j \end{pmatrix},$$

$$(5.38) \quad \eta = -P_{ij}^{-1} \begin{pmatrix} d_i w_i^{d_i} \psi_i + d_i^2 w_i^{d_i} \chi_i^2 + \text{Hess}(f_i)(\xi, \xi) \\ d_j w_j^{d_j} \psi_j + d_j^2 w_j^{d_j} \chi_j^2 + \text{Hess}(f_j)(\xi, \xi) \end{pmatrix}.$$

Both expressions are holomorphic. Substitution in the mixed field shows that the induced T and A in the logarithmic frame remain regular at the corner.

The Semple boundary. Finally put

$$W = xY - yX, \quad p = y/x, \quad q = W/x^3, \quad T = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

A direct differentiation gives

$$(5.39) \quad \delta p = c + (d - a)p - bp^2,$$

$$(5.40) \quad \delta q = A_2(1, p) - pA_1(1, p) + (\text{tr } T - 3(a + bp))q.$$

In the reciprocal second-stage chart $r_2 = q^{-1}$ one has

$$(5.41) \quad \delta r_2 = -(A_2(1, p) - pA_1(1, p))r_2^2 - (\text{tr } T - 3(a + bp))r_2.$$

These expressions are polynomial. Interchanging x and y gives the other two charts, so the four standard charts cover the entire second Semple fiber boundary. Constant deck transformations add constants to $\log w_i$ and therefore fix χ_i, ψ_i ; the root components are proportional to w_i , so both root boundary components are preserved.

Twist and role ledger. The four types of fields have different logical roles.

Reduced mixed field.: Before homogenization it has base pole at most 3 and preserves both tangency systems while changing the jet variables. Its parameter multidegree is $(0, 0)$, except for the line-compatible kernel-acceleration fields, which have multidegree $(1, 0)$.

Third difference in block i .: It has coefficient degree -1 in that block and base pole at most 3; it cancels an element of $\mathfrak{m}_z^3$. This field is absent when $d_i \leq 2$.

Homogenized field.: It has common parameter multidegree $(1, 1)$ and base pole at most 3; this is the regular global field on the universal two-component Semple tower.

Generic combination D_T .: Its coefficients lie in K_0 . It is used only to detect a generic normal direction and prove rigidity, and is not claimed to be global.

Thus every geometric differentiation is performed by a regular global twisted field. The rational functions in K_0 are used only to describe the span of their values at the generic point.

For the differentiation step it is useful to keep the homogeneous data before taking the projective quotients. Let $\widehat{\mathcal{J}} \rightarrow \mathcal{X}_2$ denote the homogeneous multicone obtained by retaining nonzero frames of the base, the two coefficient tautological lines, and the two Semple tautological directions; on a root chart we keep the variables w_1, w_2 as well. The transition rules in Phase IV glue these charts, and the quotient by the corresponding $\mathbb{G}_m$ -gauges and finite root-deck groups is $\mathcal{X}_2$. A section of a tensor product of the displayed tautological line bundles is equivalently a regular semi-invariant function on $\widehat{\mathcal{J}}$; its character records the line bundle.

Proposition 5.4 (The mixed $\mathcal{O}(3)$ package). *Assume $d_1 \geq 1$ and $d_2 \geq 2$. On a nonempty Zariski-open subset of the universal family of ordered transverse pairs of degrees (d_1, d_2) , the finite constructions above define regular global logarithmic slanted vector fields on the relative second Semple tower with common twist*

$$(5.42) \quad \mathcal{O}_{\mathbb{P}^2}(3) \boxtimes \mathcal{O}_{\mathbb{P}(V_{d_1})}(1) \boxtimes \mathcal{O}_{\mathbb{P}(V_{d_2})}(1).$$

Their span over the generic incidence field contains the three residual derivations D_T in (5.21), for $T \in \mathfrak{sl}_2$. Also, each global twisted field $\mathcal{V}$ is the descent of a polynomial homogeneous

derivation $\widehat{\mathcal{V}}$ on $\widehat{\mathcal{J}}$, equivariant under the root-deck groups, whose character is the common twisting line in (5.42). Thus $\widehat{\mathcal{V}}$ sends a semi-invariant of character χ to one of character equal to χ times the common twisting character.

Proof. The ideal identities (5.13)–(5.15) prove tangency on every affine coefficient chart. Lemma 5.3 and (5.26) prove simultaneous descent through the two projective coefficient spaces and the two root gauges. Equations (5.27)–(5.36) give base pole order three, while (5.37)–(5.41) prove regularity at double logarithmic corners and on all Semple fiber charts. Indeed, the corner formulas express the ordinary jets polynomially in the two independent logarithmic root jets, and every root component has the boundary-preserving form $w_i \widetilde{b}_i \partial_{w_i}$. The polynomial Semple formulas, together with those obtained by interchanging the two first-jet coordinates, cover the four standard charts.

The third-difference coefficients are monomials z^β with $|\beta| \leq 3$. Thus these fields, as well as the reduced mixed fields, have base pole order at most three and acquire no twist beyond $\mathcal{O}_{\mathbb{P}^2}(3)$. When both degrees are at least two, there are only finitely many constant data: three matrix directions T , six components of the symmetric map C , and four normal-coordinate directions in N_1, N_2 . When $d_1 = 1$, Lemma 5.2 replaces these by ten line-compatible mixed generators. In either case there are only finitely many third differences, none in a block of degree at most two. Their global sections therefore span a finite-dimensional complex vector space $\mathcal{V}$ with the common twist (5.42). The blockwise Taylor cancellation, performed after all summands have this common twist, says that the value of a suitable K_0 -linear combination of elements of $\mathcal{V}$ is D_T . It does not assert that the generic rational coefficients themselves define global vector fields. This is exactly the generic-span statement required later.

Finally, the fields were constructed before quotienting as polynomial derivations of the coefficient–root two-jet incidence ring. The tangency identities (5.13)–(5.15) show that these derivations preserve the incidence ideal. The coefficient normalization in Lemma 5.3 changes a lift only by the coefficient–root Euler fields, while (5.23)–(5.25) give exactly the claimed homogeneous character. Base homogenization supplies the factor $\mathcal{O}_{\mathbb{P}^2}(3)$. The corner and Semple formulas already checked above show that the homogeneous lifts remain regular in the boundary charts. Since every root component has the form $w_i \widetilde{b}_i \partial_{w_i}$, the lift commutes with the finite deck transformations. The polynomial homogeneous derivations therefore glue on $\widehat{\mathcal{J}}$ and descend to the global twisted fields just constructed. $\square$

Output used later. The finite-dimensional space $\mathcal{V}$ consists of genuine global twisted fields. At the generic incidence point its span contains D_H , D_E , and D_F . Therefore, if every element of $\mathcal{V}$ were tangent to the chosen prime divisor, then these three residual derivations would preserve its generic principal ideal. Phase V proves that this is impossible.

5.5. Phase V: residual curvature and rigidity.

Goal. Show that no positive-weight prime in the invariant fiber ring is preserved by every residual direction.

We now adapt the residual-rigidity computation of [10]. The invariant ring $K_0[u, v, W]$ has the same formal shape, but the W -components of the derivations are different: they contain the simultaneous solution of the two Hessian equations (5.17). The commutator calculation below verifies that this coupled correction still supplies the normal ∂_W direction needed for rigidity.

Choose coordinates over K_0 in which $g_1 = du$ and $g_2 = dv$, and write

$$K_1 = \begin{pmatrix} a & b \\ b & c \end{pmatrix}, \quad K_2 = \begin{pmatrix} d & e \\ e & f \end{pmatrix}.$$

For the standard basis

$$H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix},$$

substitution in (5.18) gives

$$(5.43) \quad \begin{aligned} [D_H, D_E] - D_{[E,H]} &= -2v^2\Lambda(u, v)\partial_W, \\ [D_H, D_F] - D_{[F,H]} &= -2u^2\Lambda(u, v)\partial_W, \\ [D_E, D_F] - D_{[F,E]} &= -2uv\Lambda(u, v)\partial_W, \end{aligned}$$

where

$$\Lambda(u, v) = (a + e)u + (b + f)v.$$

This linear form is generically nonzero. When $d_1, d_2 \geq 2$, for example, the two local equations at $z = 0$

$$f_1 = 1 + z_1 + z_1^2, \quad f_2 = 1 + z_2 + z_2^2$$

give $K_1 = \text{diag}(1, 0)$, $K_2 = \text{diag}(0, 1)$, and $\Lambda = u + v$. When $d_1 = 1$ and $d_2 \geq 2$, use instead

$$f_1 = 1 + z_1, \quad f_2 = 1 + z_2 + z_2^2.$$

Then $K_1 = \text{diag}(-1, 0)$, $K_2 = \text{diag}(0, 1)$, and $\Lambda = -u + v \neq 0$. These local Taylor data are realized by homogeneous equations of the prescribed degrees: after choosing the point $[0 : 0 : 1]$, homogenize the displayed terms with the required powers of Z and choose all remaining coefficients generally. The prescribed finite Taylor jet is an affine linear condition on the coefficient space, while smoothness and transversality are nonempty open conditions; a general completion, by Bertini away from the prescribed point and the displayed independent linear terms at that point, therefore lies in the SNC parameter locus. Therefore the specializations above test the actual global family, not only a formal local model. To formulate the generic conclusion without suppressing denominators, let Δ be a common denominator for the entries of G^{-1} , K_1, K_2 in the coordinate ring of $\mathcal{B}^\circ$. Then

$$\Delta\Lambda = \Lambda_1 u + \Lambda_2 v, \quad \Lambda_1, \Lambda_2 \in \mathcal{O}(\mathcal{B}^\circ).$$

The preceding specializations show that the pair (Λ_1, Λ_2) is not identically zero. Because $\mathcal{B}^\circ$ is irreducible, the locus

$$(5.44) \quad \mathcal{B}_\Lambda^\circ = \{\Delta\Lambda_1 \neq 0\} \cup \{\Delta\Lambda_2 \neq 0\}$$

is a nonempty Zariski-open subset, and $\Lambda \neq 0$ in $K_0[u, v]$. After clearing the joint variables z , the assertion that both numerator coefficients vanish identically is a closed condition on the two parameters. Since it is not the whole parameter space, its complement is a nonempty parameter open. Intersecting with the dense smooth transverse locus gives the open subset of $\mathcal{S}_{d_1, d_2}^{\text{SNC}}$ used below.

Proposition 5.5 (Residual rigidity). *Let L/K_0 be a characteristic-zero extension. There is no nonconstant weighted-homogeneous polynomial $P \in L[u, v, W]$ of positive weight whose principal ideal is preserved by D_H, D_E, D_F .*

Proof. The ideal (P) is then preserved by commutators and linear combinations. Equation (5.43) supplies a nonzero derivation $S(u, v)\partial_W$ preserving (P) . If $\deg_W P > 0$, the equality $S\partial_W P = AP$ is impossible by comparison of the W -degrees unless $A = 0$; since the polynomial ring is a domain, $\partial_W P = 0$. The same conclusion is immediate when $\deg_W P = 0$. Thus $P \in L[u, v]$.

The W -parts of the three D_T now disappear, so the full binary $\mathfrak{sl}_2$ -action preserves (P) . In a fixed ordinary degree the quotients are scalars. The two nilpotent operators $v\partial_u$ and $u\partial_v$ can have only the scalar zero, so $vP_u = uP_v = 0$. Therefore P is constant, contradicting its positive weight. $\square$

Output used later. Therefore the generic prime divisor cannot be tangent to every global twisted field in the mixed package: otherwise its principal ideal would be preserved by all three residual directions, contrary to Proposition 5.5. Phase VI differentiates the full first equation on the homogeneous multicone and descends the result directly to the relative Semple tower.

5.6. Phase VI: homogeneous Siu differentiation and the second global equation.

Goal. Differentiate the full relative invariant equation by a regular global slanted field and obtain an independent invariant two-jet differential without first constructing a section on $\mathcal{Z}$.

There is no canonical Lie derivative of a section of an arbitrary line bundle by an arbitrary twisted vector field. Siu's method avoids this ambiguity by working with homogeneous functions and homogeneous derivations before taking the projective quotient; see [19, 20, 21, 6]. In our notation the remaining logic is

$$\begin{aligned} \text{generic span of global cone fields} &\implies \text{rigidity forbids preservation of } (P), \\ &\implies \widehat{V}(\widehat{\omega}_1) \notin (P) \implies \omega_{2,a}. \end{aligned}$$

The prime $\mathcal{Z}$ is used only in the middle implication, as an ideal-theoretic test of independence.

Lemma 5.6 (Homogeneous Siu differentiation). *Let $\widehat{Y} \rightarrow Y$ be a homogeneous multicone with quotient torus G , possibly equipped with an additional finite deck group. Let B and M be line bundles on Y corresponding to characters χ and δ of G . Suppose that a section $\sigma \in H^0(Y, B)$ is represented by a regular deck-invariant semi-invariant function $\widehat{\sigma}$ of character χ , and that a twisted field $V \in H^0(Y, T_Y \otimes M)$ has a deck-equivariant homogeneous lift $\widehat{V}$ of character δ . Then*

$$(5.45) \quad D_V \sigma := \text{desc}(\widehat{V}(\widehat{\sigma})) \in H^0(Y, B \otimes M).$$

If $\widehat{V}'$ is another homogeneous lift of the same twisted field, then

$$(5.46) \quad \widehat{V}'(\widehat{\sigma}) - \widehat{V}(\widehat{\sigma}) = \widehat{h} \widehat{\sigma}$$

for a semi-invariant function $\widehat{h}$ of character δ . Therefore the restriction of $D_V \sigma$ to $(\sigma = 0)$, and in particular whether the derivative cuts a given component of this divisor, is independent of the chosen homogeneous lift.

Proof. By homogeneity, $\widehat{V}$ sends a function of character χ to one of character $\chi\delta$. Deck equivariance preserves deck invariance. Thus $\widehat{V}(\widehat{\sigma})$ descends to the line bundle $B \otimes M$, proving (5.45).

Two lifts of the same quotient field differ vertically. On a homogeneous frame chart their difference is

$$\widehat{V}' - \widehat{V} = \sum_j \widehat{h}_j E_j,$$

where the E_j are the Euler fields of the torus factors and the $\widehat{h}_j$ have character δ . If χ_j is the integer weight of $\widehat{\sigma}$ for the j -th factor, then $E_j(\widehat{\sigma}) = \chi_j \widehat{\sigma}$. Thus

$$(\widehat{V}' - \widehat{V})(\widehat{\sigma}) = \left(\sum_j \chi_j \widehat{h}_j \right) \widehat{\sigma},$$

which is (5.46). This expression is compatible on overlaps because both sides are semi-invariants of character $\chi\delta$. The finite deck group has no infinitesimal vertical direction, and equivariance has already been imposed. Reduction modulo $(\widehat{\sigma})$ proves the last assertion. $\square$

Lemma 5.7 (Generic equation and preservation of the prime ideal). *Let $\mathcal{Z} \subset \mathcal{X}_2$ be the relative integral horizontal divisor supplied by Proposition 4.3, and assume $b_2 > 0$. On the dense incidence chart used in (5.6), after a characteristic-zero algebraic field extension L/K_0 , the affine cone over the generic fiber of $\mathcal{Z}$ is defined by an irreducible weighted-homogeneous polynomial*

$$P \in L[u, v, W]$$

of positive weight. Let $\mathcal{V}$ be a global twisted field from Proposition 5.4. After rationally trivializing its common twist, let $\widetilde{V}$ be the induced derivation. Let

$$\Omega_{\mathcal{V}} = D_{\mathcal{V}} \omega_1$$

be the homogeneous derivative of the full relative equation given by Lemma 5.6 and the cone lift in Proposition 5.4. Then

$$(5.47) \quad \Omega_V|_{\mathcal{Z}} = 0 \text{ at the generic point} \iff \tilde{V}(P) \in (P).$$

Therefore, if every homogeneous derivative vanishes generically on $\mathcal{Z}$, then each of D_H, D_E, D_F preserves the principal prime ideal (P) .

Proof. The divisor $\mathcal{Z}$ is integral. If it were contained in the complement of the chart $f_1 f_2 \neq 0$ or in the Semple boundary, it would equal one of those irreducible boundary divisors. This is impossible by classes: a base boundary has positive class $d_i h$, the second Semple boundary has class $-u_1 + u_2$, while $[\mathcal{Z}] = b_1 u_1 + b_2 u_2 - t h$ with $t > 0$ and $b_1 \geq 2b_2 > 0$. Thus the displayed chart meets the generic point of $\mathcal{Z}$. More generally, $b_2 > 0$ implies that $\mathcal{Z}$ dominates X_1 and thus $\mathbb{P}^2$; it therefore cannot be contained in the pullback of the determinant divisor $g_1 \wedge g_2 = 0$, an affine transition divisor, or a chosen Semple-chart boundary. The coefficient-chart linear forms are units at the generic parameter after the corresponding affine parameter chart is chosen. This accounts for every denominator listed in (5.8). The invariant affine two-jet cone over the generic incidence point has coordinate ring $L[u, v, W]$ by (5.6). The pullback of the generic point of $\mathcal{Z}$ is a height-one prime in this unique factorization domain. It is therefore generated by an irreducible polynomial P . The prime is stable under the weighted $\mathbb{G}_m$ action. Since the units of $L[u, v, W]$ are the elements of L^* , its generator is a semi-invariant for this action and may be chosen weighted-homogeneous. Its weight is positive because all three variables have positive weight, while a nonzero weight-zero element of $L[u, v, W]$ is a unit and cannot generate a height-one prime. Every derivation of K_0 extends uniquely through the algebraic extension L/K_0 in characteristic zero, so all the residual fields act on this same ring.

At the generic point, choose rational homogeneous frames of $\mathcal{L}$, the vertical line, and the common twisting line. Up to a unit e , the homogeneous representative of the saturated equation is eP , and thus

$$\hat{\omega}_1 = \hat{\gamma}^{b_1} eP.$$

The generic point of $\mathcal{Z}$ does not lie on Γ_2 , so $\hat{\gamma}$ is a unit there. The Leibniz rule gives

$$(5.48) \quad \hat{\mathcal{V}}(\hat{\omega}_1) \equiv \hat{\gamma}^{b_1} e \tilde{V}(P) \pmod{(P)}.$$

This proves (5.47). A change of homogeneous lift alters the left-hand side by a multiple of $\hat{\omega}_1$ by Lemma 5.6, so the reduction is independent of that choice as well as of the rational frames. The set of derivations preserving a fixed ideal is a vector space over the generic incidence field. Proposition 5.4 says that the generic span of the global twisted fields contains D_H, D_E, D_F , which proves the last assertion. $\square$

Proposition 5.8 ($\mathcal{O}(3)$ slanted derivative). *After shrinking the parameter open in Proposition 4.3, write the irreducible reduced horizontal divisor of $\omega_{1,a}$ as*

$$Z_a \sim b_1 u_1 + b_2 u_2 - t h, \quad b_1 \geq 2b_2 > 0, \quad m = b_1 + b_2.$$

After a further nonempty Zariski-open shrinking, for every remaining a there is

$$\omega_{2,a} \in H^0(\mathbb{P}^2, E_{2,m} T_{\mathbb{P}^2}^*(\log D_a) \otimes \mathcal{O}_{\mathbb{P}^2}(-t+3))$$

such that the common zero set of $\omega_{1,a}$ and $\omega_{2,a}$ has codimension at least two away from $\Gamma_{2,a}$.

Proof. The open-set bookkeeping has three steps: the factor family and the global field package are defined on the first open; the curvature form Λ is nonzero on the second; and the last shrinking below makes one homogeneous derivative nonzero along Z_a on every remaining fiber. Each condition defines a nonempty Zariski-open subset, so their finite intersection is nonempty.

Use the relative notation of Proposition 4.3. On the parameter open under consideration set

$$\mathcal{B} = \mathcal{O}_{\mathcal{X}_2}(m) \otimes \Pi^* \mathrm{pr}_1^* \mathcal{O}_{\mathbb{P}^2}(-t), \quad \omega_1 = \gamma^{b_1} f \in H^0(\mathcal{X}_2, \mathcal{B}),$$

and

$$\mathcal{A} = \mathcal{O}_S(1, 1)|_{\mathcal{U}}, \quad \mathcal{M} = \Pi^* \operatorname{pr}_1^* \mathcal{O}_{\mathbb{P}^2}(3) \otimes p_{\mathcal{U}}^* \mathcal{A}.$$

Every global twisted field $\mathcal{V}$ in Proposition 5.4 is a section of $T_{\mathcal{X}_2} \otimes \mathcal{M}$ and comes with the homogeneous lift required by Lemma 5.6. It may have a component in the parameter directions because both ω_1 and $\mathcal{V}$ live on the total relative tower. Direct homogeneous differentiation gives

$$(5.49) \quad \Omega_{\mathcal{V}} := D_{\mathcal{V}} \omega_1 \in H^0(\mathcal{X}_2, \mathcal{B} \otimes \mathcal{M}).$$

If all these sections vanished along the generic point of $\mathcal{Z}$, then Lemma 5.7 would give a positive-weight irreducible $P \in L[u, v, W]$ whose ideal is preserved by D_H, D_E, D_F . This contradicts Proposition 5.5. Thus some global twisted field $\mathcal{V}$ satisfies

$$(5.50) \quad \Omega_{\mathcal{V}}|_{\mathcal{Z}} \neq 0 \quad \text{at the generic point.}$$

Put $q = p_{\mathcal{U}}|_{\mathcal{Z}}$. After shrinking $\mathcal{U}$, cohomology and base change for the proper map q identifies the fiber of

$$q_*((\mathcal{B} \otimes \mathcal{M})|_{\mathcal{Z}})$$

with the corresponding space of sections on Z_a . The section in (5.50) is nonzero on the generic fiber, so the locus on which its restriction is the zero section is a proper closed subset. Shrinking once more, its restriction to every remaining fiber is nonzero. Trivializing the one-dimensional factor $\mathcal{A}|_a$ by a nonzero scalar, the fiber restriction of (5.49) becomes

$$(5.51) \quad 0 \neq \Omega_{2,a} \in H^0(X_{2,a}, \mathcal{O}_{X_{2,a}}(m) \otimes \pi_{2,0}^* \mathcal{O}_{\mathbb{P}^2}(-t+3)), \quad \Omega_{2,a}|_{Z_a} \neq 0.$$

The direct-image formula (2.3) identifies $\Omega_{2,a}$ with the required invariant jet differential $\omega_{2,a}$. Outside $\Gamma_{2,a}$ the zero divisor of $\omega_{1,a} = \gamma_a^{b_1} s_a$ has the single prime component Z_a . The last inequality in (5.51) says exactly that Z_a is not a component of the zero divisor of $\omega_{2,a}$. Thus the two equations have no common divisorial component away from $\Gamma_{2,a}$, and their common zero set there has codimension at least two. $\square$

Corollary 5.9 ($\mathcal{O}(3)$ slanted-field alternative). *Let $f : \mathbb{C} \rightarrow \mathbb{P}^2$ be algebraically nondegenerate and suppose $t > 3$ and $f^* \omega_{1,a} \equiv 0$. Then either $f^* \omega_{2,a} \not\equiv 0$ for the differential in Proposition 5.8, or the first logarithmic lift $f_{[1]}$ is algebraically degenerate.*

Proof. Assume that both pullbacks vanish, and let $Y = \overline{f_{[2]}(\mathbb{C})}^{\text{Zar}} \subset X_2$. By Lemma 2.4, Y is not contained in Γ_2 . Thus $Y \setminus \Gamma_2$ is dense in Y and lies in the codimension-two common zero set of $\omega_{1,a}$ and $\omega_{2,a}$. Since $\dim X_2 = 4$, it follows that $\dim Y \leq 2$. Therefore $\pi_{2,1}(Y)$ is a proper algebraic subset of the threefold X_1 and contains the image of $f_{[1]}$. $\square$

Remark 5.10 (Output of the $\mathcal{O}(3)$ section). On one nonempty parameter open, the selected factor $\omega_{1,a}$ has an independent companion

$$\omega_{2,a} \in H^0(\mathbb{P}^2, E_{2,m} T_{\mathbb{P}^2}^*(\log D_a) \otimes \mathcal{O}_{\mathbb{P}^2}(-t+3)).$$

Away from the vertical divisor their common zero set has codimension at least two. This is the only output from the six-phase construction used in the Second Main Theorem argument.

6. CASE ANALYSIS AND PROOFS OF THE MAIN RESULTS

The argument now follows a fixed decision tree. If the chosen factor does not vanish on the lifted curve, its original twist gives the estimate. If it vanishes, the case $b_2 = 0$ is handled on X_1 ; for $b_2 > 0$, a pole-three derivative treats $t > 3$, while the key vanishing lemma and the zero-locus theorem treat $t \leq 3$.

The coefficient contributed by each analytic branch is summarized here; the endpoint subsections below replace the coarse differentiated bound by the sharper integral estimates when available.

branch	coefficient	reason
$f^*\omega_1 \neq 0$	$m/t \leq \rho_0^{-1}$	initial slope
$t > 3$, derivative nonzero	$m/(t-3) \leq 4\rho_0^{-1}$	$t/(t-3) \leq 4$
zero-locus evaluation nonzero	$3/\tau$	weight $3p$, twist $p\tau$
$f_{[1]}$ algebraically degenerate	$1/\kappa$	McQuillan

For one of the six boundary pairs in Lemma 7.1, let $\mathcal{U}_{\text{key}} = \mathcal{U}_{\text{van}}(d_1, d_2)$; for every other degree pair in our range, put $\mathcal{U}_{\text{key}} = \mathcal{S}_{d_1, d_2}^{\text{nc}}$. Let $\mathcal{U}_0$ be the intersection of $\mathcal{U}_{\text{key}}$ with the parameter open given by Proposition 4.3. It is nonempty because the parameter space is irreducible. Work first over $a \in \mathcal{U}_0$, and put $D = D_a$ and $\omega_1 = \omega_{1,a}$. Let $f : \mathbb{C} \rightarrow \mathbb{P}^2$ be algebraically nondegenerate and suppose $f(\mathbb{C}) \not\subset D$. If $f^*\omega_1 \neq 0$, Theorem 2.5 and $m/t \leq \rho_0^{-1}$ give the required estimate directly. It remains to treat the vanishing case

$$(6.1) \quad f^*\omega_1 \equiv 0.$$

Write the selected horizontal component as

$$Z \sim b_1 u_1 + b_2 u_2 - th.$$

If $b_2 = 0$, then $Z = \pi_{2,1}^{-1}(Z_1)$ for a proper divisor $Z_1 \subset X_1$. By Lemma 2.4, the regular lift is not contained in Γ_2 . Thus (6.1) implies

$$f_{[1]}(\mathbb{C}) \subset Z_1,$$

so $f_{[1]}$ is algebraically degenerate and Theorem 2.6 applies. This branch uses no slanted field. Thus in the two cases below we may assume $b_2 > 0$. Proposition 5.8, applied after restricting the factor family to $\mathcal{U}_0$, supplies a nonempty open subset $\mathcal{U}_1 \subset \mathcal{U}_0$ on which the chosen differential and its slanted derivative are defined simultaneously. In the remainder of this section we work over $a \in \mathcal{U}_1$.

6.1. Case 1: $t > 3$. Corollary 5.9 supplies an independent

$$\omega_{2,a} \in H^0(\mathbb{P}^2, E_{2,m} \bar{T}^* \otimes \mathcal{O}(-t+3)).$$

If $f^*\omega_{2,a} \neq 0$, Theorem 2.5 gives

$$T_f(r) \leq \frac{m}{t-3} N_f^{[1]}(r, D) + o(T_f(r)) \parallel.$$

Since $m/t \leq \rho_0^{-1}$ and $t \geq 4$,

$$(6.2) \quad \frac{m}{t-3} \leq \frac{1}{\rho_0} \frac{t}{t-3} \leq \frac{4}{\rho_0}.$$

If $f^*\omega_{2,a} \equiv 0$, Corollary 5.9 makes $f_{[1]}$ algebraically degenerate, and Theorem 2.6 applies.

6.2. Case 2: $t \leq 3$. Since $b_2 > 0$ and $b_1 \geq 2b_2$, one has $m \geq 3$. If $\rho_{\text{DEG}} > 1$, then automatically $t/m \leq 1 < \rho_{\text{DEG}}$. Otherwise the degree pair is one of the six boundary pairs, and Lemma 7.1 implies that the existing nonzero ω_1 cannot have $t/m \geq \rho_{\text{DEG}}$. Thus in every case

$$(6.3) \quad \rho = \frac{t}{m} < \rho_{\text{DEG}}(d_1, d_2).$$

Theorem 2.11 applies. For every rational $0 < \tau < \min\{3, \tau_1(t/m)\}$ it gives either

$$T_f(r) \leq \frac{3}{\tau} N_f^{[1]}(r, D) + o(T_f(r)) \parallel$$

or the algebraic degeneracy of $f_{[1]}$, followed by (2.9).

6.3. The effective constant. The discrete gap below the zero-locus threshold is explicit. Set

$$(6.4) \quad \rho_* = \max_{1 \leq q \leq 3} \frac{q}{\max\{3, \lfloor q/\rho_{\text{DEG}} \rfloor + 1\}}, \quad \tau_* = \min\{3, \tau_1(\rho_*)\}.$$

For integers $1 \leq t \leq 3$, $m \geq 3$ satisfying $t/m < \rho_{\text{DEG}}$, one has $t/m \leq \rho_*$. The function $3/\tau_1(\rho)$ is increasing on the relevant interval. Therefore any integer strictly larger than

$$(6.5) \quad \max \left\{ \rho_0^{-1}, \frac{4}{\rho_0}, \frac{3}{\tau_*}, \frac{1}{\kappa} \right\}$$

is an admissible $\mathcal{A}_{d_1, d_2}$ in Theorem 1.1. In the order displayed, the four entries account for the original section, its pole-three derivative, the zero-locus section, and McQuillan's branch.

6.4. A sharper split for a conic and a quintic. For $(d_1, d_2) = (2, 5)$ one has

$$\kappa = 4, \quad \bar{c}_1^2 = 16, \quad \bar{c}_2 = 21, \quad \rho_{\text{DEG}} = \frac{19}{48}.$$

The extra case $(m, t) = (8, 3)$ in $\mathfrak{F}_{45}(2, 5)$ changes the useful split from $3/8$ to $1/3$. Put $\rho = t/m$ and assume $b_2 > 0$.

If $\rho \leq 1/3$, then $\rho < 19/48$, so Theorem 2.11 applies. Since $3/\tau_1(\rho)$ is increasing on this interval, it is enough to evaluate the endpoint. Formula (2.13) gives

$$(6.6) \quad \tau_1(1/3) = \frac{45 - \sqrt{1989}}{6}, \quad \frac{3}{\tau_1(1/3)} = \frac{45 + \sqrt{1989}}{2} < 45.$$

The last inequality follows from $1989 < 45^2$. For the actual ratio $\rho \leq 1/3$, one may therefore choose a rational number $1/15 < \tau < \tau_1(\rho)$; the zero-locus coefficient $3/\tau$ is then strictly smaller than 45.

Suppose instead that $\rho > 1/3$. If $t \leq 3$, then $m < 3t$ and thus $(m, t) \in \mathfrak{F}_{45}(2, 5)$, contradicting the nonzero choice of ω_1 and Lemma 7.1. Thus $t \geq 4$, and the pole-three derivative is available. Integrality gives $m \leq 3t - 1$, so

$$(6.7) \quad \frac{m}{t-3} \leq \frac{3t-1}{t-3} = 3 + \frac{8}{t-3} \leq 11.$$

If the differentiated equation vanishes as well, the first lift is algebraically degenerate and McQuillan's estimate applies.

Finally,

$$(6.8) \quad \rho_{\text{RR}}(2, 5) = \frac{2}{9} \left(8 - \sqrt{\frac{455}{8}} \right) > \frac{1}{10};$$

indeed, after squaring, the last inequality is $22750 < 22801$. Choose a rational number

$$\frac{1}{10} < \rho_0 < \rho_{\text{RR}}(2, 5).$$

The direct branch then has coefficient $\rho_0^{-1} < 10$, (6.6) bounds the zero-locus branch by a number strictly smaller than 45, and (6.7) bounds the differentiated branch by 11. McQuillan's coefficient is $1/4$. Therefore the integer 45 covers every branch for a conic and a quintic.

6.5. The explicit line-octic constant. For $(d_1, d_2) = (1, 8)$ one has $\kappa = 6$ and $\rho_{\text{DEG}} = 3/8$. Put $\rho = t/m$ and assume $b_2 > 0$.

Suppose first that $\rho \leq 1/3$. Then $\rho < \rho_{\text{DEG}}$, so the zero-locus theorem applies. Since $3/\tau_1(\rho)$ is increasing on this interval, substitution of the endpoint into (2.13) gives

$$(6.9) \quad \tau_1(1/3) = \frac{23 - 5\sqrt{21}}{2}, \quad \frac{3}{\tau_1(1/3)} = \frac{3}{2}(23 + 5\sqrt{21}) < 69.$$

For the actual ratio one may therefore choose a rational number $3/69 < \tau < \tau_1(\rho)$, and the zero-locus coefficient is strictly smaller than 69.

Suppose now that $\rho > 1/3$. If $t \leq 3$, then $m < 3t$. For $t = 1$ there is no genuine second-level weight $m \geq 3$. For $t = 2$ the possibilities are $m = 3, 4, 5$, and for $t = 3$ they are $m = 3, \dots, 8$. The cases with $m = 3, 4, 5$ vanish already at twist 2, thus also at twist 3 by multiplication with a nonzero linear form; the cases $m = 6, 7, 8$ vanish at twist 3. These are exactly the line–octic assertions in Lemma 7.1. Thus $t \geq 4$. Integrality gives $m \leq 3t - 1$, and therefore

$$(6.10) \quad \frac{m}{t-3} \leq \frac{3t-1}{t-3} = 3 + \frac{8}{t-3} \leq 11.$$

Finally,

$$(6.11) \quad \rho_{\text{RR}}(1, 8) = \frac{1}{3} \left(8 - \sqrt{\frac{119}{2}} \right) > \frac{2}{23};$$

indeed, after squaring, it reduces to $62951 < 63368$. Choose a rational number

$$\frac{2}{23} < \rho_0 < \rho_{\text{RR}}(1, 8).$$

The direct branch has coefficient $\rho_0^{-1} < 23/2$, the differentiated branch is bounded by (6.10), the zero-locus branch is bounded by (6.9), and McQuillan’s coefficient is $1/6$. Thus the integer 69 covers every branch for a line and an octic.

6.6. A sharper ratio split for two cubics. For $(d_1, d_2) = (3, 3)$, the stronger part of Lemma 7.1 improves the preceding general estimate. Put $\rho = t/m$ and keep the assumption $b_2 > 0$.

Suppose first that $\rho \leq 1/5$. Since $1/5 < \rho_{\text{DEG}}(3, 3) = 1/4$, Theorem 2.11 applies, regardless of the absolute size of t . The function $3/\tau_1(\rho)$ is increasing on $[0, 1/4)$, and

$$(6.12) \quad \tau_1(1/5) = \frac{57 - \sqrt{3189}}{10}, \quad \frac{3}{\tau_1(1/5)} = \frac{57 + \sqrt{3189}}{2} \approx 56.73561582 < 57.$$

Thus one can choose a rational number $3/57 < \tau < \tau_1(\rho)$, and the zero-locus branch has coefficient strictly smaller than 57.

Suppose next that $\rho > 1/5$. If $t \leq 3$, integrality gives $m \leq 5t - 1$, contradicting the cubic finite vanishing in Lemma 7.1. Thus $t \geq 4$, so the mixed $\mathcal{O}(3)$ derivative is available. Since $m \leq 5t - 1$,

$$(6.13) \quad \frac{m}{t-3} \leq \frac{5t-1}{t-3} \leq 19.$$

If the differentiated equation also vanishes on the lifted curve, Corollary 5.9 makes the first lift algebraically degenerate, and Theorem 2.6 applies. Therefore the two branches under $f^*\omega_1 \equiv 0$ and $b_2 > 0$ are bounded by 57; the direct and $b_2 = 0$ branches are included in the final assembly.

6.7. Comparison of the three endpoint constants. The preceding calculations can be read from one table. Every entry is the coefficient of $N_f^{[1]}(r, D)$ in the corresponding branch.

pair	ratio split	first	zero locus	derivative	McQuillan	final
(3, 3)	1/5	< 16	< 57	≤ 19	1/3	57
(2, 5)	1/3	< 10	< 45	≤ 11	1/4	45
(1, 8)	1/3	$< 23/2$	< 69	≤ 11	1/6	69

The last column is the smallest displayed integer that covers all four branches. The proof below only assembles these already proved estimates.

6.8. Proofs of the main statements. We now assemble the preceding alternatives. No new numerical or geometric input is needed at this stage.

Proof of Theorem 1.1. The fixed pair (M, T) of Proposition 4.1 works throughout $\mathcal{S}_{d_1, d_2}^{\text{nc}}$. Intersect the parameter open of Proposition 4.3 with $\mathcal{U}_{\text{key}}$ and call the resulting nonempty open set $\mathcal{U}_0$. The integers b_1, b_2, m, t are fixed on this family. If $b_2 = 0$, put $\mathcal{U}_{\text{SMT}}(d_1, d_2) = \mathcal{U}_0$. If $b_2 > 0$, apply Proposition 5.8 to the restricted family and let $\mathcal{U}_{\text{SMT}}(d_1, d_2) \subset \mathcal{U}_0$ be the nonempty open set it provides.

Fix $a \in \mathcal{U}_{\text{SMT}}(d_1, d_2)$ and choose $\omega_{1,a}$ as in Lemma 4.2 and Proposition 4.3. If $f^*\omega_{1,a} \neq 0$, the jet Second Main Theorem gives the estimate with constant at most ρ_0^{-1} . Under (6.1), the branch $b_2 = 0$ makes $f_{[1]}$ algebraically degenerate and is handled directly by Theorem 2.6. Suppose therefore that $b_2 > 0$. For a degree pair other than $(3, 3)$, Case 1 applies when $t > 3$ and gives (6.2) or McQuillan's stronger estimate. When $t \leq 3$, Case 2 and (6.4) give the zero-locus estimate or McQuillan's estimate. Thus any integer strictly larger than (6.5) is admissible.

For $(d_1, d_2) = (1, 8)$ choose ρ_0 as in (6.11); the preceding line–octic calculation shows that the integer 69 is admissible.

For $(d_1, d_2) = (2, 5)$ choose ρ_0 as in (6.8); use the sharper $1/3$ dichotomy of the preceding conic–quintic subsection in place of the coarser general split. The direct, zero-locus, differentiated, and algebraically degenerate branches are then bounded respectively by 10, 45, 11, and $1/4$, with strict inequality where needed. Thus the integer 45 is admissible.

For $(d_1, d_2) = (3, 3)$ choose $1/16 < \rho_0 < \rho_{\text{RR}}(3, 3)$, so the direct branch has coefficient less than 16. Under (6.1), use instead the sharper ratio dichotomy of the preceding subsection: equation (6.12) gives a coefficient less than 57 when $t/m \leq 1/5$, and (6.13) gives a coefficient at most 19 when $t/m > 1/5$. Thus the integer 57 works for two cubics. This proves (1.1). $\square$

Proof of Theorem 1.2. The proof rests on one elementary observation. Let $q, s > 0$ and let

$$\eta \in H^0(\mathbb{P}^2, E_{2,q}T_{\mathbb{P}^2}^*(\log D_a) \otimes \mathcal{O}_{\mathbb{P}^2}(-s)).$$

For a curve $\Gamma \not\subset D_a$ with normalization ν and $B_\nu = \text{Supp } \nu^*D_a$, functoriality of logarithmic jets identifies the evaluation of η along the canonical second logarithmic lift $\nu_{[2]}$ of ν with a section of

$$E_{2,q}T_{\tilde{\Gamma}}^*(\log B_\nu) \otimes \nu^*\mathcal{O}_{\mathbb{P}^2}(-s) \simeq (K_{\tilde{\Gamma}} + B_\nu)^{\otimes q} \otimes \nu^*\mathcal{O}_{\mathbb{P}^2}(-s).$$

Its degree is

$$q(2g(\tilde{\Gamma}) - 2 + \#B_\nu) - s \deg \Gamma < 0$$

under the hypothesis of the theorem. The evaluation therefore vanishes identically. We apply this observation to the equations constructed above.

Fix the relative factor $\mathcal{Z} \subset \mathcal{X}_2$ and the integers (m, t, b_1, b_2) supplied by Proposition 4.3, after intersecting its parameter open with the open set used in the proof of Theorem 1.1; write $\mathcal{X}_2 \rightarrow \mathcal{X}_1$ for the last projection of the relative Semple tower. We construct a relative closed set in the first Semple level in the same three cases as in that proof.

First case: $b_2 = 0$. The divisor $\mathcal{Z}$ is pulled back, after shrinking the base, from a relative divisor $\mathcal{R} \subset \mathcal{X}_1$. The preceding vanishing applied to $\omega_{1,a}$ places the second lift in the zero divisor of $\omega_{1,a}$. Since a regular second lift is not contained in Γ_2 by Lemma 2.4, it lies in $\mathcal{Z}_a$; hence its first lift is contained in $\mathcal{R}_a$.

Second case: $b_2 > 0$ and $t > 3$. Proposition 5.8 provides the second relative equation ω_2 , whose base twist is $-(t-3) < 0$. In $\mathcal{X}_2$, set

$$\mathcal{Y} := \overline{((\omega_1 = 0) \cap (\omega_2 = 0))} \setminus \Gamma_2.$$

After shrinking the base, every fiber of $\mathcal{Y}$ has dimension at most two. Both equations vanish along $\nu_{[2]}$ by the degree observation. Since $\nu_{[2]}(\tilde{\Gamma})$ is not contained in Γ_2 , its dense open part outside Γ_2 , and therefore its closure, lies in $\mathcal{Y}_a$.

Third case: $b_2 > 0$ and $t \leq 3$. The argument of Case 2 above gives $t/m < \rho_{\text{DEG}}$. Choose a rational number $0 < \tau < \min\{3, \tau_1(t/m)\}$. The proof of Theorem 2.11, first on the generic fiber and then by cohomology and base change after shrinking the parameter space, gives a fixed sufficiently divisible integer p , with $p\tau \in \mathbb{Z}$, and a relative section

$$\Sigma \in H^0(\mathcal{Z}, (\mathcal{O}_{\mathcal{X}_2}(2p, p) \otimes \Pi^* \text{pr}_1^* \mathcal{O}_{\mathbb{P}^2}(-p\tau))|_{\mathcal{Z}})$$

whose restriction to every fiber is nonzero. Put $\mathcal{Y} = \text{div}_{\mathcal{Z}}(\Sigma)$. Its fibers have dimension at most two. The first equation vanishes along $\nu_{[2]}$, so Lemma 2.4 gives $\nu_{[2]}(\tilde{\Gamma}) \subset \mathcal{Z}_a$. On this lift, the tautological evaluation of Lemma 2.9 is a holomorphic section of

$$K_{\tilde{\Gamma}}^{3p}(3pB_{\nu}) \otimes \nu^* \mathcal{O}_{\mathbb{P}^2}(-p\tau).$$

Its degree is

$$3p(2g(\tilde{\Gamma}) - 2 + \#B_{\nu}) - p\tau \deg \Gamma < 0,$$

so the evaluation vanishes. The tautological derivative is nonzero at the generic point of $\tilde{\Gamma}$; consequently $\nu_{[2]}(\tilde{\Gamma}) \subset \mathcal{Y}_a$.

In the last two cases, let $\mathcal{R}$ be the scheme-theoretic image of $\mathcal{Y}$ under the proper projection $\mathcal{X}_2 \rightarrow \mathcal{X}_1$. It is closed and projective over the parameter space. After one final nonempty Zariski-open shrinking, upper semicontinuity of fiber dimension gives $\dim \mathcal{R}_a \leq 2$ for every remaining a . The preceding construction gives $\nu_{[1]}(\tilde{\Gamma}) \subset \mathcal{R}_a$ in all three cases, proving the theorem. $\square$

Proof of Corollary 1.3. Fix

$$a \in \mathcal{U}_{\text{SMT}}(d_1, d_2) \cap \mathcal{S}_{\text{Chen}}(d_1, d_2)$$

and let

$$f : \mathbb{C} \longrightarrow \mathbb{P}^2 \setminus D_a.$$

If f is algebraically nondegenerate, then $N_f^{[1]}(r, D_a) = 0$, and Theorem 1.1 gives $T_f(r) \leq o(T_f(r))$, impossible for a nonconstant curve. If f is algebraically degenerate, Corollary 2.15 makes it constant. Thus the complement is Brody hyperbolic.

It remains to pass from Brody hyperbolicity to the stated metric statement. The one-dimensional boundary strata are

$$C_1 \setminus C_2, \quad C_2 \setminus C_1.$$

Since the curves meet transversally in $d_1 d_2$ points, their logarithmic Euler characteristics satisfy

$$2g(C_i) - 2 + d_1 d_2 > 0 \quad (i = 1, 2)$$

throughout the stated degree range. Thus both punctured curves are Brody hyperbolic, and the zero-dimensional stratum $C_1 \cap C_2$ is finite. We use the following form of Green's criterion: if X is projective, $D = \bigcup_{i=1}^r D_i$ is a simple normal crossing divisor, and both $X \setminus D$ and every open stratum

$$\bigcap_{i \in I} D_i \setminus \bigcup_{j \notin I} D_j \quad (\emptyset \neq I \subset \{1, \dots, r\})$$

are Brody hyperbolic, then $X \setminus D$ is hyperbolically embedded in X ; see [11] and the precise restatement [12, Theorem 4.1]. The strata in the present case are exactly the complement, the two punctured curves above, and the finite set $C_1 \cap C_2$. The criterion therefore shows that the complement is hyperbolically embedded in $\mathbb{P}^2$; in particular it is Kobayashi hyperbolic. The set $\mathcal{U}_{\text{SMT}}(d_1, d_2) \cap \mathcal{S}_{\text{Chen}}(d_1, d_2)$ is very general by the discussion following Corollary 1.3. $\square$

At this point the geometric and analytic proof is complete, subject only to the finite Key Vanishing Lemma used above. Section 7 reduces that lemma to integral matrices, and Section 8 proves their exact ranks. No new geometric branch is introduced in those two sections.

7. FINITE KEY VANISHING: REDUCTION TO INTEGRAL MATRICES

We first state the exact finite result. The threshold ρ_{DEG} was computed in (3.15). For the five boundary pairs other than $(3, 3)$, put

$$\mathfrak{F}_{\text{bd}}(d_1, d_2) = \left\{ (m, t) \in \mathbb{Z}_{\geq 3} \times \{1, 2, 3\} : \frac{t}{m} \geq \rho_{\text{DEG}}(d_1, d_2) \right\}.$$

For a conic and a quintic, we also need the endpoint case $(8, 3)$:

$$\mathfrak{F}_{45}(2, 5) = \left\{ (m, t) \in \mathbb{Z}_{\geq 3} \times \{1, 2, 3\} : \frac{t}{m} > \frac{1}{3} \right\} = \mathfrak{F}_{\text{bd}}(2, 5) \cup \{(8, 3)\}.$$

For two cubics, put

$$\mathfrak{F}_{57}(3, 3) = \{(m, t) \in \mathbb{Z}_{\geq 3} \times \{1, 2, 3\} : 3 \leq m \leq 5t - 1\}.$$

Lemma 7.1 (Finite key vanishing). *For each pair*

$$(d_1, d_2) \in \{(3, 3), (3, 4), (2, 6), (2, 5), (1, 9), (1, 8)\}$$

there is a nonempty Zariski-open subset $\mathcal{U}_{\text{van}}(d_1, d_2) \subset \mathcal{S}_{d_1, d_2}^{\text{snc}}$ such that, for every parameter a in this subset,

$$(7.1) \quad H^0(\mathbb{P}^2, E_{2,m} T_{\mathbb{P}^2}^*(\log D_a) \otimes \mathcal{O}_{\mathbb{P}^2}(-t)) = 0$$

for every $(m, t) \in \mathfrak{F}_{57}(3, 3)$ in the cubic–cubic case, for every $(m, t) \in \mathfrak{F}_{45}(2, 5)$ in the conic–quintic case, and for every $(m, t) \in \mathfrak{F}_{\text{bd}}(d_1, d_2)$ in the other four cases.

After keeping only the least twist for each m , the required cases are as follows.

(d_1, d_2)	ρ_{DEG}	minimal pairs (m, t)
$(3, 3)$	1/4	$3 \leq m \leq 4$ for $t = 1$; $5 \leq m \leq 9$ for $t = 2$; $10 \leq m \leq 14$ for $t = 3$
$(3, 4)$	37/48	$(3, 3)$
$(2, 6)$	23/30	$(3, 3)$
$(2, 5)$	19/48	$(3, 2), (4, 2), (5, 2), (6, 3), (7, 3), (8, 3)$
$(1, 9)$	61/84	$(3, 3), (4, 3)$
$(1, 8)$	3/8	$(3, 2), (4, 2), (5, 2), (6, 3), (7, 3), (8, 3)$

The proof follows the chain

$$\begin{aligned} \text{local coefficients} &\implies \text{transition and divisibility equations} \implies \text{integral matrices} \\ &\implies \text{symmetry blocks} \implies \text{exact ranks.} \end{aligned}$$

This section proves the first four arrows and identifies global sections with matrix kernels. Section 8 proves that all these kernels are zero.

The main symbols used in the reduction are listed here.

symbol	meaning
U_0	the ordinary affine chart where candidate sections are written
U_X	the mixed logarithmic chart where regularity and twisting are tested
a, b	the two local equations on U_0
q, Δ_X	the mixed logarithmic one-form and its frame determinant
W_{xy}, W_X	the ordinary and mixed invariant Wronskians
V_m	the finite space of ordinary-chart coefficients
$\Phi_{m,t}$	the linear map formed by all transition and divisibility equations

We now prove Lemma 7.1. It is enough to establish the vanishing at one transverse pair, because upper semicontinuity will then give the required nonempty parameter open. Multiplication by a nonzero homogeneous polynomial allows us to keep only the least twist for each m ; these minimal pairs are listed in the table above. The guiding principle is to generate candidate two-jet differentials on one large ordinary chart, test their logarithmic regularity on one mixed logarithmic chart, and impose the available symmetries before constructing any

matrix. This chart-and-divisibility strategy starts from the one-component normal form in [8, Proposition 3.1 and Section 3]. The argument below is not a restatement of that calculation: two defining polynomials produce two residue directions, their projective weights must be balanced when $d_1 \neq d_2$, and the overlap equations are coupled. For $(3, 3)$ we impose the $(\mu_3)^2$ action and coordinate exchange before constructing the matrices. The pairs $(2, e)$, $e = 5, 6$, have a different cyclic action with coordinate exchange. The line pairs $(1, e)$, $e = 8, 9$, keep the cyclic action but not the involution, while $(3, 4)$ is small enough to run as one unsplit block. The cubic organization is related to the symmetry method developed in [10]. That citation concerns the organizational idea of splitting a finite calculation by symmetry; the two-component normal form, divisibility criterion, matrices, and verification program used here are derived independently in this section. All operations are performed with integral polynomials. Every characteristic-zero conclusion is certified by full column rank at an explicitly recorded good prime; a second prime is used as an independent reproducibility check whenever practical.

The proof has four stages. We first choose explicit SNC test pairs. We then derive a finite ambient normal form and translate regularity on the mixed chart into divisibility equations. Next we decompose these equations by the available finite symmetries. Finally, exact modular row reduction certifies that every resulting block has zero kernel.

Mathematical reduction. For every test pair and every relevant (m, t) , the ordinary-chart normal form defines a finite coefficient space V_m , and all forbidden overlap and twisting remainders define a linear map

$$\Phi_{m,t}: V_m \longrightarrow W_{m,t}.$$

Proposition 7.8 proves the natural identification

$$H^0(\mathbb{P}^2, E_{2,m} T_{\mathbb{P}^2}^*(\log D) \otimes \mathcal{O}_{\mathbb{P}^2}(-t)) \simeq \ker \Phi_{m,t}.$$

The symmetry decompositions identify this kernel with the direct sum of the kernels of the integral block matrices. Thus the geometric vanishing and the exact rank calculation are connected by a proved equivalence, not by a numerical heuristic.

7.1. Degree-uniform test pairs. We use the following integral pairs; in each row, $A = 0$ and $B = 0$ are the two components of the test divisor:

$$(7.2) \quad \begin{array}{c|cc} (d_1, d_2) & A & B \\ \hline (3, 3) & X^3 + 2Y^3 + Z^3 & 2X^3 + Y^3 + Z^3 \\ (3, 4) & X^3 + Y^3 + Z^3 & X^4 + Y^4 + Z^4 \\ (2, e), e = 5, 6 & XY + Z^2 & X^e + Y^e + Z^e \\ (1, 8) & Y & X^8 + Y^8 + Z^8 + X^4 Y^4 \\ (1, 9) & Y & X^9 + Y^9 + Z^9 + X^3 Y^3 Z^3 \end{array}$$

Each row gives an SNC divisor and remains SNC after adjoining the line $L = \{Z = 0\}$. The elementary gradient and transversality checks are recorded in Appendix A; no computational rank assertion is used in those checks.

7.2. A special symmetric pair of cubic curves. We work with the following special transverse pair:

$$(7.3) \quad A = X^3 + 2Y^3 + Z^3, \quad B = 2X^3 + Y^3 + Z^3.$$

Let

$$D = \{AB = 0\}.$$

Both cubics are smooth, and the two curves meet transversally. In fact, Appendix A verifies the stronger statement that $A + B + L$ is SNC.

The pair has two useful symmetries. First, the involution

$$(7.4) \quad \sigma(X, Y, Z) = (Y, X, Z)$$

satisfies

$$\sigma^* A = B, \quad \sigma^* B = A.$$

Second, for $\alpha^3 = \beta^3 = 1$, the diagonal transformation

$$(7.5) \quad [X : Y : Z] \mapsto [\alpha X : \beta Y : Z]$$

fixes both A and B . Thus the finite-dimensional spaces appearing in the computation decompose into characters of $(\mu_3)^2$, and the involution σ exchanges the two character indices.

For the pairs $(2, e)$ in (7.2), the corresponding symmetry is

$$[X : Y : Z] \mapsto [\zeta X : \zeta^{-1} Y : Z], \quad \zeta^e = 1,$$

together with the involution $X \leftrightarrow Y$. The monomial

$$x^r y^s (x')^i (y')^j W_{xy}^k$$

has character $r - s + i - j \pmod{e}$, because W_{xy} has character zero. Coordinate exchange sends the character c to $-c$ and multiplies the k -th Wronskian layer by $(-1)^k$. Therefore the $e = 5$ candidate space splits into six blocks indexed by $\{0\}, \{1, 4\}, \{2, 3\}$ and the two involution signs; for $e = 6$ it splits into eight blocks indexed by $\{0\}, \{1, 5\}, \{2, 4\}, \{3\}$ and the two signs.

For either line pair $(1, e)$ the same cyclic transformation

$$[X : Y : Z] \mapsto [\zeta X : \zeta^{-1} Y : Z], \quad \zeta^e = 1,$$

preserves both components. There is no coordinate-exchange involution, because the distinguished component is the line $Y = 0$. Since $a = y$ has character -1 on the Z -chart and b is invariant, a source monomial in the k -th Wronskian layer has the shifted character

$$(7.6) \quad r - s + i - j + n_k \pmod{e}, \quad n_k = m - 2k.$$

Thus the complete candidate space splits into exactly e cyclic character blocks. The shift by n_k is essential: it is contributed by the factor a^{-n_k} in the normal form.

7.3. The ordinary and mixed logarithmic charts. The ordinary chart below supports the finite ansatz and makes the finite symmetries transparent. The mixed chart is where we test logarithmic regularity, the negative twist, and the overlaps between the two boundary components.

Let

$$(7.7) \quad U_0 = \{ZAB \neq 0\}.$$

On U_0 , put

$$x = \frac{X}{Z}, \quad y = \frac{Y}{Z},$$

and

$$(7.8) \quad a = x^3 + 2y^3 + 1, \quad b = 2x^3 + y^3 + 1.$$

The ordinary invariant two-jet frame on U_0 is

$$(7.9) \quad x', \quad y', \quad W_{xy} := x'y'' - x''y'.$$

The mixed logarithmic chart is taken in $\{X \neq 0\}$. Put

$$u = \frac{Y}{X}, \quad z = \frac{Z}{X}.$$

Then

$$(7.10) \quad a_X = 1 + 2u^3 + z^3, \quad b_X = 2 + u^3 + z^3.$$

Set

$$(7.11) \quad q = \left(\log \frac{a_X}{b_X} \right)'.$$

A direct computation gives

$$(7.12) \quad q = \frac{3u^2(3 + z^3)}{a_X b_X} u' + \frac{3z^2(1 - u^3)}{a_X b_X} z'.$$

The determinant of the frame (z', q) is therefore

$$(7.13) \quad \Delta_X := a_{X,u}b_X - a_Xb_{X,u} = 3u^2(3 + z^3).$$

We define

$$(7.14) \quad U_X = \{X \neq 0, u(3 + z^3) \neq 0\}.$$

On this chart, the logarithmic invariant two-jet frame is

$$(7.15) \quad z', \quad q, \quad W_X := qz'' - q'z'.$$

The two sets U_0 and U_X cover the complement of a finite set. Indeed,

$$\mathbb{P}^2 \setminus (U_0 \cup U_X) \subset \{ZAB = 0\} \cap \{XY(3X^3 + Z^3) = 0\},$$

and the right-hand side is finite because none of the curves $Z = 0$, $A = 0$, or $B = 0$ is a component of $XY(3X^3 + Z^3) = 0$. Since the jet differential sheaves used here are locally free on the normal crossing locus, it is enough to test regularity on $U_0 \cup U_X$.

For the unequal-degree rows of (7.2), the same construction uses the projectively invariant logarithmic form. Put

$$g = \gcd(d_1, d_2), \quad \alpha = \frac{d_2}{g}, \quad \beta = \frac{d_1}{g},$$

and on the X -chart define

$$(7.16) \quad q = \alpha d \log a_X - \beta d \log b_X, \quad \Delta = \alpha a_{X,u}b_X - \beta a_Xb_{X,u}, \quad E = \alpha a_{X,z}b_X - \beta a_Xb_{X,z}.$$

Then

$$(7.17) \quad q = \frac{\Delta u' + E z'}{a_X b_X}, \quad u' = \frac{a_X b_X}{\Delta} q - \frac{E}{\Delta} z'.$$

The frame generator may replace q by any nonzero rational scalar multiple without changing the kernel problem over $\mathbb{Q}$. The implementation uses this freedom to keep the mixed-chart numerators integral. For the diagonal nonlinear models it uses

$$q_{\text{cmp}} = \frac{1}{d_1} d \log a_X - \frac{1}{d_2} d \log b_X = \frac{g}{d_1 d_2} q.$$

For the mixed conic model $a_X = u + z^2$, of bidegree $(2, e)$, it instead uses

$$q_{\text{cmp}} = d \log a_X - \frac{2}{e} d \log b_X = \frac{g}{e} q, \quad g = \gcd(2, e).$$

Thus $q_{\text{cmp}} = q/5$ for $e = 5$, while $q_{\text{cmp}} = q/3$ for $e = 6$. This is the normalization whose degree factors cancel in the corresponding mixed numerators. For the two line models the program uses the integral form q displayed below. Thus in every model the resulting Δ_{cmp} , E_{cmp} , and every matrix entry produced by the program are integers. Each rescaling is an invertible change of frame over $\mathbb{Q}$; it need not be invertible after reduction modulo every prime. Instead, the program first constructs its own integral matrix in this normalized frame. Full column rank of that integral matrix modulo one prime implies full column rank over $\mathbb{Q}$, and the rational change of frame then identifies its kernel with the kernel of the invariant form used in (7.16). In particular, the proof-effective prime 5 for the conic–quintic case is legitimate even though 5 divides the denominator in the rational rescaling $q_{\text{cmp}} = q/5$. The order is important: the change of frame is made over characteristic zero, the normalized matrix is then reconstructed with integral entries, and only this integral matrix is reduced modulo 5. No occurrence of the rational scalar $1/5$ is reduced in $\mathbb{F}_5$.

Likewise, the program's cyclic blocks are defined directly by congruences of integer monomial exponents and, where present, by the explicit exchange involution. Their construction does not invoke semisimplicity of a finite-group representation over the proof field. Semisimplicity is used only in characteristic zero to show that these blocks exhaust the candidate space. Thus a proof prime may divide the order of the displayed symmetry group without invalidating

the integral block computation. For the two line models these formulas are especially simple. When $e = 8$,

$$(7.18) \quad \begin{aligned} a_X &= u, & b_X &= 1 + u^8 + z^8 + u^4, & q &= 8d \log a_X - d \log b_X, \\ \Delta &= 8 + 8z^8 + 4u^4, & E &= -8uz^7. \end{aligned}$$

When $e = 9$,

$$(7.19) \quad \begin{aligned} a_X &= u, & b_X &= 1 + u^9 + z^9 + u^3z^3, & q &= 9d \log a_X - d \log b_X, \\ \Delta &= 9 + 9z^9 + 6u^3z^3, & E &= -9uz^8 - 3u^4z^2. \end{aligned}$$

In both cases z, u, b_X, Δ are pairwise coprime. For $e = 8$, put $\delta_0 = \Delta/4$. The identity

$$2b_X - \delta_0 = u^4(2u^4 + 1)$$

reduces the only nonobvious gcd to u and the factors of $2u^4 + 1$; the former does not divide b_X , and no nonconstant polynomial in u divides the polynomial b_X , which is monic in z . For $e = 9$, a common factor of b_X and $\Delta = 9b_X - u(b_X)_u$ would, after excluding u , divide $3u^6 + z^3$. On each component $z = \lambda u^2$, $\lambda^3 = -3$, the restriction of b_X is

$$1 - 2u^9 - 27u^{18},$$

which is not identically zero. This proves the required coprimality.

The coefficients α, β are exactly what make q independent of the projective trivialization. For every test pair in (7.2), none of Z, A, B is a component of the homogeneous polynomial $X\Delta$. Thus

$$U_0 = \{ZAB \neq 0\}, \quad U_X = \{X\Delta \neq 0\}$$

again cover the complement of a finite set. The general logarithmic frame is (z', q, W_X) with $W_X = qz'' - q'z'$. Formulas (7.11)–(7.13) are the specialization $d_1 = d_2 = 3$.

7.4. An auxiliary logarithmic compactification and a finite ambient normal form.

The ordinary frame on U_0 is convenient for symmetry, but it is not adapted to the line at infinity. In particular, the negative twist cannot be encoded by simply subtracting t from the degree of each numerator in the ordinary frame: cancellations between different ordinary basis terms may remove the apparent poles at infinity. We therefore separate the finite ambient bound from the twisting condition.

Let

$$L := \{Z = 0\}, \quad D^+ := D + L.$$

For every test pair in (7.2), the divisor D^+ is simple normal crossing. Multiplication by the section $Z^t \in H^0(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(t))$ gives an injection

$$(7.20) \quad E_{2,m}T_{\mathbb{P}^2}^*(\log D) \otimes \mathcal{O}_{\mathbb{P}^2}(-t) \xrightarrow{\cdot Z^t} E_{2,m}T_{\mathbb{P}^2}^*(\log D) \hookrightarrow E_{2,m}T_{\mathbb{P}^2}^*(\log D^+).$$

The image of the first arrow consists of logarithmic two-jet differentials for D that vanish to order at least t along L . Since $Z = 1$ on the affine chart used to define U_0 , multiplication by Z^t does not change the local expression on U_0 . Thus we may first generate a finite ambient space inside the larger logarithmic bundle for D^+ and impose the removal of the extra logarithmic pole along L , together with the order- t vanishing, on U_X .

We begin with the local pole estimate used for all three components of D^+ .

Lemma 7.2 (Layerwise logarithmic pole bound). *Let $s = 0$ be a smooth local branch of a logarithmic divisor on a surface, and let v be a transverse coordinate. Put*

$$\lambda = \frac{s'}{s}, \quad W_{sv} = s'v'' - s''v'.$$

Then the logarithmic Wronskian satisfies the triangular identity

$$(7.21) \quad \lambda v'' - \lambda' v' = \frac{1}{s} W_{sv} + \frac{(s')^2 v'}{s^2}.$$

Therefore, in weighted degree m , the coefficient of the ordinary layer W_{sv}^k has pole order at most

$$n_k := m - 2k$$

along $s = 0$. More specifically, the coefficient of

$$(s')^p (v')^{m-3k-p} W_{sv}^k$$

has pole order at most $p + k$.

Proof. Consider a logarithmic monomial

$$\lambda^a (v')^b (\lambda v'' - \lambda' v')^r, \quad a + b = m - 3r.$$

A contribution to the ordinary Wronskian layer W_{sv}^k can occur only when $r \geq k$. Choose the leading term $s^{-1} W_{sv}$ in exactly k of the r Wronskian factors and the lower term $s^{-2} (s')^2 v'$ in the remaining $r - k$ factors. The pole order is then

$$a + k + 2(r - k).$$

The resulting number of factors s' outside W_{sv}^k is

$$p = a + 2(r - k),$$

so the pole order equals $p + k$. Since $p \leq m - 3k$, this is at most $m - 2k$. The statement follows by linearity. $\square$

Write

$$k_{\max} = \lfloor m/3 \rfloor, \quad n_k = m - 2k, \quad s_k = m - 3k, \quad d_k^+ = (d_{\text{tot}} - 1)m - (2d_{\text{tot}} - 1)k.$$

Proposition 7.3 (Finite ambient normal form on U_0). *Every global section of*

$$E_{2,m} T_{\mathbb{P}^2}^* (\log D^+)$$

restricts to U_0 in the form

$$(7.22) \quad \omega_0 = \sum_{k=0}^{k_{\max}} \sum_{i+j=s_k} \frac{P_{i,j,k}(x, y)}{a^{n_k} b^{n_k}} (x')^i (y')^j W_{xy}^k,$$

where

$$(7.23) \quad P_{i,j,k} \in \mathbb{C}[x, y], \quad \deg P_{i,j,k} \leq d_k^+ = (d_{\text{tot}} - 1)m - (2d_{\text{tot}} - 1)k.$$

The exponent $n_k = m - 2k$ and the degree bound d_k^+ are necessary ambient bounds. Arbitrary polynomials satisfying (7.23) need not define a global section; the missing compatibility conditions are imposed on U_X in the next subsection.

Proof. Apply Lemma 7.2 to the branches $a = 0$ and $b = 0$. The coefficient of W_{xy}^k has pole order at most n_k along each branch. After multiplication by $a^{n_k} b^{n_k}$, every coefficient is regular on the affine plane, thus is a polynomial. This proves the denominator assertion.

It remains to control the degree by using the extra logarithmic component L . On the X -chart write

$$z = \frac{Z}{X}, \quad u = \frac{Y}{X}, \quad x = \frac{1}{z}, \quad y = \frac{u}{z}.$$

The ordinary jets satisfy

$$(7.24) \quad x' = -\frac{z'}{z^2}, \quad y' = \frac{zu' - uz'}{z^2}, \quad W_{xy} = -\frac{W_{zu}}{z^3},$$

where $W_{zu} = z'u'' - z''u'$. Also,

$$a = \frac{a_X}{z^{d_1}}, \quad b = \frac{b_X}{z^{d_2}}.$$

Thus the k -th layer of (7.22) becomes

$$(7.25) \quad \frac{z^{(d_{\text{tot}}-2)m-(2d_{\text{tot}}-3)k}}{a_X^{n_k} b_X^{n_k}} \sum_{i+j=s_k} (-1)^{i+k} P_{i,j,k}(z^{-1}, uz^{-1}) (z')^i (zu' - uz')^j W_{zu}^k.$$

For $0 \leq \ell \leq s_k$, the coefficient of

$$(z')^{s_k-\ell} (u')^\ell W_{zu}^k$$

is a triangular linear combination of the polynomials $P_{s_k-j,j,k}(z^{-1}, uz^{-1})$ with $j \geq \ell$; its diagonal term is a nonzero scalar multiple of

$$z^{(d_{\text{tot}}-2)m-(2d_{\text{tot}}-3)k+\ell} P_{s_k-\ell,\ell,k}(z^{-1}, uz^{-1}).$$

By the refined part of Lemma 7.2, a section logarithmic along $L = \{z = 0\}$ may have pole order at most

$$(s_k - \ell) + k$$

in this coefficient. Starting with $\ell = s_k$ and descending in ℓ , the triangularity shows that no polynomial $P_{s_k-\ell,\ell,k}$ can have degree exceeding

$$(d_{\text{tot}} - 2)m - (2d_{\text{tot}} - 3)k + \ell + (s_k - \ell) + k = (d_{\text{tot}} - 1)m - (2d_{\text{tot}} - 1)k.$$

This proves (7.23). $\square$

The important point is that the negative twist t does not appear in (7.23). The degree bound describes a finite ambient space with an allowed logarithmic pole along L . The twist is imposed by requiring the transformed coefficients to lose that pole and to vanish to order t along L .

7.5. Regularity and twisting test on U_X . On $U_0 \cap U_X$ one has

$$(7.26) \quad x = \frac{1}{z}, \quad y = \frac{u}{z}, \quad a = \frac{a_X}{z^{d_1}}, \quad b = \frac{b_X}{z^{d_2}}.$$

The first derivatives transform as

$$(7.27) \quad x' = -\frac{z'}{z^2}, \quad y' = \frac{u'z - uz'}{z^2}.$$

Set

$$H_0 := a_X b_X, \quad E_0 := \alpha a_{X,z} b_X - \beta a_X b_{X,z}, \quad \Delta_X := \alpha a_{X,u} b_X - \beta a_X b_{X,u}.$$

Equation (7.12) gives

$$(7.28) \quad u' = \frac{H_0}{\Delta_X} q - \frac{E_0}{\Delta_X} z'.$$

The pair (z', q) is a logarithmic one-jet frame for D on U_X , and

$$W_X = qz'' - q'z'$$

is the associated invariant Wronskian. Substituting (7.26), (7.27), and (7.28) into (7.22) and collecting in the mixed frame gives

$$(7.29) \quad \omega_0|_{U_X} = \sum_{k=0}^{k_{\max}} \sum_{j=0}^{s_k} R_{j,k}(u, z) (z')^{s_k-j} q^j W_X^k.$$

Proposition 7.4 (Exact extension and twisting criterion). *A candidate (7.22) is the restriction of a section*

$$\omega \in H^0(\mathbb{P}^2, E_{2,m} T_{\mathbb{P}^2}^*(\log D) \otimes \mathcal{O}_{\mathbb{P}^2}(-t))$$

if and only if every coefficient in (7.29) satisfies

$$(7.30) \quad R_{j,k} \in z^t \mathbb{C}[u, z, \Delta_X^{-1}].$$

Equivalently, after clearing a fixed bounded denominator, the condition is a finite collection of polynomial divisibility equations that are homogeneous and linear in the coefficients of the $P_{i,j,k}$.

Proof. The mixed frame (z', q, W_X) is regular for the logarithmic jet bundle associated with D on U_X . Thus a section of the untwisted bundle is regular on U_X exactly when its scalar coefficients lie in

$$\mathcal{O}(U_X) = \mathbb{C}[u, z, \Delta_X^{-1}].$$

Under the injection (7.20), the image of a section twisted by $\mathcal{O}(-t)$ vanishes to order at least t along $L = \{z = 0\}$. This is exactly condition (7.30).

Since $U_0 \cup U_X$ misses only finitely many points and the jet bundle is locally free, a section satisfying this condition extends uniquely across the missing codimension-two set. This proves the equivalence. $\square$

For computation, the denominator clearing can be made uniform and explicit. Write

$$u' = \mathbf{a}q + \mathbf{b}z', \quad \mathbf{a} = H_0/\Delta_X, \quad \mathbf{b} = -E_0/\Delta_X.$$

Differentiating with respect to the source variable and substituting $u' = \mathbf{a}q + \mathbf{b}z'$ gives the exact identity

$$(7.31) \quad u'' = \mathbf{a}q' + \mathbf{b}z'' + \mathbf{a}\mathbf{a}_u q^2$$

$$(7.32) \quad + (\mathbf{a}_u \mathbf{b} + \mathbf{a}_z + \mathbf{a}\mathbf{b}_u)qz' + (\mathbf{b}\mathbf{b}_u + \mathbf{b}_z)(z')^2.$$

Therefore,

$$(7.33) \quad \begin{aligned} W_{zu} = & -\mathbf{a}W_X + \mathbf{a}\mathbf{a}_u q^2 z' \\ & + (\mathbf{a}_u \mathbf{b} + \mathbf{a}_z + \mathbf{a}\mathbf{b}_u)q(z')^2 + (\mathbf{b}\mathbf{b}_u + \mathbf{b}_z)(z')^3. \end{aligned}$$

This is an identity in the localized differential-polynomial ring. Since $\mathbf{a}, \mathbf{b}$ have one power of Δ_X in the denominator, their first partial derivatives have at most two, and every coefficient in (7.33) has Δ_X -denominator order at most three. A source layer W_{zu}^r can contribute to the target layer W_X^k only when $r \geq k$. The $m - 3r$ first-jet factors contribute at most one power of Δ_X^{-1} each, the k leading Wronskian factors contribute one power each, and the remaining $r - k$ lower Wronskian factors contribute at most three powers each. Therefore the total exponent of Δ_X is bounded by

$$(7.34) \quad (m - 3r) + k + 3(r - k) = m - 2k = n_k.$$

No substitution introduces a_X or b_X in a denominator beyond the original source exponent, and $n_r \leq n_k$ for $r \geq k$.

We also record the z -bound explicitly. Before replacing u' and W_{zu} by the mixed frame, a monomial from source layer r with a numerator of degree d carries the factor

$$z^{d_{\text{tot}}n_r - 2(m - 3r) - 3r - d} = z^{(d_{\text{tot}} - 2)m - (2d_{\text{tot}} - 3)r - d}.$$

Since $d \leq (d_{\text{tot}} - 1)m - (2d_{\text{tot}} - 1)r$, this exponent is at least

$$(d_{\text{tot}} - 2)m - (2d_{\text{tot}} - 3)r - ((d_{\text{tot}} - 1)m - (2d_{\text{tot}} - 1)r) = -(m - 2r) = -n_r.$$

The substitutions (7.28) and (7.33) introduce no additional powers of z^{-1} . Thus the pole order in z is at most $n_r \leq n_k$. Therefore,

$$(7.35) \quad \mathcal{N}_{j,k} := z^{n_k} a_X^{n_k} b_X^{n_k} \Delta_X^{n_k} R_{j,k} \in \mathbb{C}[u, z].$$

The same valuation argument applies to every row of (7.2), including the unequal-degree rows. Indeed, $A + B + L$ is SNC, so z, a_X, b_X are pairwise coprime, and the check following (7.17) says exactly that

$$\gcd(z a_X b_X, \Delta_X) = 1.$$

Thus localizing at Δ_X changes none of the valuations along the prime factors of $z a_X b_X$. This is the required bridge from the general coprimality $\gcd(A, B) = 1$ to the concrete divisibility test.

For example, for the special pair (7.3), the four factors

$$z, \quad a_X, \quad b_X, \quad \Delta_X$$

are pairwise coprime in $\mathbb{C}[u, z]$. Indeed, z divides none of the other three. If an irreducible factor divided both a_X and b_X , it would divide $a_X - b_X = u^3 - 1$; after setting $u^3 = 1$, however, both polynomials become $3 + z^3$, so no factor of $u^3 - 1$ divides them identically. Also, u divides neither a_X nor b_X . Finally, for every root ζ of $\zeta^3 = -3$ one has

$$a_X(u, \zeta) = 2(u^3 - 1), \quad b_X(u, \zeta) = u^3 - 1,$$

which are not the zero polynomial in u ; thus no irreducible factor of $3 + z^3$ divides a_X or b_X . Since $\mathbb{C}[u, z]$ is a unique factorization domain, this verifies the general coprimality statement directly in the cubic case. The valuation conditions along $z = 0$, $a_X = 0$, and $b_X = 0$ can therefore be combined for every test pair. Thus (7.30) is equivalent to the concrete divisibility condition

$$(7.36) \quad z^{n_k+t} a_X^{n_k} b_X^{n_k} \mid \mathcal{N}_{j,k} \quad \text{in } \mathbb{C}[u, z].$$

For the conic and cubic models, the fixed product in (7.36) may be handled by the corresponding exact polynomial-remainder test. For a line pair, however, $a_X = u$, and treating the whole product as a univariate polynomial in z would give a nonconstant leading coefficient. We instead use pairwise coprimality and impose the following three equivalent conditions on every cleared numerator $\mathcal{N}_{j,k}$:

- (i) no monomial has z -degree below $n_k + t$;
- (ii) no monomial has u -degree below n_k ;
- (iii) the remainder modulo the monic polynomial $b_X^{n_k}$ in $K[u][z]$ is zero, where $K = \mathbb{C}$ for the characteristic-zero system and $K = \mathbb{F}_p$ for a certificate matrix.

The three row families are kept disjoint in the certificate. Thus all divisibility conditions are imposed simultaneously, and each is homogeneous and linear in the coefficients of the $P_{i,j,k}$. The triangularity in the Wronskian degree allows the equations to be generated from the largest k downward.

How one matrix row is formed. Write a cleared numerator as

$$\mathcal{N}_{j,k} = \sum_{r,s} c_{r,s} u^r z^s.$$

Each coefficient $c_{r,s}$ is a linear combination of the unknown coefficients in the polynomials $P_{i,j,k}$. Condition (i), for example, sets $c_{r,s} = 0$ whenever $s < n_k + t$. This one linear equation is one row of the matrix. Conditions (ii) and (iii) give rows in the same way, using low u -degree coefficients and coefficients of the remainder modulo $b_X^{n_k}$. Thus the matrix is only a compact record of the proved divisibility conditions; it is not a numerical approximation.

7.6. The involution and character reductions. The involution $\sigma(X, Y, Z) = (Y, X, Z)$ preserves U_0 and acts on the ordinary frame by

$$x \leftrightarrow y, \quad x' \leftrightarrow y', \quad W_{xy} \mapsto -W_{xy}.$$

If one fixes an eigenvalue $\varepsilon = \pm 1$ and imposes

$$(7.37) \quad \sigma^* \omega_0 = \varepsilon \omega_0,$$

then the coefficients in (7.22) satisfy

$$(7.38) \quad P_{i,j,k}(x, y) = \varepsilon(-1)^k P_{j,i,k}(y, x).$$

Thus only one representative from each orbit

$$(i, j, x^r y^s) \longleftrightarrow (j, i, x^s y^r)$$

needs to be generated. If an orbit is fixed and $\varepsilon(-1)^k = -1$, its coefficient is forced to be zero.

The diagonal $(\mu_3)^2$ -action gives a further decomposition. For

$$[X : Y : Z] \mapsto [\alpha X : \beta Y : Z], \quad \alpha^3 = \beta^3 = 1,$$

one has

$$x \mapsto \alpha x, \quad y \mapsto \beta y, \quad x' \mapsto \alpha x', \quad y' \mapsto \beta y', \quad W_{xy} \mapsto \alpha \beta W_{xy}.$$

Therefore the monomial

$$x^r y^s (x')^i (y')^j W_{xy}^k$$

has character

$$(7.39) \quad (r + i + k, s + j + k) \pmod{3}.$$

For a fixed character $(\chi_1, \chi_2) \in (\mathbb{Z}/3\mathbb{Z})^2$, we generate only the monomials satisfying

$$(7.40) \quad r + i + k \equiv \chi_1 \pmod{3}, \quad s + j + k \equiv \chi_2 \pmod{3}.$$

The involution σ exchanges the character (χ_1, χ_2) with (χ_2, χ_1) . Thus the nine characters split into six orbits:

$$(7.41) \quad (0, 0), (1, 1), (2, 2), (0, 1) \leftrightarrow (1, 0), (0, 2) \leftrightarrow (2, 0), (1, 2) \leftrightarrow (2, 1).$$

For diagonal characters one imposes σ -eigenconditions inside the character space. For an off-diagonal orbit, an eigenvector is determined by its component in either one of the two exchanged character spaces.

Lemma 7.5 (The cubic symmetry blocks exhaust the candidate space). *Let V be any finite-dimensional candidate space in the above normal form, and let $K \subset V$ be the kernel of the extension and twisting equations. Then $K = 0$ if and only if its intersection with every one of the twelve character-eigenvalue blocks indexed by (7.41) and $\varepsilon \in \{+1, -1\}$ is zero.*

Proof. Over $\mathbb{C}$, the finite abelian group $(\mu_3)^2$ acts semisimply, so $V = \bigoplus_{\chi} V_{\chi}$. The equations are equivariant and thus their kernel has the same decomposition. The involution sends V_{χ} to $V_{\sigma\chi}$. If $\chi = \sigma\chi$, the involution is diagonalizable with eigenvalues ± 1 . If $\chi \neq \sigma\chi$ and $0 \neq v \in K \cap V_{\chi}$, then at least one of $v + \sigma v$ and $v - \sigma v$ is nonzero; these vectors lie respectively in the $+1$ and -1 eigenspaces of $V_{\chi} \oplus V_{\sigma\chi}$. Thus every nonzero kernel contains a nonzero vector in one of the listed blocks. The converse is immediate. $\square$

Lemma 7.6 (The conic-pair symmetry blocks exhaust the candidate space). *For either model $(2, e)$, let V be the finite candidate space and let $K \subset V$ be the kernel of the extension and twisting equations. Then $K = 0$ if and only if its intersection with every character-eigenvalue block indexed by an orbit of $c \mapsto -c$ in $\mathbb{Z}/e\mathbb{Z}$ and by $\varepsilon \in \{+1, -1\}$ is zero.*

Proof. Over $\mathbb{C}$, the cyclic group μ_e acts semisimply, and equivariance of the overlap and divisibility equations gives $K = \bigoplus_c (K \cap V_c)$. Coordinate exchange sends V_c to V_{-c} and preserves K . On a fixed character ($c = -c$), it is diagonalizable with eigenvalues ± 1 . On a two-element orbit $\{c, -c\}$, every nonzero $v \in K \cap V_c$ yields a nonzero vector among $v + \sigma v$ and $v - \sigma v$ in the corresponding $+1$ or -1 block. Thus a nonzero kernel meets one of the enumerated blocks, and the converse is immediate. $\square$

Lemma 7.7 (The line-pair character blocks exhaust the candidate space). *For either line model $(1, e)$, let V be the finite candidate space and let $K \subset V$ be the kernel of the extension and twisting equations. Then $K = 0$ if and only if its intersection with each of the e character spaces defined by (7.6) is zero.*

Proof. The cyclic group μ_e acts semisimply in characteristic zero, and all overlap and divisibility equations are equivariant. Both V and K are therefore direct sums of their e character spaces. $\square$

Proposition 7.8 (Finite matrix equivalence). *Fix one of the test pairs in (7.2) and integers $m, t \geq 1$. Let V_m be the finite coefficient space in (7.22)–(7.23), and let*

$$\Phi_{m,t}: V_m \longrightarrow W_{m,t}$$

record the coefficients of all forbidden remainders in (7.36); for a line pair, $\Phi_{m,t}$ records the three row families listed after that equation. Restriction to U_0 induces a natural isomorphism

$$H^0(\mathbb{P}^2, E_{2,m} T_{\mathbb{P}^2}^*(\log D) \otimes \mathcal{O}_{\mathbb{P}^2}(-t)) \simeq \ker \Phi_{m,t}.$$

Also, the symmetry decompositions above identify this kernel with the direct sum of the kernels of the corresponding block matrices.

Proof. Proposition 7.3 sends every global section to a unique vector of V_m . Proposition 7.4, together with the coprimality and denominator bounds proved above, says that such a vector comes from a twisted global section exactly when all the remainders recorded by $\Phi_{m,t}$ vanish. The resulting section on $U_0 \cup U_X$ extends uniquely across the finite complement. This proves the first assertion. Equivariance of the equations and Lemmas 7.5, 7.6, and 7.7 prove the second. $\square$

The implementation also checks equivariance after the full chart transformation. If a cleared target row contains the monomial $u^U z^Z$, has q -degree Q , and lies in target Wronskian layer k , its character in the source convention is

$$(7.42) \quad -2U - Z + 2(m - 2k) + Q \pmod{e}.$$

For every complete transformed column in the $m = 3$ self-test, each nonzero target row has the same character as the source column. This detects, in particular, omission of the denominator shift in (7.6).

7.7. Counting the ambient variables. Let

$$M(d) = \binom{d+2}{2}$$

be the number of monomials in two variables of total degree at most d . Since the negative twist is imposed by the equations on U_X , rather than by deleting monomials on U_0 , the ambient variable count depends on m but not on t :

$$(7.43) \quad N_{\text{raw}}^+(m; d_{\text{tot}}) = \sum_{k=0}^{\lfloor m/3 \rfloor} (m - 3k + 1) \binom{(d_{\text{tot}} - 1)m - (2d_{\text{tot}} - 1)k + 2}{2}.$$

For a line pair $(1, e)$, so that $d_{\text{tot}} = e + 1$, this specializes to

$$(7.44) \quad N_e(m) = \sum_{0 \leq k \leq \lfloor m/3 \rfloor} (m - 3k + 1) \binom{em - (2e + 1)k + 2}{2}.$$

The exact counts for the eight line-pair cases are

(d_1, d_2)	(m, t)	$N_e(m)$	blocks	largest block
(1, 8)	(3, 2)	1,336	8	174
(1, 8)	(4, 2)	3,077	8	398
(1, 8)	(5, 2)	6,066	8	779
(1, 8)	(6, 3)	10,807	8	1,382
(1, 8)	(7, 3)	17,876	8	2,279
(1, 8)	(8, 3)	27,849	8	3,543
(1, 9)	(3, 3)	1,669	9	186
(1, 9)	(4, 3)	3,857	9	429

Thus the line-octic calculation has 48 blocks and 67,011 ambient columns in total; the line-nonic calculation has 18 blocks and 5,526 columns.

The remainder of this subsection specializes to $(d_1, d_2) = (3, 3)$, so $d_{\text{tot}} = 6$ and the degree bound becomes $5m - 11k$.

For a fixed character (χ_1, χ_2) , define

$$(7.46) \quad M_d(a, b) := \#\{(r, s) \in \mathbb{Z}_{\geq 0}^2 : r + s \leq d, r \equiv a \pmod{3}, s \equiv b \pmod{3}\}.$$

Then

$$(7.47) \quad N_{\chi_1, \chi_2}^+(m) = \sum_{k=0}^{\lfloor m/3 \rfloor} \sum_{i+j=m-3k} M_{5m-11k}(\chi_1 - i - k, \chi_2 - j - k),$$

where all congruences are understood modulo 3. For an off-diagonal character orbit, the number of variables in a σ -eigensystem is the number in either exchanged character. For a diagonal character (χ, χ) , the σ -even and σ -odd dimensions are

$$(7.48) \quad N_{\chi}^{\pm,+}(m) = \frac{1}{2} (N_{\chi,\chi}^+(m) \pm T_{\chi}^+(m)),$$

where

$$(7.49) \quad T_{\chi}^+(m) = \sum_{\substack{0 \leq k \leq \lfloor m/3 \rfloor \\ m-3k \text{ even}}} (-1)^k \# \left\{ r \geq 0 : 2r \leq 5m - 11k, \ r + \frac{m-3k}{2} + k \equiv \chi \pmod{3} \right\}.$$

For each twist t , the following table records the endpoint $m = 5t - 1$ of the certified cubic range. The last column is the largest block after both the character and involution decompositions.

$$(7.50) \quad \begin{array}{c|c|c|c} t & m & N_{\text{raw}}^+ & \max N_{\text{sym}}^+ \\ \hline 1 & 4 & 1,265 & 156 \\ 2 & 9 & 16,511 & 1,924 \\ 3 & 14 & 76,905 & 8,830 \end{array}$$

Thus the largest ambient candidate space occurs at $(m, t) = (14, 3)$ and has 76,905 variables. After the $(\mu_3)^2$ -character decomposition and the involution, the largest block has 8,830 variables. The raw ambient total over the fourteen implemented cases

$$1 \leq m \leq 14, \quad t = t_0(m) = \left\lceil \frac{m+1}{5} \right\rceil$$

is 280,798. These consist of the twelve theorem-relevant cases $3 \leq m \leq 14$ and the two low-weight self-tests $m = 1, 2$. These variables are never placed in one matrix: the computation splits by (m, t) , character orbit, and involution eigenvalue. Within each such block, the Wronskian layers organize the downward generation of equations, but the rank test is performed on the full coupled matrix containing all layers. Although the ambient variables do not depend on t , the divisibility equations (7.36) become stronger as t increases.

Output of the reduction. For every listed pair (m, t) , Proposition 7.8 gives a proved isomorphism

$$H^0(\mathbb{P}^2, E_{2,m} T_{\mathbb{P}^2}^*(\log D) \otimes \mathcal{O}(-t)) \simeq \ker \Phi_{m,t}.$$

The finite symmetry blocks exhaust this kernel. No numerical claim has been used so far; the next section supplies the exact ranks.

8. EXACT RANK CERTIFICATES FOR THE FINITE MATRICES

The preceding section reduced the geometric vanishing problem to the kernels of explicit integral matrices. This section proves that those kernels are zero. Full column rank modulo one prime is enough: a maximal minor that is nonzero modulo that prime is a nonzero integer, so it is also nonzero over $\mathbb{Q}$ and $\mathbb{C}$. Readers interested only in the mathematical proof may read the procedure, the rank table, and the final semicontinuity argument. The certificate-format paragraphs are included for independent checking.

8.1. Modular rank verification. For each degree pair and each (m, t) in its finite range, the computation is as follows.

- (1) Use the injection (7.20) and generate the finite ambient normal form (7.22) on U_0 , with denominator exponent $m - 2k$ and degree bound $(d_{\text{tot}} - 1)m - (2d_{\text{tot}} - 1)k$. Do not subtract t from this degree bound.
- (2) For $(3, 3)$, split the ambient variables according to the $(\mu_3)^2$ -characters and impose the involution eigencondition. For $(2, e)$ use the cyclic character and involution blocks described above. For $(1, e)$ use exactly the e shifted cyclic characters in (7.6), without an involution. Keep one full unsplit sparse basis for $(3, 4)$.

- (3) Transform the candidate to U_X using (7.26), (7.27), (7.28), and the triangular Wronskian formula (7.33).
- (4) Collect the expression in the regular logarithmic frame (z', q, W_X) and form the cleared polynomials $\mathcal{N}_{j,k}$ in (7.35).
- (5) Impose the exact divisibility conditions (7.36). For a line pair use the three separate row families listed after that equation. These conditions simultaneously remove the spurious logarithmic pole along L , impose the twist $\mathcal{O}(-t)$, and cancel all forbidden poles along A and B .
- (6) Choose a prime for which the adopted normalized integral matrix is defined, reduce that integral matrix to the corresponding finite field, and verify full column rank. If the frame was obtained by a rational rescaling such as $q_{\text{cmp}} = q/5$, perform that rescaling over characteristic zero and reconstruct the integral matrix before reduction; do not reduce the rational change-of-frame scalar itself. Repeat over a second prime as an implementation check; only one proof-effective prime is logically necessary.

For (3, 3) it suffices to check the twelve genuine second-level pairs

$$(8.1) \quad 3 \leq m \leq 14, \quad t = t_0(m) := \left\lceil \frac{m+1}{5} \right\rceil.$$

Indeed, if $t \geq t_0(m)$ and a section with twist $-t$ existed, multiplication by a nonzero form of degree $t - t_0(m)$ would inject it into the space with twist $-t_0(m)$. The implementation also checks $m = 1, 2$ as low-degree self-tests. The remaining minimal pairs are exactly those displayed after Lemma 7.1.

The passage from modular rank to characteristic zero is elementary but decisive. Every cleared matrix has integral entries. Full column rank modulo a prime p exhibits a maximal minor that is nonzero modulo p ; that minor is a nonzero integer and remains nonzero over $\mathbb{Q}$.

The certificate program constructs the matrices from (7.36). For every cubic pair in (8.1), all twelve character–eigenvalue matrices have full column rank. Lemma 7.5 then gives zero kernel for the entire cubic candidate space. For (2, e), every cyclic character–eigenvalue block has full column rank; the (3, 4) unsplit matrix does as well. For (1, e), every shifted cyclic character block has full column rank; Lemma 7.7 therefore gives zero kernel. The proof primes are recorded in Table 1. For each block, the program writes the degree pair, (m, t) , row and column numbers, rank, nullity, nonzero pivot product and pivot hash, modulus, and elapsed time, both as plain text and in JSON Lines format. The complete source, machine-readable certificates, checksums, and reproduction instructions are included in the supplementary archive cited above. A separate verifier audits the expected case and block coverage and the internal consistency of every certificate record. The frozen source and binaries allow the rank computations to be reproduced.

Remark 8.1 (Certificate contract). For each independent block, the archive records the degree pair, (m, t) , character label, involution sign when present, proof modulus, row and column counts, rank, nullity, matrix-construction version, source hash, certificate hash, and pivot/checker data. The separate checker verifies case coverage and the internal consistency of every record. The matrix generator and the checker are frozen in the archived version used for this paper.

The proof moduli displayed in the table already suffice. The archive also contains reruns at other primes to test the implementation independently; those reruns are reproducibility checks, not additional mathematical hypotheses. For (3, 3), the newly required case $(m, t) = (9, 2)$ was replayed in full at prime 7. For (2, 5), the additional case $(m, t) = (8, 3)$ has 15,849 columns distributed among six character–eigenvalue blocks. All six blocks have full column rank modulo 5, and an independent replay gives full column rank modulo 7 as well.

Prime 5 is a bad reduction for one (2, 6) block: there the rank is 92/93. It will be kept in the archive only as a diagnostic and is not used in the proof. The proof-effective (2, 6) certificate is the full-rank prime-7 run, independently reproduced at prime 11.

TABLE 1. Finite-rank certificate summary. “Columns” is summed over the indicated independent blocks; in every row the summed rank equals the summed number of columns, so the kernel dimension is zero. The proof modulus is the prime used for the characteristic-zero inference. For $(3, 3)$, the totals include the 12 theorem-relevant pairs in (8.1) and the two low-weight implementation self-tests $m = 1, 2$.

pair	cases	blocks	columns	rank	kernel dim.	proof modulus
$(3, 3)$	14	168	280,798	280,798	0	5
$(3, 4)$	1	1	781	781	0	5
$(2, 6)$	1	8	1,040	1,040	0	7
$(2, 5)$	6	36	38,286	38,286	0	5
$(1, 9)$	2	18	5,526	5,526	0	5
$(1, 8)$	6	48	67,011	67,011	0	5

For both line pairs, prime 5 supplies the proof-effective full-pivot certificates and prime 11 supplies an independent full-rank replay. Prime 7 is excluded: some relevant integral minors vanish modulo 7, and the degree-nine test curve has bad reduction there because $1^3 = -27$ in $\mathbb{F}_7$. None of the prime-7 diagnostics is used in the proof.

By the maximal-minor argument above, the proof-effective ranks in Table 1 give zero kernel over $\mathbb{Q}$, and thus over $\mathbb{C}$. This proves the required vanishing for every test pair in (7.2).

To pass to general pairs of the fixed degrees, let $\mathcal{U}$ be the Zariski-open parameter space of ordered smooth transverse pairs, let $\mathcal{D} \subset \mathbb{P}^2 \times \mathcal{U}$ be the relative simple normal crossing divisor, and let $\mathcal{E}_{m,t}$ be the relative logarithmic invariant two-jet bundle twisted by $\mathcal{O}_{\mathbb{P}^2}(-t)$. Properness of the projection $\mathbb{P}^2 \times \mathcal{U} \rightarrow \mathcal{U}$ and upper semicontinuity of fiberwise cohomology imply that

$$(C_1, C_2) \mapsto h^0(\mathbb{P}^2, E_{2,m}T_{\mathbb{P}^2}^*(\log(C_1 + C_2)) \otimes \mathcal{O}(-t))$$

is upper semicontinuous. Vanishing at the special pair therefore holds on a nonempty Zariski-open subset of $\mathcal{U}$. There are only finitely many pairs (m, t) under consideration; intersecting their nonempty open loci proves Lemma 7.1.

Output of the certificate section. The matrices have zero kernel over $\mathbb{C}$ for one explicit SNC pair in each degree family. Upper semicontinuity then gives the nonempty open sets $\mathcal{U}_{\text{van}}(d_1, d_2)$ used in the main proof.

9. POSSIBLE IMPROVEMENTS OF THE SMT CONSTANTS

This section is not used in the proof of Theorem 1.1 and may be skipped. The completed certificates already prove the constants 57, 45, and 69 without any extra assumption. Here we only explain how more finite vanishings could improve those constants. The related resource estimates are moved to Appendix B.

9.1. The limiting constants of the present method. If the additional vanishings listed below are established, the smallest integer constants allowed by the present two-branch organization would be 27, 17, and 18 for $(d_1, d_2) = (3, 3), (2, 5), (1, 8)$, respectively. The corresponding ratio splits and zero-locus endpoints are

(d_1, d_2)	ρ_*	$3/\tau_1(\rho_*)$
$(3, 3)$	$1/7$	$(27 + \sqrt{701})/2 < 27,$
$(2, 5)$	$2/9$	$(207 + 54\sqrt{14})/25 < 17,$
$(1, 8)$	$4/19$	$(222 + \sqrt{47859})/25 < 18.$

Relative to Lemma 7.1, these improvements require exactly the following additional minimal vanishings:

$$\begin{array}{l|l} (3, 3) & (5, 1), (6, 1), (10, 2), (11, 2), (12, 2), (13, 2), (15, 3), \\ & (16, 3), (17, 3), (18, 3), (19, 3), (20, 3), \\ (2, 5) & (3, 1), (4, 1), (6, 2), (7, 2), (8, 2), (9, 3), \\ & (10, 3), (11, 3), (12, 3), (13, 3), \\ (1, 8) & (3, 1), (4, 1), (6, 2), (7, 2), (8, 2), (9, 2), (10, 3), \\ & (11, 3), (12, 3), (13, 3), (14, 3). \end{array}$$

Indeed, these are exactly the integral pairs with $1 \leq t \leq 3$ above the indicated split, after twist monotonicity removes all nonminimal twists. Thus this list, rather than a change in the analytic or geometric argument, is the remaining obstruction to the limiting constants.

9.2. The unresolved line–septic case.

Remark 9.1. For $(d_1, d_2) = (1, e)$ one has

$$\bar{c}_1^2 = (e - 2)^2, \quad \bar{c}_2 = (e - 1)^2, \quad A(1, e) = 4e^2 - 34e + 43.$$

At $e = 10$ this gives

$$A(1, 10) = 103, \quad \rho_{\text{DEG}}(1, 10) = \frac{103}{96} > 1,$$

so no finite Key Vanishing Lemma is needed for $e \geq 10$. For $e = 9$ one has $\rho_{\text{DEG}} = 61/84$, leaving only $(m, t) = (3, 3), (4, 3)$; for $e = 8$ one has $\rho_{\text{DEG}} = 3/8$, leaving

$$(3, 2), (4, 2), (5, 2), (6, 3), (7, 3), (8, 3).$$

Lemma 7.1 certifies exactly these eight cases, and the degree-one construction in Section 5 supplies the required pole-three differentiation package. This is why the theorem begins at $e = 8$.

At $e = 7$ one has $A(1, 7) = 1$ and $\rho_{\text{DEG}} = 1/60$; the required minimal profile is

t	m
1	$3 \leq m \leq 60$
2	$61 \leq m \leq 120$
3	$121 \leq m \leq 180$.

Since $A(1, 6) = -17$, the pair $(1, 7)$ is the formal Riemann–Roch endpoint of this two-jet architecture. It should not be confused with a predicted Key Vanishing endpoint: $A(1, 7)$ is barely positive, the list contains 178 minimal cases, and the corresponding finite computation is not part of the present theorem. No claim for $(1, 7)$ is made here.

APPENDIX A. SNC VERIFICATION FOR THE TEST PAIRS

We verify here that every row of (7.2) is an SNC divisor and remains SNC after adjoining $L = \{Z = 0\}$. These elementary checks are separate from the modular rank computation.

The result of the checks is summarized first.

test family	verified conclusion
cubic–cubic and cubic–quartic	both curves are smooth and meet each other and L transversally
conic–quintic and conic–sextic	the conic and Fermat curve are smooth and $A + B + L$ is SNC
line–octic and line–nonic	the nonlinear curve is smooth and $A + B + L$ is SNC

A.1. The cubic–quartic pair. Both Fermat curves are smooth. If their gradients were proportional at a common point, comparison of the nonzero coordinate components would force all nonzero coordinates to be equal. Neither two nor three equal nonzero coordinates can satisfy the cubic equation, and a point with only one nonzero coordinate cannot lie on it. Thus the curves meet transversally. On L , a common point would satisfy simultaneously $(X/Y)^3 = -1$ and $(X/Y)^4 = -1$, which is impossible. Smoothness of the Fermat curves also gives their transversality with L .

A.2. The conic pairs. For $(2, e)$, $e = 5, 6$, the conic $A = XY + Z^2$ and the Fermat curve $B_e = X^e + Y^e + Z^e$ are smooth. Suppose that they meet nontransversally. If $Z = 0$, then $A = 0$ forces $XY = 0$, which is incompatible with $B_e = 0$. Thus we may scale $Z = 1$, so $XY = -1$. Writing $\nabla B_e = \lambda \nabla A$, the third component gives $\lambda = e/2$, and the first two components give

$$2X^{e-1} = Y, \quad 2Y^{e-1} = X.$$

Multiplication by X and Y , respectively, yields $X^e = Y^e = -1/2$. Thus

$$(XY)^e = (-1)^e = X^e Y^e = \frac{1}{4},$$

a contradiction. On L , the conic meets the line at $[1 : 0 : 0]$ and $[0 : 1 : 0]$, neither of which lies on B_e ; the Fermat curve also meets L transversally. Thus $A + B_e + L$ is SNC.

A.3. The line–octic pair. Let

$$B_8 = X^8 + Y^8 + Z^8 + X^4 Y^4.$$

At a singular point of B_8 , the equation $(B_8)_Z = 8Z^7 = 0$ gives $Z = 0$. Neither X nor Y can vanish, while the other two partial derivatives give

$$2X^4 + Y^4 = 0, \quad X^4 + 2Y^4 = 0.$$

Their determinant is 3, a contradiction. Thus B_8 is smooth. On the line $A = \{Y = 0\}$ its intersection equation is $X^8 + Z^8 = 0$, so X and Z are nonzero and the intersection is transverse. Along L , smoothness and $(B_8)_Z = 0$ imply that at least one of the other two partial derivatives is nonzero, so the intersection with L is transverse. Finally, $A \cap L = [1 : 0 : 0] \notin B_8$, and thus $A + B_8 + L$ is SNC.

A.4. The line–nonic pair. Put

$$B_9 = X^9 + Y^9 + Z^9 + X^3 Y^3 Z^3.$$

A singular point with a zero coordinate is impossible from the remaining partial derivatives. If all coordinates are nonzero, set $a = X^3$, $b = Y^3$, and $c = Z^3$. The three singularity equations become

$$3a^2 + bc = 3b^2 + ac = 3c^2 + ab = 0.$$

Multiplication gives $27 = -1$, a contradiction. Thus B_9 is smooth. Its intersections with $A = \{Y = 0\}$ and L are transverse by restriction, and $A \cap L = [1 : 0 : 0] \notin B_9$. Thus $A + B_9 + L$ is SNC.

A.5. The symmetric cubic pair. For (7.3), the two cubics are smooth. If $A = B = 0$, then

$$X^3 = Y^3, \quad 3X^3 + Z^3 = 0,$$

so all intersection coordinates are nonzero. Their gradients cannot be proportional: the Z -components would force the scalar to be 1, while the X -components would then require $3X^2 = 6X^2$. Also, $A = B = Z = 0$ has no projective solution, and each cubic meets L transversally. Thus $A + B + L$ is SNC.

APPENDIX B. COMPUTATIONAL SCALE AND FEASIBILITY

This appendix is not part of the proof. It records only the expected size and practical organization of possible future computations.

All the additional targets are governed by the same finite map $\Phi_{m,t}$ and the same symmetry blocks as in Section 7. In particular, no new symbolic normal form is required: only the weight range is enlarged. The following counts are produced by the frozen program’s matrix-free `--estimate-only` path, using exactly the basis and symmetry conventions of the certificate computation:

(d_1, d_2)	new targets	$\sum N_{\text{raw}}$	shards	N_{raw} at the endpoint	B_{max}
(3, 3)	12	1,248,727	144	280,686	31,931
(2, 5)	10	288,129	60	84,865	16,973
(1, 8)	11	704,491	88	195,750	24,725
total	33	2,241,347	292	—	—

Here N_{raw} is the ambient number of coefficient variables before the finite-group decomposition, while B_{max} is the actual number of unknowns in the largest single modular-rank problem. The largest individual Wronskian layers have respectively 12,257, 8,848, and 12,184 unknowns. Thus none of the endpoint calculations forms one matrix with hundreds of thousands of columns: the complete frontier consists of 292 independent symmetry shards.

If B denotes the column dimension of one shard, the dense envelope for exact Gaussian elimination is $O(B^2)$ field elements of storage and $O(B^3)$ field operations. The implemented sparse elimination is usually smaller, but fill-in makes this dense envelope the safer planning model. For orientation, the completed cubic case $(m, t) = (14, 3)$ has $B_{\text{max}} = 8,830$. Its slowest block took 2,423 seconds in the historical endpoint run and 2,920 seconds in the later complete-profile rerun; monitoring of the historical run observed a 19.33 GiB working set on the recorded laptop. Using the completed cases in each family and a cubic-time dimensional extrapolation gives approximately 32–39 hours for the largest cubic shard, 40–50 hours for the largest conic–quintic shard, and 1.4 hours for the largest line–octic shard. These are deliberately rough resource estimates, not measured frontier running times or mathematical inputs; the sparsity and fill patterns differ among the three families. Nor does the sampled historical working set determine a per-shard peak. The matrix-free estimator does not construct the target rows, so exact row counts, nonzero counts, fill-in, and peak memory for the unrun frontier blocks are not yet known. We therefore make no claim here about a minimum hardware configuration; those quantities will be recorded with the completed certificates.

A cluster implementation remains straightforward. Each job is indexed by the degree pair, (m, t) , prime, character orbit, and, when present, involution eigenvalue. It generates its columns on demand, performs exact finite-field elimination, and writes an independent rank certificate; no communication with another shard is required. On each node, replacing tree-based sparse vectors by sorted packed rows and switching to blocked dense updates after fill-in should substantially reduce both memory traffic and overhead, while preserving the same exact modular proof. For scale, a square array for the largest block in each family occupies respectively 3.80, 1.07, and 2.28 GiB with 32-bit field entries, before row-space, polynomial-generation, fill-in, and elimination workspace. These figures are dimensional baselines, not peak-memory predictions. The existing C++ interface already exposes the orbit and eigenvalue shards, so the 292 primary-prime jobs can be dispatched directly by an ordinary cluster scheduler. Full column rank at one good prime is logically sufficient; selected large blocks should also be replayed at a second prime as an implementation check.

Therefore, the limiting cases form a concrete, communication-free target for parallel C++ computation, but this feasibility analysis is neither a hardware guarantee nor a proof of any unperformed vanishing. If every listed normalized integral block has full column rank, the same argument gives the prospective constants above. As the blocks are completed, their row

and nonzero counts, exact ranks, hashes, peak-memory records, and independently replayable certificates will be posted on Song-Yan Xie’s homepage. ²

AUTHOR CONTRIBUTIONS

Song-Yan Xie formulated the problem investigated in this paper. Lei Hou and Song-Yan Xie developed the $\mathcal{O}(3)$ slanted-vector-field argument, including the new fields required in the two-component setting. The theoretical foundation of the Key Vanishing Lemma is the exponent-bound and finite-linear-algebra method developed by Hou and Xie in the preceding two papers and adapted here to two components. For the present coupled verification, Pengchao Wang introduced a new algorithmic idea and used it to substantially improve the exact finite algorithm. This conceptual improvement, rather than a mere code implementation, is what makes the required verification feasible on a standard laptop. All authors checked the mathematical arguments and computations and jointly revised and polished the manuscript.

AI-ASSISTED PREPARATION

Generative artificial-intelligence tools were used for language polishing. Under detailed author direction, they also assisted in drafting portions of the $\mathcal{O}(3)$ argument; the authors supplied the core ideas, precise mathematical constraints, and iterative guidance. Apart from that author-directed drafting assistance, the ideas and arguments are due to the authors. The authors independently verified every argument in its final form and accept full responsibility for the manuscript.

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²The computational evidence and subsequent updates are maintained at <https://xiesongyan.github.io/>.

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ACADEMY OF MATHEMATICS AND SYSTEMS SCIENCE, CHINESE ACADEMY OF SCIENCES, BEIJING 100190, CHINA

Email address: houlei@amss.ac.cn

ACADEMY OF MATHEMATICS AND SYSTEMS SCIENCE, CHINESE ACADEMY OF SCIENCES, BEIJING 100190, CHINA

Email address: 2589355972@qq.com

STATE KEY LABORATORY OF MATHEMATICAL SCIENCES, ACADEMY OF MATHEMATICS AND SYSTEMS SCIENCE, CHINESE ACADEMY OF SCIENCES, BEIJING 100190, CHINA

SCHOOL OF MATHEMATICAL SCIENCES, UNIVERSITY OF CHINESE ACADEMY OF SCIENCES, BEIJING 100049, CHINA

Email address: xiesongyan@amss.ac.cn